





Bachelor Thesis

Quantization and Spike-Based Explainability in Spiking Neural Networks for Gravitational-Wave Glitch Classification

Mucaj Deva
Artificial Intelligence

Academic Year 2025/2026

Bachelor in Artificial Intelligence
Pavia 2026

Bachelor in Artificial Intelligence







Mucaj Deva

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Supervisors: **Claudio Zandron, Department of Informatics, Systems and Communication, UniMiB**
Davide Gerosa, Department of Physics "Giuseppe Occhialini", UniPv

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Abstract

Gravitational-wave detectors record a constant stream of short noise transients, the glitches, which can mimic or mask astrophysical signals and impair results. Gravity Spy sorts them by their time–frequency morphology, which is the first step towards tracing each class of glitch to its instrumental cause.

This thesis builds a spiking neural network (SNN) for the classification task, with a membrane decay realizable as a bit shift, and which classifies four Gravity Spy classes at 0.9833 ± 0.0014 validation accuracy over three seeds. Deploying it efficiently requires quantization, the core instrument of the thesis and its research question: whether a quantized SNN bases its decisions on the same evidence as its full-precision counterpart, and at what energy cost. In convolutional networks accuracy has been shown to survive quantization while explanations need not: an SNN adds the membrane potential as a second quantization target, and its explanations unfold in time.

The working hypothesis is that quantizing the membrane disrupts the timing of an explanation before quantizing the weights disrupts its spatial layout.

Explanations are computed with the Spike Activation Map (SAM), which is a gradient-free method that reads the spike trains directly. The divergence metrics implemented are first calibrated: maps of networks differing only in their seed correlate at 0.153, so each quantized model is compared only with the full-precision parent it is fine-tuned from. Weights, membrane and both arms are quantized at 8, 4 and 2 bits each.

Results show that down to 4 bits accuracy and explanations hold. At 2 bits weights the readout falls silent on most images and accuracy collapses to about 0.52, while a 2-bit membrane keeps 0.982 but shares a top-20 % overlap of only 0.438 with its parent, below the noise floor of every class, on maps that mostly remain faithful by the deletion metric. The latter configuration saves no energy, while a joint 4-bit model keeps accuracy with one eighth of the memory and an estimated energy between one sixtieth and about one third of the parent's.

The predicted order is not observed, as the temporal metric of the explanation does not move before the spatial ones. The membrane does act on timing, in the dynamics: at 2 bits every first-layer neuron that fires at least three times within an image does so at exactly regular intervals, against about half of them in the full-precision parents.

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Chapter 1
Introduction

1.1 Motivation

In September 2015 the two Advanced LIGO (Laser Interferometer Gravitational-Wave Observatory) detectors made the first direct detection of a gravitational wave, GW150914, which was produced by the merger of two black holes [1]. Two years later, LIGO and Virgo jointly observed GW170817, the first signal from a binary neutron star merger [2]. Its electromagnetic counterpart, from a short gamma-ray burst to a kilonova, was then followed by telescopes around the world, an event that marked the beginning of multi-messenger astronomy [3].

Since then, detections have gone from exceptional events to routine ones, leading to a continuous stream of data in the field of gravitational-wave astronomy, data that must be analyzed, often in real time. In this setting, the main obstacle is not the faintness of the astrophysical signals but the noise. The detectors are sensitive enough to record a constant stream of short, non-Gaussian transients of instrumental and environmental origin, known as *glitches*. What makes glitches both interesting and detrimental to astronomical analysis is that they can mimic genuine signals or overlap with them, which raises the false-alarm rate and reduces the effective sensitivity of searches.

Gravity Spy, the source of the data used in this work, is an Open Source project specifically created to address this problem. It combines citizen-science classification with machine learning, aiming to sort glitches into morphological families based on their time–frequency spectrograms [44]. Knowing which family a glitch belongs to is the first step towards tracing it back to its physical cause and, where possible, addressing the source of the problem and ultimately removing that cause at the instrument.

More broadly, machine learning has become indispensable to perform data-quality work, but it comes at a cost. The demand for computing power is indeed growing faster than the resources available, public alerts must be issued within minutes of a detection, and data volumes are expected to grow further with future observing runs and next-generation detectors. Making models lighter and cheaper to run is therefore a natural goal. Two approaches are particularly relevant: quantization, which compresses an existing network, and spiking neural networks (SNNs), which are designed from the outset for energy-efficient, and event-driven computation.

Running a neural network outside a data center generally means making it smaller, and the standard tool for this is indeed quantization. Quantization is a practice implemented to represent a network's weights and activations with fewer bits, trading numerical precision for savings in memory and energy. A quantized model is routinely validated by checking that its accuracy has not dropped, and for many applications this is a sufficient test.

It is not sufficient, however, when the model is also expected to explain its decisions. Rabbi et al. [33] evaluated five convolutional architectures at INT8 and INT4, comparing the attribution maps of the quantized models with those of their full-precision counterparts. Their core finding was that explanation similarity degrades measurably between

INT8 and INT4, even where accuracy differences are negligible. They also discovered that when quantization does cause a misclassification, the quantized model is generally attending to *different* regions of the input, rather than weighing the same evidence differently.

In other words, accuracy and interpretability do not degrade together.

This finding, however, is limited to conventional artificial neural networks (ANNs), and spiking neural networks (SNNs), the architecture most closely associated with low-power deployment [36], differ from ANNs in ways that make the question both harder and more interesting.

1.2 Research question

The question this thesis addresses is:

> *Does a quantized spiking neural network under edge-realistic constraints base its decisions on the same evidence as its full-precision counterpart, and at what energy cost?*

Spiking networks possess three properties that are absent in convolutional networks, and each of them is crucial for framing the research question.

A second quantization target. A spiking neuron carries a membrane potential which is a state variable read and written at every time step, and whose storage scales with the number of neurons, rather than the number of parameters. In the network studied here it occupies 367.5 KiB against 1 152.7 KiB of weights at full precision, and it becomes the larger share of the footprint as soon as the weights are compressed. It is a quantization target of its own [43], and it can and will be quantized independently of the weights [38], whereas a conventional network offers only weights and activations.

A temporal axis in the explanation. A spiking explanation is a *sequence* of maps, one per time step, which effectively builds a temporal axis in the explanations. Quantization may preserve where a network looks while displacing when it decides, a change to which spatial metrics are blind by construction.

A native explainer. The Spike Activation Map (SAM) [22] reads spike activity directly. It requires neither gradients, which side-steps the approximation error a surrogate gradient injects into a transplanted Grad-CAM-style method [37], nor ground-truth labels.

The working hypothesis follows from the first two properties:

> **Quantizing the membrane potential degrades the temporal structure of the explanation before quantizing the weights degrades its spatial structure**,

a hypothesis that originates from the principle that the membrane potential is the variable that integrates evidence over time, while a weight error is fixed for a whole forward pass. The hypothesis is falsifiable, and testing it precisely requires quantizing the two targets separately: the membrane arm should displace the temporal centre of mass of the explanation at a *higher* bit-width than the one at which the weight arm lowers its spatial correlation.

1.3 Contributions

To address the research question, it is necessary to employ a measurement tool. However, since the metrics used throughout the work yield values between zero and one whatever input they are given, a reference baseline is needed: before the results can be interpreted, one must know the values these metrics produce in the absence of any change. The thesis

is therefore structured into two parts: the first dedicated to the design and calibration the first dedicated to the design and calibration of the tool, and the second to its operational application to the quantization grid.

The core contributions are therefore:

1. **A hardware-realizable full-precision baseline.** The input resolution of 112 px was chosen based on network health, measured as dead and saturated neuron fractions, and on the resolution of the explanation maps, not on accuracy: at 64 px the maps of the second and third layers shrink to 16×16 and 8×8, too coarse for spatial metrics beyond the first layers. The membrane decay was also searched only among values realizable with a shift, alone or followed by a subtraction, and the search returned a value of $\beta = 0.75$. Over three seeds the reference reaches 0.9833 ± 0.0014 validation accuracy.

2. **A corrected implementation of the Spike Activation Map.** The decay parameter γ is fixed on the temporal axis it actually governs. The neuron contribution score now sums over strictly earlier time steps ($t' < t$), which removes a constant pedestal that had inflated map correlations.

3. **A calibration of the explanation-divergence metrics** at four levels: numerical noise, continued training with no bits removed, a thirtyfold change in γ, and independently initialized networks of equal accuracy. Spatial metrics are validated only between a model and its own parent, so every quantized model is compared with the full-precision counterpart model of its own seed.

4. **A bit-width characterization of explanation fidelity.** Weight and membrane quantization are applied separately and then together, at 8, 4 and 2 bits and three seeds, alongside a full-precision fine-tuning control.

5. **An analytical energy and memory account** of every operating point, with synaptic operations computed from per-layer firing rates, in order that accuracy, explanation fidelity, and energy can be read on a common bit-width axis.

The central result is a configuration that the accuracy criterion alone cannot discriminate from its neighbours. Quantizing the membrane potential to 2 bits leaves accuracy at parity, 0.982, yet the region its explanations highlight overlaps the full-precision one by 0.44, below the numerical floor of every class (0.70 to 0.84); a 4-bit weight model of the same accuracy keeps an overlap of 0.74.

The change is real but bounded. Where SAM is faithful, the deletion metric shows that in five of seven cases the new explanations still point at evidence the model uses, and that the parent's maps explain the quantized model about as well as its own: the quantized model highlights a different subset of evidence it still relies on.

The hypothesis stated above is not supported. Read against the floor and the fine-tuning level of each class, the two effects it orders appear at the same bit-width. At 8 bits neither crosses both levels in any class-and-seed cell. At 4 bits the membrane arm moves the temporal centre of mass beyond both levels in every class, and in nine of the twelve cells. The weight arm lowers the spatial correlation beyond both in every class, and in ten of the twelve cells, and moves the centre of mass in the direction in which fine-tuning alone already moves it. At 2 bits the membrane modifies *where* the explanation points more than *when*.

The membrane does reach the timing of the network's own dynamics, at 2 bits. In all six models with a 2-bit membrane, every first-layer neuron that fires at least three times within an image does so at exactly regular intervals: the coefficient of variation measured within the image is zero on all of the 20 to 53 % of neuron-image pairs that reach three spikes. In the three full-precision parents, measured the same way, 42 to 53 % of those pairs are exactly regular and the mean coefficient is 0.153 to 0.188. At 4 bits the regularity of layer 1, pooled over the partition, stays within 0.01 of the parent, so the dynamics do

not change before the spatial explanation does. The temporal metric of the explanation does not register the change at 2 bits either.

The configuration with the changed explanation is also dominated on energy and memory, so in this network there is no trade-off to make. A joint 4-bit model keeps accuracy at parity and an explanation that moves only slightly beyond what continued training alone produces, with one eighth of the memory and about 1/60 of the estimated energy if the input image and the accumulators are quantized to the width of the weights. With the first layer in floating point and 32-bit integer accumulators the factor is between 1/2.4 and 1/3.2 across seeds.

1.4 Application domain

The classification task is the identification of transient noise — known as *glitches* — in the data of the LIGO gravitational-wave detectors, using the Gravity Spy release [44, 16, 15].

The domain was chosen for three reasons. Explanation fidelity particularly has an *operational* consequence: glitch classification guides detector characterization, where a commissioner can act physically on the instrument: a compressed model reaching the same label from different evidence would preserve the label but would be corrupting the diagnostic cue.

The classes have physically interpretable morphology, so a domain expert can judge whether an attention map falls where the glitch actually is, which is not necessarily the case of a generic image benchmark. Moreover, the data is public and large, with every trigger labelled by the Gravity Spy classifier. The labels are machine labels (not a verified ground truth), and in Section 2.4 it is reported how far the scores released with them support them.

1.5 Structure

Chapter 2 reviews spiking neural networks, their training by surrogate gradients, quantization for spiking architectures, explainability methods applicable to them, and the gravitational-wave classification task. Chapter 3 describes the data pipeline, the model, the selection of the operating configuration, the Spike Activation Map and the protocol used to calibrate the divergence metrics. Chapter 4 describes the quantization study: the quantizers, the grid, the fine-tuning protocol, the divergence metrics and the energy model. Chapter 5 reports the full-precision reference, the physical validation of the network dynamics, the explainability analyses, the calibration, and the results of the quantization grid. Chapter 6 discusses the findings and their limits, and outlines future work.

Chapter 2
Background and Related Work

2.1 Spiking neural networks

Spiking neural networks (SNNs) are often described as the third generation of neural networks [27]. They take their inspiration from biological neurons, which communicate through brief, all-or-nothing electrical impulses called *spikes*. In a conventional ANN, each neuron outputs a continuous value computed once per input, while in an SNN, information is carried by sequences of binary events distributed over time: at each of T discrete time steps, a neuron either emits a spike or stays silent. The network therefore processes an input as a short temporal sequence. This holds even when the input is static, as with a spectrogram, our specific case, which is presented repeatedly or encoded as spikes over the T steps.

This representation is exactly what makes SNNs attractive for low-power computing. Because activations are binary, the multiply–accumulate operations that dominate the cost of an ANN reduce to simple accumulations, performed only when a spike actually is fired. Computation is therefore sparse and event-driven, with a silent neuron costing almost nothing. Neuromorphic processors such as Intel's Loihi [10] are built to exploit this sparsity directly, and the energy advantage of an SNN is set to grow as its firing activity decreases.

The most widely used neuron model in deep SNNs is the leaky integrate-and-fire (LIF) neuron [14]. In this configuration, each neuron maintains an internal state, its membrane potential U, which integrates incoming input, decays ("leaks") towards rest over time, and triggers a spike when it crosses a threshold θ.

As shown in Figure 2.1, the LIF model sits between biologically detailed neuron models and the artificial neurons of conventional deep learning (ANNs).

In the discrete-time formulation used by snnTorch [12], the dynamics of a neuron at time step t are the following

$$U[t] = \beta\, U[t-1] + WX[t] - S[t-1]\,\theta, \tag{2.1}$$

$$S[t] = \begin{cases} 1 & \text{if } U[t] > \theta, \\ 0 & \text{otherwise,} \end{cases} \tag{2.2}$$

where $WX[t]$ is the weighted input from the previous layer, $S[t]$ is the binary output spike, and the subtraction term resets the potential by θ after each spike. The decay factor $\beta \in [0, 1)$ sets how much of its past state the neuron retains: values close to 1 give a long memory, so the neuron integrates evidence over many time steps; smaller values make it respond mainly to recent input.

The property of the membrane potential to give the LIF neuron a memory is what separates it from a stateless activation function: the output at time t depends on the entire history of inputs up to t. The same mechanism, nonetheless, also creates the main difficulty in training. The spike function in Eq. (2.2) is a step function, whose derivative is zero

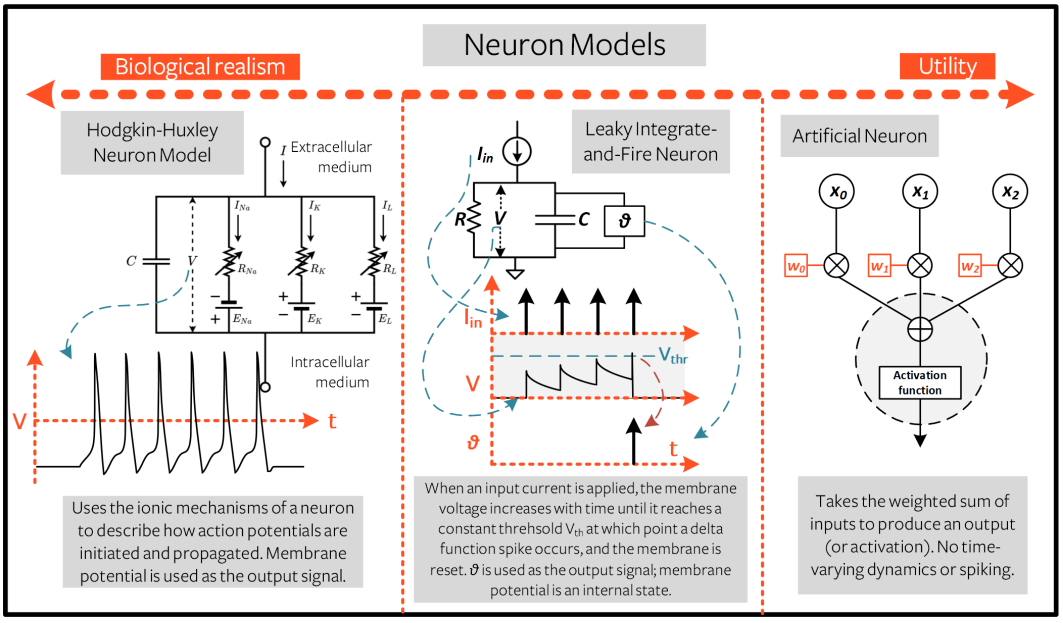


Figure 2.1: Neuron models ordered from biological realism to computational utility. Left: the Hodgkin–Huxley model describes the generation of action potentials through voltage-dependent ionic conductances, at a high computational cost. Centre: the leaky integrate-and-fire (LIF) neuron reduces the membrane to an RC circuit: the capacitor C integrates the input current I_{in}, while the resistor R makes the stored charge leak away over time, and a spike is emitted when the membrane potential reaches the threshold (denoted ϑ in the figure and θ in the text), after which the potential is reset. Unlike the Hodgkin–Huxley model, the membrane potential of a LIF neuron is an internal state and the spike is the output. Right: the artificial neuron of conventional ANNs, which computes a weighted sum of its inputs followed by an activation function, with no temporal dynamics and no spikes. Reproduced from the snnTorch documentation [11] (MIT License).

almost everywhere and undefined at the threshold, so standard backpropagation cannot pass through it. The usual solution is the *surrogate gradient* approach [30]: the step function is kept in the forward pass, but in the backward pass its derivative is replaced with that of a smooth approximation, such as an arctangent. Combined with backpropagation through time over the T steps, this allows deep SNNs to be trained with the same gradient-based tools as conventional networks.

These properties change the quantization problem in three main direct ways. First, SNN activations are already binary, so the precision that quantization removes lies in the weights and in the membrane potential, a state variable with no direct equivalent in an ANN. Second, the threshold makes the network highly sensitive to small numerical changes. Indeed, a perturbation that moves U slightly above or below θ turns a spike into silence, or the reverse. Because the membrane carries its state forward, such an error does not stay local, but can propagate through later time steps and layers. Third, explanations of an SNN can be temporal as well as spatial. The Spike Activation Map (SAM) [22] builds attribution maps directly from the timing of spikes, without computing gradients, and therefore shows not only *where* the network finds evidence in the input but also *when* during the T time steps it does so. Whether quantization preserves this temporal reasoning, and not only accuracy, is the question this thesis sets out to answer.

2.1.1 Training by surrogate gradients

Unrolled over its T time steps, a network of LIF neurons is effectively a recurrent network: the membrane potential is a hidden state that each step reads and updates (Equation 2.1). Training it by gradient descent requires solving two credit-assignment problems at once [30]: a *spatial* one, across layers, as in any multi-layer network, and a *temporal* one, since a weight influences the loss through every later state of the neurons it feeds. Backpropagation through time (BPTT) solves both by applying the chain rule to the unrolled graph. For a weight W feeding a neuron with potential $U[t]$ and output $S[t]$,

$$\frac{\partial \mathcal{L}}{\partial W} = \sum_t \frac{\partial \mathcal{L}}{\partial S[t]} \frac{\partial S[t]}{\partial U[t]} \frac{\partial U[t]}{\partial W}, \qquad \frac{\partial S[t]}{\partial U[t]} = \Theta'(U[t] - \theta), \tag{2.3}$$

where Θ is the Heaviside step of Equation 2.2 and $\partial U[t]/\partial W$ is the total derivative through all earlier states. The price is memory: every state of every time step must be kept until the backward pass, a cost that grows as $O(NT)$ for a layer of N neurons.

The obstacle. Every path through Equation 2.3 contains the factor Θ', which is zero everywhere except at the threshold, where it is undefined. The gradient is therefore either zero or meaningless: a small change of a weight almost never flips a spike, so the loss is piecewise constant in the weights, with flat plateaus on which gradient descent cannot move. This is the same problem met by networks with binary activations, and the solutions developed for spiking networks are inspired by those [30].

The surrogate. A surrogate gradient leaves the model unchanged and alters only its derivative. The forward pass keeps the exact binary spike, while in the backward pass every occurrence of Θ' is replaced by the derivative σ' of a smooth function peaked at the threshold,

$$\frac{\partial S[t]}{\partial U[t]} \longrightarrow \sigma'(U[t] - \theta). \tag{2.4}$$

Neurons close to threshold, meaning those for which a small change could flip a spike, receive the largest gradient, and gradient flows through silent neurons as well. The resulting update is the gradient of a *virtual* surrogate loss, never computed explicitly, which follows the true one while remaining continuous and finite [30]. This distinguishes the approach from *smoothed* spiking networks, which involve replacing the spike itself with a soft or probabilistic function and so change the model that is deployed.

The choice of surrogate. Piecewise-linear, fast-sigmoid and exponential surrogates have all been used successfully, all three nonlinear and monotonically increasing towards the threshold. The diversity of working choices suggests that the method does not depend critically on the exact shape [30]. This work uses the arctangent surrogate [13] as implemented in snnTorch [12],

$$\sigma'(x) = \frac{\alpha/2}{1 + \left(\frac{\pi}{2}\,\alpha\, x\right)^2}, \qquad \alpha = 2, \tag{2.5}$$

which equals 1 at the threshold and falls off as the inverse square of the distance from it, so that neurons far from threshold still receive a small, non-zero gradient. The reset term $-S[t-1]\,\theta$ of Equation 2.1 is treated as a constant in the backward pass: the recurrence then contributes $\partial U[t]/\partial U[t-1] = \beta$ alone, without a second surrogate factor through the reset, a simplification that has been found empirically to improve training [30]. With this training configuration, spiking networks can manage to reach accuracies comparable to conventional architectures on image benchmarks [12].

The same device reappears in Section 2.2: the straight-through estimator used to train through a rounding operation [7] replaces the zero derivative of a step function with that of a smooth one, exactly as the surrogate gradient does for the spike.

A consequence for explainability. The substitution motivates the choice of explanation method in Section 2.3: any gradient-based attribution computed on such a network inherits the approximation error of the surrogate, because the gradients it uses are not the gradients of the function the network actually computes.

2.1.2 Input encoding

A spiking network processes its input over T discrete time steps, while an image is a single static frame, so the frame must be turned into a sequence. In this context, two strategies dominate. In *rate coding* each normalized pixel intensity is used as the probability of emitting a spike at every step, and the value is carried by the spike count; in *direct* (or *constant*) encoding, the analog frame itself is presented, identical, at every time step.

Rate coding pays its simplicity in time steps: a pixel is represented by counting spikes, so with T steps it can take only $T+1$ distinct values, and the sampling makes the representation stochastic: the discretization error shrinks only as T grows, which is why rate-coded and converted networks typically require between 100 and 2 000 time steps to approach the accuracy of their non-spiking counterparts [34].

Direct encoding, as introduced by DIET-SNN [34], removes that error at the source. "The pixel intensities of an image are applied directly to the input layer of the SNN at each timestep", and the first convolutional layer, followed by LIF neurons, becomes at once a feature extractor and a learned spike generator: it integrates the weighted pixel values and fires when its membrane potential crosses threshold. Since the input carries its full precision at every step, the number of steps is no longer bounding the precision of the input. DIET-SNN combines this encoding with a trained membrane leak and firing threshold per layer, which increase activation sparsity, and reaches 69 % top-1 accuracy on ImageNet with 5 time steps and about 12× less compute energy than the equivalent conventional network [34]. This work adopts the encoding but not the trained neuron parameters: β and the threshold are shared by all layers, and β is chosen by the constrained search of Section 3.4.2.

Direct encoding has three remarkable consequences that the rest of this thesis relies on.

Temporal structure is generated inside the network. Since the stimulus does not vary, **all temporal structure in the network originates from the internal LIF dynamics** — leak, integration and reset — and not from variation in the input. The time axis of a spike

train, and therefore of a Spike Activation Map, is the time the network takes to integrate evidence, which is not the time of the glitch: in an Omega-scan spectrogram, physical time runs along the horizontal axis of the image and is, for the network, a spatial dimension. Every temporal analysis in this thesis is to be read in that light.

The first layer is the only one that multiplies. Every other layer receives binary spikes and only accumulates, but the first convolution receives analog pixel values and performs full multiply-accumulates. Because its input is identical at every step, so is its output current, which can be computed once rather than T times. Section 4.8 shows how this choice decides whether the spiking network is cheaper than its conventional equivalent.

The input is deterministic. Without sampling, two forward passes on the same image produce identical spike trains, which is what allows two models to be compared spike for spike on the same input, as the calibration of Section 5.6 requires.

2.2 Quantization

Quantization represents the parameters and intermediate values of a network with fewer bits than the 32 of floating point. Storing a value in b bits divides its memory by $32/b$, and integer operations on narrow operands cost far less energy than floating-point ones (Section 2.2.2), which is why quantization is the standard practice adopted between a trained model and its deployment on constrained hardware [20, 29].

The uniform quantizer. A uniform quantizer maps a real value x to one of at most 2^b equally spaced levels: it divides by a step Δ, rounds to the nearest integer, clips to the representable range, and multiplies back,

$$\hat{x} = \Delta \cdot \mathrm{clip}\Big(\mathrm{round}\Big(\frac{x}{\Delta}\Big),\, q_{\min},\, q_{\max}\Big) \tag{2.6}$$

where $q_{\min}$ and $q_{\max}$ are the smallest and largest integer levels representable with b bits: $-(2^{b-1}-1)$ and $2^{b-1}-1$ for a symmetric signed grid, which keeps zero exactly representable at the cost of one unused code, 2^b-1 levels in all; 0 and 2^b-1 for an unsigned one, which uses all 2^b. The largest representable value is therefore $\Delta\, q_{\max}$.

For a symmetric grid over $[-w_{\max}, +w_{\max}]$ the largest level must reach $w_{\max}$, so the step is

$$\Delta = \frac{w_{\max}}{2^{b-1}-1} = \frac{2w_{\max}}{2^b-2}, \tag{2.7}$$

the step the weight quantizer of Chapter 4 computes for each output channel. The unit in which the calibration expresses a perturbation, the 8-bit step of a single grid set by the largest weight of the whole network, is defined in Section 4.3.

Two main errors follow from Equation 2.6, with the choice of range trades one against the other: values inside the range are moved by at most $\Delta/2$ (*rounding* error), values outside it are cut to the nearest bound (*clipping* error). For values spread over many steps the rounding error is close to uniform on $[-\Delta/2, \Delta/2]$, with a root-mean-square of $\Delta/\sqrt{12}$; a Gaussian perturbation whose standard deviation is one step, the unit of Section 5.6.3, is therefore about 3.5 times larger than the rounding it stands for. A wide range avoids clipping but makes Δ coarse; a narrow one refines Δ at the cost of saturating the tails.

When the quantity is non-negative, or its negative values carry no information for the task, an *unsigned* grid over $[0, x_{\max}]$, with step $x_{\max}/(2^b-1)$, spends no levels on values the network does not use. The membrane potential of this network turns out to be of the second kind (Section 5.7).

More specifically, the step can be shared by a whole tensor (*per-tensor*) or computed separately for each output channel (*per-channel*). A per-tensor step is set by the largest

value in the tensor, so channels whose weights are systematically smaller use only a fraction of the available levels; the per-channel scheme avoids this at the price of one scale factor per channel [24, 29].

Given a quantizer, two families of method decide *when* it is applied: after training, or during it.

Post-training quantization (PTQ). PTQ converts an already trained full-precision model: the weights are rounded to their grid, and the ranges of the activations, which depend on the data, are estimated on a small unlabelled calibration set [29]. It is a relatively fast method and needs neither labelled data nor a training pipeline, which makes it the "push-button" option and the one most used for rapid deployment. Its limit is precision: at 8 bits the rounding error is small compared with the spread of the weights, and per-channel PTQ keeps classification accuracy within about 2 % of floating point on standard convolutional networks. At 4 bits the step is sixteen times coarser, and accuracy typically degrades unless more elaborate correction methods are used [24, 29]. Rabbi et al. [33] use static weight-only PTQ at INT8 and INT4, the setting in which they observe explanations degrading before accuracy.

Quantization-aware training (QAT). QAT includes the quantizer in training, so that the network can adapt its weights to the grid. Specifically, in the forward pass every quantized tensor is passed through Equation 2.6 — *simulated* or *fake* quantization, since the values are rounded but still stored and processed in floating point — so that the loss is computed on the network as it will be deployed [20]. The difficulty lies in the backward pass: rounding is a step function, whose derivative is zero almost everywhere, and no gradient would reach the weights. The *straight-through estimator* solves it by treating rounding as the identity in the backward pass [7]: the forward pass sees the quantized value, the gradient flows as if it had not been quantized. This is the same device, in a different guise, as the surrogate gradient of Section 2.1.1, which also keeps a step function in the forward pass and replaces its derivative in the backward one. QAT needs labelled data and additional training, usually as a fine-tuning of the full-precision model, and in exchange reaches lower bit-widths: it narrows the 8-bit gap to about 1 % and makes 4-bit weights usable, with losses between 2 and 10 % depending on the size of the network [24, 29].

The two families therefore answer different needs: PTQ when the model must be compressed cheaply and 8 bits suffice, QAT when the target precision is lower than PTQ can reach. In a spiking network both apply not only to the weights but also to the membrane potential, as the next section describes.

2.2.1 Quantization for spiking networks

Spiking architectures add constraints of their own, as could already be inferred. The membrane decay β multiplies a stored state at every time step, so on integer hardware it is cheapest when realizable as a bit shift: $\beta = 2^{-k}$ is a right shift, and $\beta = 1 - 2^{-k}$ is a shift plus a subtraction. MINT [43] quantizes both targets uniformly and shares a single scale factor between weights and membrane potentials. Since the weighted input and the stored state are then expressed in the same integer units, they can be added directly, and the multipliers that conventional uniform quantization needs to rescale one into the other disappear from the inference loop. At 2 bits, MINT reports 90.6 % accuracy on CIFAR-10 with a VGG-16 and a memory reduction of 93.8 % with respect to full precision. HardFSNN [26] shows the cost of that sharing: weights have a narrow, centred distribution, whereas membrane potentials have a much broader one with a long tail, so a scale fitted to the weights compresses the dynamic range of the membrane and collapses many distinct potentials onto the same level. It therefore scales the membrane in proportion to the weights rather than identically, replaces batch normalization with an integer, shift-based version, and verifies the result on an FPGA accelerator, noting that most quantized spiking

networks are still evaluated with simulated quantization, in which part of the computation remains in floating point.

Smith et al. [38] argue independently that accuracy is an insufficient criterion for quantized spiking networks. They in fact propose an Earth Mover's Distance between per-neuron firing-rate distributions as a diagnostic of behavioural divergence. They separate weight from membrane quantization as this thesis does, and report that membrane-potential quantization requires at least four bits to preserve both accuracy and dynamics while learned weight quantization remains viable at two. They propose, without testing it, a mechanism for the asymmetry: a weight quantization error is fixed for an entire forward pass, whereas a membrane error propagates through the recurrent update at every time step.

This thesis reaches the opposite result at 2 bits: the membrane arm keeps accuracy at parity (Section 5.9). A plausible reason is where the quantizer acts. Here it rounds the state carried into the next step, after the spike decision (Section 4.4). At every step the threshold is compared with β times the quantized previous state plus a full-precision input current, so the comparison itself is never made on a 2-bit value.

2.2.2 Energy and memory accounting

Since no model is deployed on hardware in this work, energy is estimated analytically, as is standard in the spiking literature: the operations an inference requires are counted and multiplied by a per-operation cost at a reference process node, conventionally the 45 nm figures of Horowitz [17]. The result is a like-for-like comparison between configurations of the same network.

Counting operations. In a conventional layer every connection performs one multiply-accumulate (MAC) per inference. In a spiking layer the input is binary, so a connection only *adds* its weight, and only when the presynaptic neuron fires. The number of these synaptic operations (SOP) is therefore the dense MAC count scaled by the firing rate f of the input layer and by the number of time steps,

$$\mathrm{SOP} = f \cdot T \cdot \mathrm{MAC}. \tag{2.8}$$

In 32-bit floating point a MAC costs 4.6 pJ (a 3.7 pJ multiplication plus a 0.9 pJ addition) and an accumulate 0.9 pJ. That factor of five is the physical reason a spiking network can be cheaper, and it also sets the limit: a spiking layer saves energy only while $f \cdot T < 4.6/0.9 \approx 5$, so sparsity and a short time window are both necessary. The first layer is an exception, since under direct encoding it receives analog values and performs full MACs (Section 2.1.2).

Quantized operations. The same table supplies the integer figures used for quantized operating points: an 8-bit multiplication costs 0.2 pJ and an 8-bit addition 0.03 pJ, thirty times less than its floating-point counterpart, so that an 8-bit MAC costs 0.23 pJ under the same convention that gives 4.6 pJ in floating point. Narrower integer operations are not tabulated, so any figure below eight bits is an extrapolation, whose rule is stated in Chapter 4.

What the count leaves out. The estimate covers arithmetic only. It ignores the update of each neuron (leak, threshold comparison, reset), control logic, and above all data movement, which on real hardware is often the larger cost: in the same 45 nm table, reading 64 bits from on-chip SRAM costs between 10 and 100 pJ depending on its size, and from off-chip DRAM between 1.3 and 2.6 nJ [17], one to three orders of magnitude more than an arithmetic operation.

Why memory is accounted separately. For this reason the memory footprint is reported alongside the energy estimate: the energy of an access depends on where the data lives, which is a property of the hardware, whereas the number of bits to be stored is a property of the model. The footprint is computed as the bits needed for the weights plus the bits needed for the membrane state of every neuron. The second term is specific to spiking networks: a conventional network can discard the activations of a layer once the next one has been computed, but a spiking network must keep the potential of every neuron from one time step to the next for the whole inference. That state scales with the number of neurons rather than the number of parameters, and it can be quantized independently of the weights (Section 2.2).

2.3 Explainability for spiking networks

Attribution methods developed for conventional networks transfer badly to spiking ones, for two different causes. Gradient-based methods such as Grad-CAM [37] weight feature maps by the gradient of the class score, and in a spiking network that gradient exists only as a surrogate (Section 2.1.1). The approximation is applied again at every layer the gradient crosses, so the error accumulates with depth: Kim and Panda [22] observe that Grad-CAM transplanted to spiking networks produces smoothed, poorly discriminative maps, worst at the shallow layers. Perturbation methods such as LIME [35] avoid gradients altogether and are architecture-agnostic, but they need hundreds to thousands of forward passes per explanation. Both families, moreover, return a single map per input, and so discard the temporal axis along which a spiking network computes.

The Spike Activation Map. The Spike Activation Map (SAM) of Kim and Panda [22] was designed for spiking networks from the start. It rests on a biological observation: spikes that follow each other at short inter-spike intervals are more likely to drive the membrane of the downstream neuron above threshold, and therefore carry more information than isolated ones. SAM numerically quantifies this into a score computed in three steps. A *temporal spike contribution score* weights a past spike at t' by its recency with respect to the current step t,

$$T(t, t') = \exp\left(-\gamma\,|t - t'|\right), \tag{2.9}$$

where γ sets how fast the contribution of a spike fades. A *neuronal contribution score* sums these weights over the spikes the neuron emitted before t,

$$N^{k}_{ij,t} = \sum_{t' \in P^{k}_{ij},\, t' < t} T(t, t'), \tag{2.10}$$

where P^{k}_{ij} is the set of firing times of the neuron in channel k at position (i, j); it is high for a neuron that has fired often and recently, and low for one that fired rarely or long ago. Finally, the map at time t keeps the score only of the neurons that are firing at t and sums over the channels of the layer,

$$M_{ij,t} = \sum_{k} N^{k}_{ij,t}\, S^{k}_{ij,t}, \tag{2.11}$$

yielding one spatial heat map per time step and per layer.

Properties. Several properties of SAM matter for this thesis. It uses solely the spike trains of the forward pass, so the explanation describes the function the network actually computes, and its cost is marginal — a store of recent spike times and a lookup table for the kernel, which Kim and Panda note could be implemented in an inference accelerator. It needs no class label: SAM shows where the network concentrates its activity for

a given input, not the evidence for a particular class, so its faithfulness must be tested against the class the model predicts. And it has a temporal axis: an explanation is a sequence of T maps, which is what makes it possible to ask whether quantization changes *when* a network decides as well as *where* it looks. Kim and Panda validate the method on Tiny-ImageNet, finding SAM more discriminative than Grad-CAM adapted to spiking networks.

Feature-attribution methods for spiking networks have also been surveyed more broadly [31], but SAM remains the method that reads the spike trains directly, and it is the explainer used throughout this work. Its one free parameter, γ, is fixed in Section 5.5.1.

2.3.1 Evaluating an explanation

A stable explanation can still be wrong, so stability requires a separate test of faithfulness. The *deletion metric* of Petsiuk et al. [32] removes pixels in decreasing order of attributed importance, and records how quickly the model's confidence falls, summarized as the area under that curve. The faster the fall, the smaller the area. The underlying hypothesis is that a faithful map degrades the prediction faster than a map that carries no information. Petsiuk et al. read the area on its own; this thesis compares it with the removal of pixels in random order on the same image (Section 5.5.2), so that a map counts as faithful only by the margin it gains over chance.

The metrics used to quantitatively compare two explanations with each other are borrowed from the saliency-model literature, where they were specifically developed to compare a predicted map against human eye fixations [8]: Pearson correlation, structural similarity [42], and the intersection-over-union of the highest-scoring regions. Bylinskii et al. observe that rankings produced by different metrics frequently disagree, which is the main reason for reporting more than one. This thesis reports the correlation and the overlap. Structural similarity, which Rabbi et al. [33] use alongside them, is left out as redundant with the two.

A separate line of work asks whether attribution methods are testing anything at all. Adebayo et al. [4] randomize model weights progressively and verify that explanations change in response, as a map that survives the destruction of the model it sets to explain is not explaining it. The perturbation experiment of Section 5.6.3 is a graded form of that test, run in the opposite direction.

2.4 Gravitational-wave glitch classification

The sensitivity of the LIGO interferometers is limited by non-Gaussian transient noise of instrumental and environmental origin. As introduced in Chapter 1, these glitches are short, can mimic astrophysical signals, and can be grouped into morphological families that correlate with distinct physical causes.

The Gravity Spy project [44, 6] combines machine classification with citizen science to label them. The release used here [15] covers observing runs O1 through O3b and provides, for each Omicron trigger, the trigger metadata, the per-class scores of a convolutional classifier, the label that classifier assigns, and the addresses of the Omega-scan spectrograms, which are downloaded separately from the Zooniverse archive. The labels used in this work are therefore machine labels, not volunteer consensus, and no threshold on the classifier's confidence was applied when the data were selected.

The scores released with the labels do not always support them. Of the 9 568 triggers selected for the four classes, the score of the assigned class is below 0.5 for 24 % and below 0.01 for 615, and for 4.4 % the label is not the class with the highest of the four scores. The disagreement is concentrated in two classes: the score falls below 0.5 for 53 % of the Violin_Mode triggers and 37 % of the Extremely_Loud ones, against 4 % of Blip and 3 % of Scattered_Light. An accuracy measured against these labels is an agreement with the Gravity Spy classifier, not with a verified truth.

The classification task is therefore image classification over time–frequency representations. Its interest for this thesis is not its difficulty — at 98 % agreement with the reference classifier it is not difficult — but that the explanations have a consumer. Glitch classification is directly connected with detector characterization, in which the morphology of a glitch family is traced back to an instrumental cause and the instrument is modified accordingly. An explanation that points at the wrong part of the spectrogram sends a commissioner to the wrong causes.

2.5 The gap this thesis occupies

There are two close works useful to frame this thesis. Rabbi et al. [33] measure explanation fidelity under quantization, but for conventional networks, implementing post-training weight-only quantization, and relying purely on spatial explanations. Smith et al. [38] address spiking networks and separate the two quantization targets, but measure divergence in *population firing statistics* aggregated over a dataset, and do not take into account explanations.

The unoccupied ground is therefore a per-sample, spatially and temporally resolved explanation fidelity for a spiking classifier with the two quantization targets separated.

Chapter 3
Methodology

This chapter describes the data pipeline, the model, the procedure by which the operating configuration was selected, and the implementation of the explainability metrics. Every measurement reported here is on the validation partition. The code, the configuration and the result file behind every table are public: Appendix C gives the repository, the environment, the seeds, and the script that produces each result.

3.1 Data

3.1.1 Acquisition

The Gravity Spy machine-learning release [15] is distributed as eight CSV files, one per detector (H1, L1) and observing run (O1, O2, O3a, O3b). Each record corresponds to an Omicron trigger with signal-to-noise ratio above 7.5 and peak frequency between 10 Hz and 2048 Hz, and carries the trigger metadata, the per-class confidence of the Gravity Spy classifier, the assigned label, and URLs to Omega-scan spectrograms.

The labels are the classifications produced by that convolutional model, neither a physical ground truth nor the volunteer-combined label of the separate citizen-science release. Accuracy reported in this thesis therefore measures agreement with that reference classifier.

3.1.2 Class selection

The original dataset featured twenty-four classes, of which two are not glitch morphologies: `No_Glitch`, the absence of a morphological trigger, and `None_of_the_Above`, the classifier's residual category. Only four classes were retained — **Blip**, **Extremely_Loud**, **Scattered_Light**, **Violin_Mode** — on three criteria established before any separability analysis, the successive multivariate checks are therefore a verification rather than the selection rule.

Abundance. These are all sufficiently populated classes, with 29 573, 24 002, 99 665 and 2 392 members respectively. The smallest sets the balanced cardinality at 2 392 per class, sufficient to train without augmentation (Figure 3.1).

Morphological diversity. The four retained classes span physically distinct families of instrumental noise [44, 16, 6]: short broadband transients of teardrop morphology and ∼0.04 s duration (Blip) [9]; high-amplitude events that saturate the spectrogram (Extremely_Loud); long-duration arches of 2–2.5 s correlated with ground motion, caused in O3 by relative motion between the optical suspension and reaction-mass chains (Scattered_Light) [40]; and narrowband lines at ∼500 Hz and harmonics, corresponding to resonances of the glass fibres suspending the mirrors (Violin_Mode). The choice of spanning the morphological range serves two purposes: it preserves class separability,

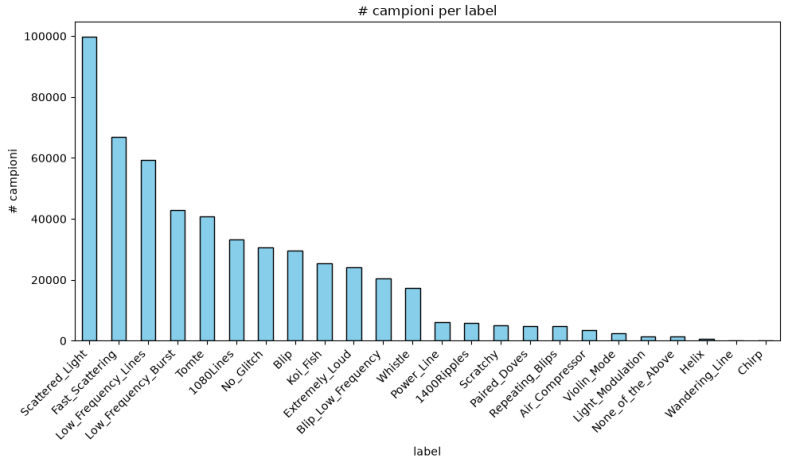


Figure 3.1: Class frequencies in the Gravity Spy machine-learning release, all observing runs and both detectors combined. The distribution is strongly imbalanced.

which keeps the classification problem itself manageable, since it is not the main scope of the thesis, and makes the explanation analysis informative, since each class should elicit a recognizably different attention pattern. An example of each is shown in Figure 3.2.

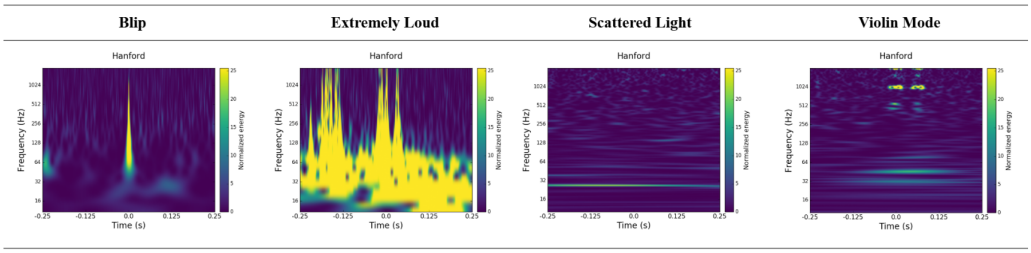


Figure 3.2: One Omega-scan spectrogram per retained class (Hanford detector, 0.5 s window). From left: a short teardrop-shaped transient, a saturated broadband event, a long low-frequency band, and narrow lines near 500 Hz and 1 kHz.

Practical constraint. Four classes keep training and the downstream quantization grid feasible on a single 6 GB consumer GPU.

Verification in the tabular domain supported the choice, as it shows that each class is isolated by a *different* descriptor, which is the relevant property: Extremely_Loud occupies $\log(\mathrm{SNR}+1)$ between 6 and 12 while the others lie between 2 and 5; Blip is separated by duration; Scattered_Light sits below roughly 100 Hz in peak frequency with a bimodal bandwidth distribution; and Violin_Mode is the only class extending to 2 kHz (Figure 3.3). A UMAP projection [28] of the standardized descriptors, computed on a stratified subsample of 500 events per class ($n_{\mathrm{neighbors}} = 15$, seed 42), confirms the separation independently (Figure 3.4).

Violin_Mode is the weakest case, and this nature along with its conformation anticipates a result that recurs throughout the thesis analysis and results: in the $\log(\mathrm{SNR}+1)$ versus $\log(\mathrm{duration}+1)$ plane it overlaps both Blip and Scattered_Light, being separated only in peak frequency. It will be the lowest-recall class, on the mean over seeds, in every configuration that classifies, and the most sensitive class whenever the kernel or the weights of a fixed network are perturbed.

As a side but dutiful note, restricting the problem to four classes keeps it tractable, but means the results are not directly comparable to the full Gravity Spy benchmark.

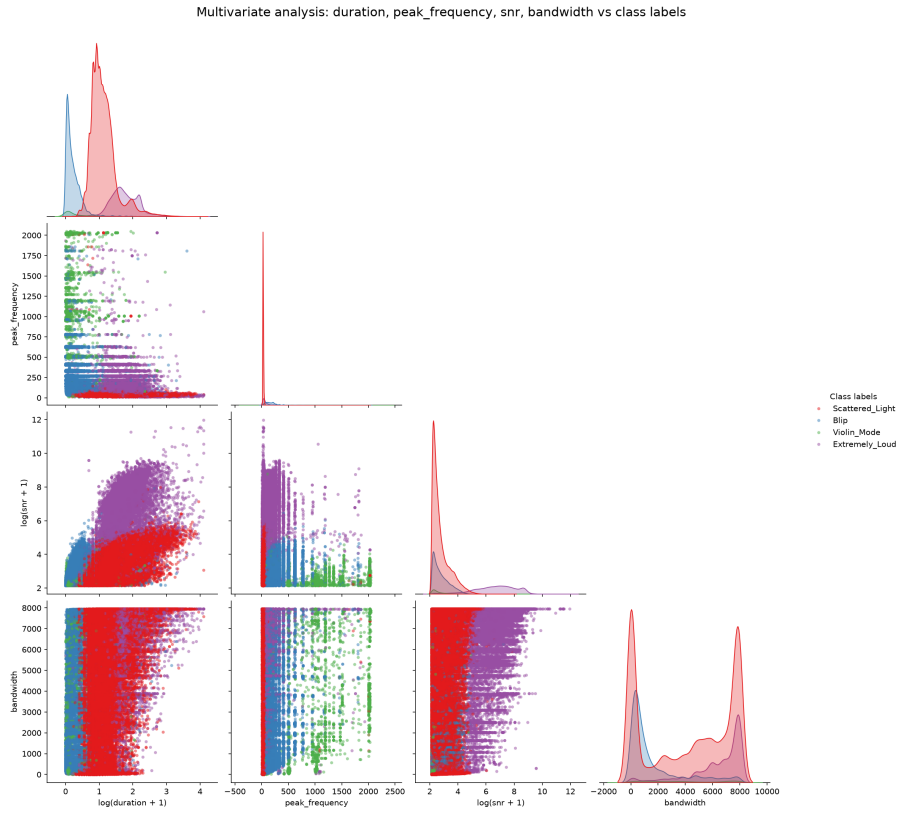


Figure 3.3: Pair plot of the four retained descriptors, coloured by class. Each class is isolated by a different descriptor; Violin_Mode overlaps Blip and Scattered_Light except in peak frequency.

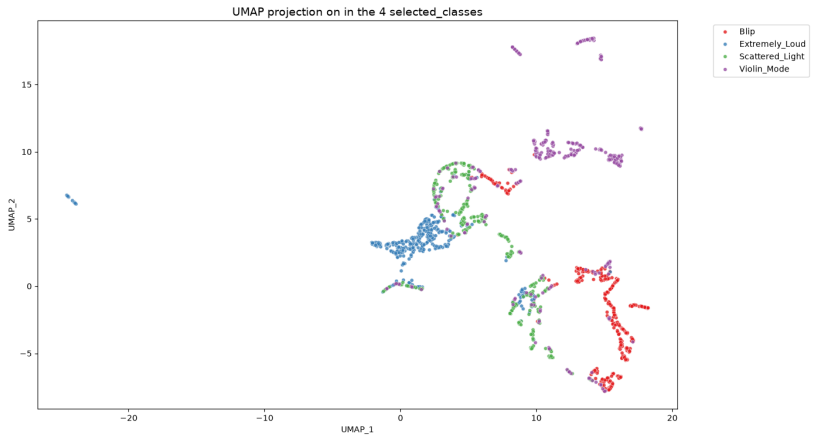


Figure 3.4: Two-dimensional UMAP projection of the standardized descriptors, 500 events per class.

3.1.3 Feature selection

Of the twelve numerical Omicron descriptors, four were retained. The selection rests on four criteria, derived from their distributions and from their correlation matrix (Figure 3.5).

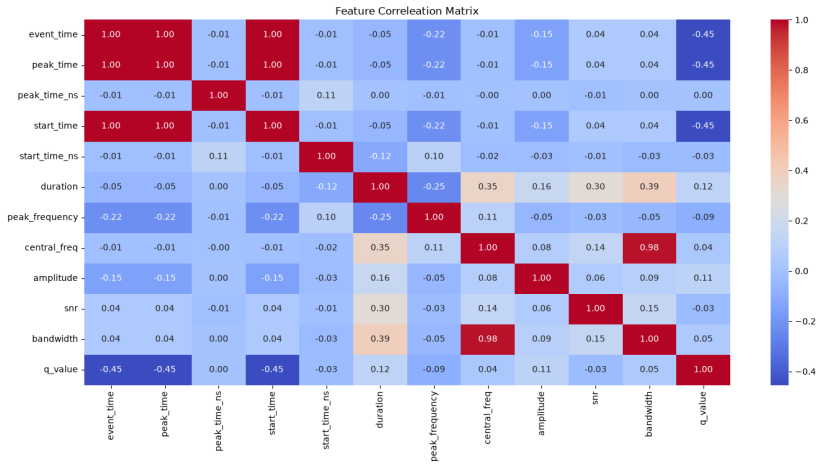


Figure 3.5: Pearson correlation between the twelve numerical Omicron descriptors. The three timestamps are perfectly correlated, and central frequency and bandwidth correlate at 0.98.

Exact redundancy. The three timestamp variables (event time, peak time, and start time) are mutually correlated at $r = 1.00$, being the same GPS time to within sub-second offsets, so at most one of them could be retained. Central frequency and bandwidth correlate at 0.98; bandwidth was retained because peak frequency already encodes spectral position, so bandwidth supplies extent rather than duplicating location.

Temporal provenance. Redundancy leaves one timestamp admissible, but none is retained. A timestamp supplies *when* a glitch occurred rather than *what* it is, and the correlation matrix shows that the shortcut is available: event time correlates at -0.22 with peak frequency, reflecting instrumental drift across observing runs. A model given the time of the trigger could separate the classes by observing period instead of by morphology, and that separation would not transfer to data recorded later.

Degenerate distributions. The nanosecond remainders are uniformly distributed and no correlation of theirs exceeds 0.12 in absolute value, consistent with timing quantization noise, while the quality factor contains a sentinel value of order 10^{20} that compresses the real distribution to the axis origin.

Reference to the noise floor. Amplitude and SNR measure the strength of the same event on two scales. Omicron reports amplitude as a strain amplitude of the trigger; the SNR is computed on whitened data and is therefore that amplitude referred to the noise spectral density of the detector at the frequency and time of the event, $\mathrm{SNR} \sim A/\sqrt{S_n(f,t)}$. The noise curve spans three to four orders of magnitude between 10 and 2048 Hz and differs between H1 and L1 and between observing runs, so across the population the denominator varies more than the numerator. Their linear correlation is 0.06, indeed the two descriptors are not statistically redundant, and neither one is the substitute for the other. Amplitude was dropped because an absolute strain amplitude is not comparable across frequency, detector and epoch, while SNR is comparable and is the quantity on which the release itself is thresholded, at 7.5. Amplitude also correlates at -0.15 with the three timestamps against 0.04 for SNR, carrying the calibration history of the instrument into the descriptor and reopening a similar shortcut as the one discussed before.

The four retained descriptors — peak frequency, duration, bandwidth, SNR — form a complete and non-redundant characterization of a transient in the time–frequency plane:

spectral position, temporal extent, spectral extent and prominence. Their largest pairwise correlation is 0.39, well below the 0.98 that forced the exclusion of central frequency.

Only one of the four image URL columns was kept. The four correspond to successive time windows of the same detection (0.5 s, 1 s, 2 s, 4 s), and only the first was retained. Because the time windows are effectively *different views*, they are not interchangeable, which is why unavailable images were excluded instead of substituted from another column.

Residual class imbalance is removed by random undersampling to the cardinality of the smallest class, giving an exactly balanced set of 9 568 images.

3.1.4 Preprocessing and partition

The downloaded spectrograms are matplotlib formatted figures comprising a frame with axes, colour bar and annotation, therefore a centre crop of 470×550 pixels isolates the spectrogram panel, which is then resized to the working resolution. The order of operation matters, since cropping the original and resizing afterwards keeps the same portion of the spectrogram framed at every resolution. Channel-wise normalization uses the standard ImageNet statistics, mean $(0.485, 0.456, 0.406)$ and standard deviation $(0.229, 0.224, 0.225)$. These are not the statistics of this corpus, and has a consequence that matters later in the analyses: zero in normalized space is the ImageNet channel mean, and that is the value the deletion metric substitutes for a removed pixel (Section 3.7.2). A removed pixel is therefore set to a neutral grey rather than to black, which would itself be a strong input.

The partition is frozen to disk. A dedicated script generates the train/validation/test assignment once, 70 / 15 / 15 under a fixed seed, and writes it as a per-file record of path, class and split, so that every subsequent run reads that file. A file listed but absent from disk is dropped from its partition with a warning. A file present on disk but absent from the assignment is never loaded, the loader iterating the assignment rather than the directory, so no sample can enter a partition without it having been assigned to one.

Freezing the partition allows for the assignment to be a function of the file list rather than of the corpus size, so removing a file removes exactly that file instead of reshuffling every remaining assignment. And the audit trail is externally verifiable: a reader can confirm that no file appears in two partitions, and the manifest records the SHA-256 digest of the assignment file.

Four downloaded files proved unreadable. Two were recovered by re-downloading; two are permanently unavailable at source (0.02 % of the corpus) and were excluded.

The final corpus is **9 566 images, partitioned** 6 695/1 435/1 436. The split is uniform and unstratified, drawn once under seed 42, so the balance of the corpus carries into the partitions by sampling: class shares lie between 23.9 % and 25.9 % in all three. The physical properties were checked afterwards. The medians of SNR, duration, peak frequency and bandwidth are 14.97/14.09/14.98, 1.25/1.25/1.10 s, 75.3/75.3/75.3 Hz and 4112/4254/4131 Hz: peak frequency is identical across the three, bandwidth agrees to 3.5 %, SNR to a little over 6 %. Duration is the exception, the test median sitting 12 % below the other two.

3.2 Model

3.2.1 Architecture

`GWGlitchSNN` is a four-block convolutional spiking network. Each block is a 5×5 convolution with stride 2 and padding 2 followed by a LIF layer. The spatial downsampling is carried by the stride rather than by pooling, which keeps it inside the convolution. Average pooling would put analog values on a path that carries only spikes; max pooling keeps them binary but adds a second operator between a LIF layer and the convolution that follows it. Table 3.1 gives the layer configuration at the working resolution.

Stage	Operation	Output shape	Notes
Input	direct encoding $\times$T	[T, B, 3, 112, 112]	analog frame repeated
Block 1	Conv 3 $\rightarrow$ 16, k5, s2, p2 + LIF	[T, B, 16, 56, 56]	exposed as `spike1`
Block 2	Conv 16 $\rightarrow$ 32, k5, s2, p2 + LIF	[T, B, 32, 28, 28]	exposed as `spike2`
Block 3	Conv 32 $\rightarrow$ 64, k5, s2, p2 + LIF	[T, B, 64, 14, 14]	exposed as `spike3`
Block 4	Conv 64 $\rightarrow$ 128, k5, s2, p2 + LIF	[T, B, 128, 7, 7]	exposed as `spike4`
Readout	Flatten (6 272) $\rightarrow$ Linear $\rightarrow$ LIF	[T, B, 4]	one per class, summed over T

Table 3.1: Layer configuration at the working resolution of 112 px. The flattened width is derived from the input size at construction (25 088 at 224 px, 6 272 at 112 px, 2 048 at 64 px).

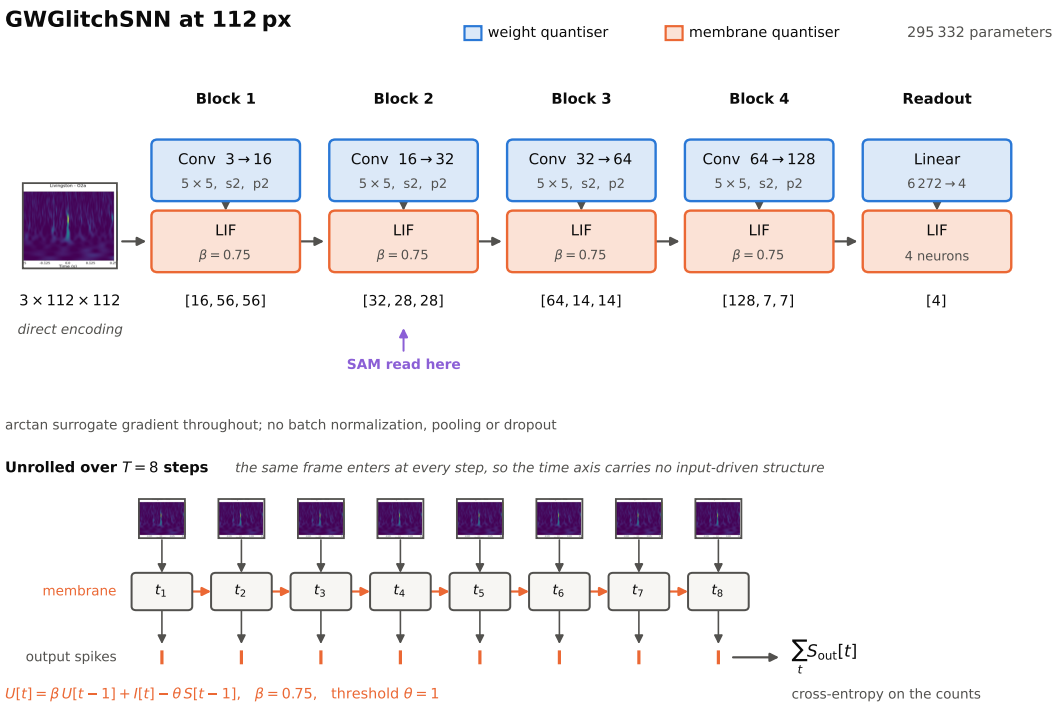


Figure 3.6: Datapath of `GWGlitchSNN` at the working resolution, and its unrolling over the T time steps.

The forward pass returns the stacked output spike train *and* the four hidden spike tensors of shape $[T, B, C, H, W]$, so that every diagnostic and explainability procedure consumes raw hidden spike trains directly from the model interface. Figure 3.6 draws the datapath and its unrolling over the T steps, the form in which those tensors are produced.

Parameter count is nearly resolution-independent: 295 332 at 112 px, of which 270 240 are convolutional and unaffected by input size, since only the linear readout shrinks with resolution, a fact with consequences for the design of the quantization grid discussed in Section 4.1.

Batch normalization, pooling and dropout are intentionally absent. Normalization layers interact non-trivially with quantization, and their treatment belongs to a later phase. The interaction is not only a quantization one: the statistics a spiking network would normalize accumulate over the time axis as well as over the batch, which is why two of the variants proposed for directly trained spiking networks redefine the operation — tdBN ties the normalization to the firing threshold [45], BNTT gives each time step its own learnable parameters [21]. On the hardware side two routes exist. The multiplication can be approximated by a shift, as in the shift-based normalization of binarized networks [19], or the layer can be removed altogether by folding its parameters into the preceding convolution before deployment [20].

3.2.2 Readout and loss

Classification uses a *rate-coded readout*: output spikes are summed over the T time steps and the resulting per-class counts are passed to a cross-entropy loss as logits. A readout that counts spikes rewards neurons that fire repeatedly, which keeps SAM in the multi-spike regime, where its inter-spike-interval formulation remains meaningful. Optimization uses Adam [23]. The predicted class is the argmax of the counts, and a tie, an all-zero count included, goes to the first class in alphabetical order, Blip. Ties concern 5 to 13 of the 1 435 validation images in every model of this thesis that classifies at parity; Section 5.9 reports the models in which they do not stay rare.

3.3 Training

3.3.1 Hyperparameter optimization

Hyperparameters are searched with Optuna [5], maximizing validation accuracy. Each trial reports validation accuracy after every epoch and is pruned by a Hyperband pruner [25] (minimum resource 1 epoch, maximum 15, reduction factor 3), which allocates epochs to promising configurations and truncating the others early. Trials are drawn by Optuna's default sampler (TPE), which is not seeded, so a repeated search would visit different configurations. The two studies used in this thesis hold 31 (v4) and 32 (v5) recorded trials, pruned ones included. The database also keeps three earlier studies of two to four trials each and a v5 search at 224 px stopped after six; none of them contributes to any result reported here. Studies are persisted to SQLite, one study name per search and resolution.

Two studies were run. The first (v4), at 224 px resolution, searched the learning rate in the interval $[10^{-4}, 3 \times 10^{-3}]$ (log-uniform), β in $(0, 1)$ and $T \in \{5, \ldots, 15\}$, and selected a learning rate of 1.036×10^{-4}, $\beta = 0.4148$ and $T = 7$. This was the configuration used for the resolution study. The second (v5), at the working resolution found after the study (112 px, as it will be better discussed in Section 3.4), imposed the deployment constraints described in Section 3.4.2 and produced the configuration used throughout the rest of the thesis.

3.3.2 Experiment tracking

Every trial is logged to Weights & Biases. Beyond validation loss and accuracy, the logged metrics also include the mean firing rate of each of the four hidden layers. Firing rate is the proxy for energy consumption in spiking hardware [36], so tracking it per layer during the search shows whether high-accuracy configurations achieve their accuracy through dense and expensive activity or through sparse spiking, and also whether any layer is silent or saturated before any explainability diagnostic is run.

A separately logged quantity, recorded under the name `avg_firing_rate`, is the mean over the four *output* neurons only. A value near 0.25 indicates that roughly one output neuron fires consistently, which is a signature of confident classification. It is relevant to note that it is *not* the global firing rate of the network and is never used for energy accounting, for which only the per-hidden-layer rates are valid.

3.3.3 Final training

The training script reads the frozen configuration and trains the model. The script is parametrized on input resolution and seed, the same code indeed serves the resolution study, the final training and the multi-seed reference. The seed is set explicitly before the model is constructed. The checkpoint is written whenever validation accuracy improves, and training stops after five consecutive epochs without improvement in validation loss, which permits early stopping, and with a ceiling set at 50 epochs. Each run writes the checkpoint together with a record of the full per-epoch history, the per-layer firing rates, the per-class validation recall, the seed and the resolution.

3.4 Selecting the operating configuration

The final operating point is defined by the input resolution, the three hyperparameters, and the layer and decay constant at which the explanations are read. Each is settled on evidence measured on the validation partition and recorded in a single frozen manifest.

3.4.1 Input resolution

Nine training runs were performed: three resolutions (224, 112, 64 px) at three seeds each, on the frozen partition, with hyperparameters held fixed and only the input size varying. Since those hyperparameters had been found at 224 px, the protocol systematically handicaps the lower resolutions, so any deficit they show is an upper bound on the deficit attributable to resolution itself.

The procedure and its outcome are reported in Section 5.1.

3.4.2 Hyperparameter search under deployment constraints

The v5 search uses the procedure of Section 3.3 over the same three hyperparameters, chosen because they govern three distinct axes: learning rate (optimization stability), β (membrane leak and temporal memory), and T (latency, computation and memory cost).

Two deployment constraints are imposed inside the search space. The first restricts β to $\{0.25, 0.5, 0.75, 0.875\}$, values whose decay is realizable as a bit shift, the set spanning both families ($\beta = 2^{-k}$ and $\beta = 1 - 2^{-k}$). The second caps T at 8. Constraining the search guarantees that whatever it returns is deployable by construction, and Section 5.2 shows that this is not equal to rounding an unconstrained optimum afterwards: the free search, run at 224 px, selects $\beta = 0.4148$, whose admissible neighbours are 0.25 and 0.5, while the constrained search selects 0.75.

The lower bound of the learning rate was moved from 10^{-4} to 10^{-5}, because the v4 optimum (1.036×10^{-4}) sat on the boundary of the earlier range, and a boundary optimum should be investigated further to clarify whether it is a consequence of the decided range.

3.4.3 Baseline manifest

The outcome of these choices is recorded in a single file, `baseline_manifest.json`. It contains the input resolution, the three hyperparameters, the originating study, the decay rate and reference layer of the explainer, the list of seeds (1234, 2345, 3456), the checkpoint paths, the sizes of the training and validation partitions, and the first sixteen hexadecimal digits of the SHA-256 digest of the partition file (`754b2515b6509075`). Every downstream script reads the model and its configuration from the manifest.

3.5 Diagnostics of the network dynamics

Two diagnostics characterize whether a trained network is a healthy spiking system or a formally correct one whose neurons are dead or saturated. The two are reused unchanged on every quantized model.

Dead and saturated neurons. For each hidden layer, the mean firing rate of every individual neuron is accumulated across the validation partition, weighted by batch size. A rate below 0.01 identifies a *dead* neuron, while a rate above 0.80 a *saturated* one. Neither cutoff is derived, since thresholds of this kind are conventions read off the distribution: the pruning literature trims neurons whose fraction of zero activations exceeds one standard deviation of the layer mean [18], and the dormant-neuron analysis treats the cutoff as a tuned hyperparameter, reporting values between 0 and 0.1 [39].

The two ends are pathological for different reasons. A neuron firing on fewer than one input in a hundred is wasted capacity and, being silent, it cannot appear in any explanation; its output is near-deterministic, carrying 0.081 of the one bit a binary unit can hold. Saturation, instead, is not an information problem — a Bernoulli variable at 0.80 retains 0.722 bits so as much as one at 0.20 — but an energy one: the energy of a spiking layer scales with the number of spikes it emits [41], so a unit near its ceiling spends the sparsity the architecture was chosen for.

Regularity of firing. The coefficient of variation of the inter-spike intervals,

$$\mathrm{CV} = \frac{\sigma_{\mathrm{ISI}}}{\mu_{\mathrm{ISI}}}, \tag{3.1}$$

distinguishes regular, clock-like firing ($\mathrm{CV} \to 0$) from irregular, Poisson-like firing ($\mathrm{CV} \approx 1$). It is relevant here because the Spike Activation Map (SAM) is exactly built on inter-spike intervals. A random sample of 200 neuron positions per layer is fixed once at the first batch, and the intervals of exactly those positions are accumulated over every validation image. The CV is then computed only where at least five intervals were observed. Since a neuron can produce at most $T - 1 = 7$ intervals within a single image, pooling over the whole partition is what makes the estimate stable.

A second coefficient is computed inside the single image, over the intervals a neuron produces within the T steps of that image, and it assesses whether firing is periodic there. Two spikes give one interval whose standard deviation is zero by construction, so the per-image coefficient is read only where a neuron produced at least three spikes, and the full-precision parents are measured under the same rule.

3.6 The Spike Activation Map

The definition is the one given in Section 2.3: the recency weight of Equation 2.9, accumulated over a neuron's preceding spikes by Equation 2.10, masked by the spikes emitted at t and summed over channels by Equation 2.11. What the definition leaves open the implementation had to settle.

The summation range is exclusive of t. The score measures the contribution of a neuron's *preceding* trajectory, as including $t' = t$ would mean adding a term of weight $\exp(0) = 1$ for every neuron firing at t, so the map produced is the correct SAM *plus the raw binary spike map*. The pedestal is independent of both γ and the spike history, and therefore inflates every correlation between maps computed at different γ. A consequence of the correct definition is that SAM at $t = 0$ is identically zero, no past history existing at the first step.

The implementation uses the recursion

$$N_{t+1} = e^{-\gamma}\,(N_t + S_t)\,, \qquad N_0 = 0, \tag{3.2}$$

algebraically identical to Equation 2.10 but $O(T)$ rather than $O(T^2)$. A reference implementation computing the explicit sum is retained and the two are asserted equal at run time.

Algorithm 1 Spike Activation Map of one image at one layer

Require: spike train $S \in \{0,1\}^{T\times C\times H\times W}$ of layer ℓ, decay γ
Ensure: maps $M_0, \ldots, M_{T-1} \in \mathbb{R}^{H\times W}$
1: $N \leftarrow 0$ {$[C, H, W]$, neuronal contribution score}
2: **for** $t = 0$ **to** $T-1$ **do**
3: $\quad M_t \leftarrow \sum_k N^k \odot S_t^k$ {Eq. 2.11; $M_0 = 0$}
4: $\quad N \leftarrow e^{-\gamma}\,(N + S_t)$ {Eq. 2.10, steps $t' < t+1$ only}
5: **end for**
6: **return** M

Visualizations overlay every time step on the denormalized input under a *shared colour scale* to assure that a change in intensity across time is genuine and is not an artefact of per-panel rescaling.

The reference configuration for every analysis below is **layer 2 with** $\gamma = 0.5$, both recorded in the manifest. The choice of γ is justified in Section 5.5.1.

3.7 Evaluating and comparing explanations

3.7.1 Multi-sample statistics

For each class, SAM is computed on 25 validation images, aggregated over time (mean over T) and stacked, and three maps are reported: the *mean*, meaning the region attended on average; the *standard deviation*, showing where the explanation varies most across instances; and their ratio, the *coefficient of variation*, which indicates whether a region is attended consistently or only in some examples. Mean and standard deviation share a colour scale across classes so that they are numerically comparable. The coefficient of variation has an extremely long tail where the mean is near zero, and its colour scale is truncated at the 95th percentile.

The same statistics are repeated for all four hidden layers, on 20 images per class. Since SAM sums over the channel axis and the four layers have 16, 32, 64 and 128 channels, raw magnitudes are not comparable across depth, and both maps are normalized by the channel count of their layer before being plotted on a shared scale.

3.7.2 Faithfulness: the deletion metric

Stability does not by itself imply correctness, so faithfulness is tested separately with the deletion metric of Petsiuk et al. [32]. Pixels are removed in decreasing order of attributed importance and the confidence of the model is read after each removal. A map that points at the evidence the model uses destroys that confidence faster than one that does not. Algorithm 2 states the procedure for a single image; `xai/deletion_metric.py` implements it.

Algorithm 2 Deletion metric for one image

Require: network f, image x, reference layer ℓ, decay γ, fractions $F = \{0, 0.05, \ldots, 1\}$
Ensure: confidence curves c^{SAM} and c^{rand} over F

$M \leftarrow \mathrm{SAM}(f, x, \ell, \gamma)$ {$[T, H, W]$, one map per time step}
$A \leftarrow \frac{1}{T} \sum_t M_t$
$A \leftarrow \mathrm{upsample}_{\mathrm{bilinear}}(A)$ {layer resolution to input resolution}
$\hat{y} \leftarrow \arg\max_c \mathrm{softmax}(f(x))_c$
for ord $\in$ {SAM, rand} **do**
  if ord $=$ SAM **then**
    $\pi \leftarrow \mathrm{argsort}(A,\ \mathrm{descending})$
  else
    $\pi \leftarrow$ uniform random permutation of $\{1, \ldots, HW\}$
  end if
  for $\phi \in F$ **do**
    $x' \leftarrow x$ with pixels $\pi_{1 \ldots \lfloor \phi HW \rfloor}$ set to 0 in all three channels
    $c_\phi^{\mathrm{ord}} \leftarrow \mathrm{softmax}(f(x'))_{\hat{y}}$
  end for
end for

SAM produces one map per time step and the metric needs one, so the maps are averaged over t, and what survives that average is an ordering of the pixels: only the ordering enters what follows. SAM is class-agnostic, so the tracked class is the one the model predicts on the unmodified image, held fixed while the pixels disappear.

The pixel ranking is computed once per image and per ordering, and the mask at each fraction takes its first $\lfloor \phi HW \rfloor$ entries. The removed sets are nested: a pixel deleted at one fraction stays deleted at every larger one. What the curve traces is the progressive degradation of a single ranking, not 21 unrelated maskings.

At every fraction the two curves delete the same number of pixels and differ only in which ones, so the gap between them cannot come from one condition removing more of the image than the other. It does not isolate the ranking alone. SAM ranks contiguous regions first, while a random permutation scatters the removed pixels over the whole image, so the two masks also differ in spatial structure, a confound shared by every comparison against random removal. Positions where the upsampled map is exactly zero are tied and are removed in the order the sort returns them: beyond the fraction at which the support of the map is exhausted, the SAM ordering carries no information about the image.

The area under a curve is the confidence the model retains, integrated over the fraction deleted: a faithful map destroys that confidence early and leaves less area behind than random deletion, which makes $\Delta = \mathrm{AUC}_{\mathrm{SAM}} - \mathrm{AUC}_{\mathrm{random}}$ negative, its magnitude the margin the map gains over chance. The 15 curves of a class are averaged point by point, and the area under the mean curve is taken by the trapezoidal rule, which joins consecutive samples by a straight segment and sums the 20 trapezia lying between the 21 fractions. The curve has no analytic form and only 21 empirical samples, so a higher-order rule would interpolate structure that was never measured.

3.7.3 Divergence metrics

Two explanations of the same image produced by two models are compared with different metrics addressing different questions. The first two read the stack summed over time, $A = \sum_t M_{\cdot,\cdot,t}$, and are spatial, while the third reads the stack unaggregated and is the only one that sees the time axis. Their provenance in the saliency literature is discussed in Section 2.3.

Spatial Pearson correlation (PCC). The linear correlation between the two summed maps, flattened over the HW positions,

$$\mathrm{PCC}(A, B) = \frac{\sum_i (A_i - \bar{A})(B_i - \bar{B})}{\sqrt{\sum_i (A_i - \bar{A})^2}\,\sqrt{\sum_i (B_i - \bar{B})^2}}. \tag{3.3}$$

Centring and dividing by the standard deviations makes it invariant to any affine rescaling $B \mapsto aB + b$ with $a > 0$, and that invariance is what allows a comparison across models whose overall spiking activity differs. Indeed, a quantized model that fires uniformly more than its parent is not charged for the offset, only for a change in the spatial pattern. The quantity is undefined when either map is constant, a case that in the implementation returns as missing rather than as zero.

Top-k intersection over union (IoU). Let A_k hold the positions whose value in A exceeds the $(1-k)$ empirical quantile of A, and B_k the same for B. The metric is the Jaccard index of the two sets,

$$\mathrm{IoU}_k = \frac{|A_k \cap B_k|}{|A_k \cup B_k|}, \tag{3.4}$$

with $k = 0.20$, the value used by Rabbi et al. [33]. Where the correlation weighs every position by its value, this reads the ranking alone and only at its top, as it asks whether the two models highlight the same region, not whether they assign it the same intensity. Two independent rankings select kHW positions each and share k^2HW of them in expectation, over a union of $2kHW - k^2HW$, so the chance level is $k/(2-k) = 0.111$. Every overlap reported in this thesis is read at this value of k; Figure 5.12 shows, on one model, how the reading changes with it.

The threshold is a strict inequality against an empirical quantile, so the selected set holds exactly kHW positions only when the values are distinct. Ties can leave it empty, which happens when k is small enough for the quantile to fall inside a plateau, and the value is then undefined.

Temporal centre of mass (CoM). The mean time step at which the energy of the SAM stack is concentrated,

$$\mathrm{CoM} = \frac{\sum_t t\, E_t}{\sum_t E_t}, \qquad E_t = \sum_{i,j} M_{ij,t}, \tag{3.5}$$

with the steps indexed from 0 and $E_0 = 0$, SAM being identically zero at the first step. It is reported both as a signed difference between the two centres of mass, whose sign says whether the energy of the explanation moves earlier or later, and as its absolute value, which is the form the numerical floors are calibrated against. Both are in units of time steps. Aggregating over the whole map discards where the activity sits and keeps when it occurs, which is what makes it the complement of the two spatial metrics.

Beyond the three, the *per-timestep correlation* applies Equation 3.3 to the pair of maps at each t separately and returns a vector of length T in place of a scalar. It localizes the step at which agreement breaks down, which provides an information that neither a summed map nor a single centre of mass can show.

3.7.4 Calibration protocol

The divergence metrics are calibrated before they are used. One check precedes the calibration: a checkpoint compared with itself must return perfect agreement, meaning that two calls to the sampling routine must return bit-identical images and two forward passes of the same model must return identical maps. What follows are four reference divergences, measured on fifteen validation images per class at the reference layer and γ — twenty for the second, matching the grid it is read against.

1. **Numerical floor.** Gaussian noise is added to the weight tensors of one checkpoint, at magnitudes expressed as fractions of its measured 8-bit quantization step, and the perturbed model is then compared with the unperturbed one. Biases and the LIF buffers are excluded, since perturbing β would mix membrane dynamics with weight noise.

2. **Fine-tuning alone.** The control arm of the quantization grid (Section 4.1) runs the whole procedure with no bits removed. The result is the divergence that continued training costs on its own, and it is the operative threshold: a grid result closer to its parent than the control arm is a result about continued training, and not about the removed bits. Section 5.6.4 reports it.

3. γ **reference.** The sensitivity analysis gives the divergence produced by changing the explainer itself thirtyfold.

4. **Independent solutions.** The three full-precision checkpoints are compared pairwise in order to measure how similar two independently trained solutions are.

Chapter 4
The Quantization Study

This chapter describes how the quantized models are built and how they are compared with their full-precision parents. Every design choice below is either inherited from the calibration of Section 5.6 or settled by a measurement reported in Section 5.7. Again, the whole analysis is performed on the validation partition.

4.1 Design principle: three separate arms

The fundamental design choice is to **quantize the two targets separately before quantizing them together**, as separating them turns the grid into a causal ablation and is what makes the hypothesis of Chapter 1 testable. Table 4.1 lists the configurations that follow; Figure 4.1 shows where in the datapath each arm acts, and that the two never act on the same tensor.

Arm	Weights	Membrane potential	Purpose
FP32 reference	32-bit float	32-bit float	baseline for every comparison
FP32 control	32-bit float	32-bit float	separates the pipeline from the bits
Weight-only (W)	8 / 4 / 2 bit	32-bit float	isolates the synaptic (spatial) effect
Membrane-only (U)	32-bit float	8 / 4 / 2 bit	isolates the state (temporal) effect
Joint (W+U)	8 / 4 / 2 bit	8 / 4 / 2 bit	deployable configuration

Table 4.1: The quantization grid: nine quantized configurations and one control, each at the three seeds of the reference (30 fine-tuning runs).

The *control* arm runs the whole fine-tuning procedure with no quantizer attached. Both bit-widths are left unset, so neither the weight parametrization nor the membrane hook is ever registered and the network trains in full precision. Its purpose is subtractive. The nine quantized configurations differ from their parent in two respects at once, the bits removed and the operations that remove them — reloading the parent, fine-tuning it, selecting a checkpoint, re-evaluating — and the control runs those operations alone, so whatever it produces is their contribution and theirs alone.

It produces two sets of weights. The first is the checkpoint the selection rule returns, and it is bit-identical to the parent: the same label on all 1 435 validation and 1 436 test images, per-layer firing rates equal to the last digit reported. The identity follows from the rule: the selection begins with the un-fine-tuned model already held as a candidate and replaces it only when an epoch exceeds its validation accuracy *strictly*. With nothing quantized, that initial candidate sits at the parent's own accuracy, so an epoch would have to beat the parent to displace it, and on none of the three seeds did it. What the control establishes is that reloading the parent, selecting a checkpoint and re-evaluating leave the

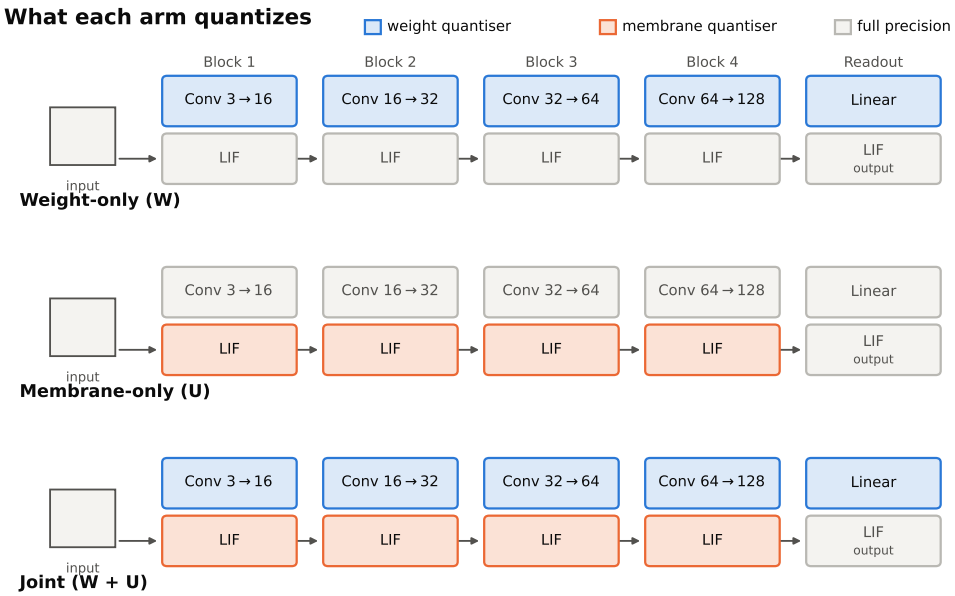


Figure 4.1: The three arms, as which parts of the datapath each one quantizes.

model bit-identical, so none of the divergence measured in the other nine configurations can be charged to the apparatus around the training. The training itself is not covered by that argument, and it is the subject of the second set of weights.

The second is the state of the weights after the last epoch, kept separately and not identical to the parent. Fine-tuning moved them even though it never succeeded in replacing the checkpoint. What that state measures is the effect of continued training alone on an explanation, and it is the fine-tuning level of the calibration (Section 5.6.4).

Two quantities are deliberately left out of the grid. Firstly the membrane decay $\beta = 0.75$, which is already shift-implementable, having been constrained inside the hyperparameter search (Section 3.4.2) and is not re-quantized. Secondly, the biases are kept in full precision: there are 244 of them, under 0.1 % of the parameters, and each enters a neuron's input once per step.

The two arms respond differently to the resolution choice. 270 240 of the 295 332 parameters are convolutional and therefore resolution-independent, so what the weight arm saves is nearly constant across resolutions. The membrane state, which is what the other arm quantizes, falls twelvefold between 224 and 64 pixels.

4.2 Quantization-aware fine-tuning from the parent

Quantization-aware training is implemented by fake quantization: in the forward pass each quantized tensor is rounded to its grid and then clamped to the grid's bounds, and in the backward pass the rounding is replaced by the identity, the straight-through estimator [7].

The two quantizers are attached by different means because they act on different kinds of object: a weight is a parameter, owned and updated by the optimizer, while a membrane potential is a state that is rebuilt at every time step.

Weights are quantized through a *parametrization*. PyTorch moves the stored tensor aside and recomputes `weight` from it on every access, so the module's weight becomes a function of a hidden full-precision tensor. The optimizer holds and updates that hidden tensor, the forward pass only ever reads its quantized image, and the straight-through estimator carries the gradient back to it unchanged. Keeping the full-precision copy is essential to

fine-tuning and makes it possible: an update smaller than half a quantization step applied to a value already on the grid would round away and leave nothing behind, whereas if accumulated in the hidden tensor many such updates eventually carry the weight to the next level.

When the training ends the parametrization is folded away and the quantized values are written into ordinary parameters, so the saved checkpoint holds weights already sitting on the grid.

The membrane is quantized through a *forward hook* on each of the four hidden LIF layers. A hook that returns a value replaces the output of the module it is attached to, and a `Leaky` layer returns the pair of spike and membrane: the hook passes the spike untouched, binary as it already is, and substitutes the quantized membrane. That substituted value is the one the forward loop carries into the following step, so the grid constrains the state as it crosses a time step and reaches the spikes only by way of the step after. The membrane grid is fixed in advance, unsigned on $[0, 2]$ from the clipping study.

Every quantized model is fine-tuned from the full-precision checkpoint of the same seed. This is the constraint that guarantees that the quantization effect is distinct from initialization variance. The numerical floor of the calibration (PCC $= 0.975$) holds for comparisons between a model and its own parent, which attends to the same regions of the input. A quantized model trained from a random initialization would be an independent solution, free to attend to different regions, for which the relevant level is 0.153, and almost no quantization effect would be distinguishable from initialization variance.

Schedule. The fine-tuning adapts the parent to its grid. Adam runs at one tenth of the reference learning rate (1.047×10^{-5}), for at most 20 epochs, with early stopping after 5 epochs without improvement of the validation loss. The reduced learning rate follows from the constraint above: the floor of the calibration holds only while the quantized model remains a perturbed copy of its parent, and a step sized for training from scratch would let it leave that neighbourhood, moving the comparison toward the independent-solutions level. How far even this schedule moves a model with nothing quantized is measured in Section 5.6.4.

Before the first epoch the model is evaluated with the quantizer attached and no training, which gives the *post-training quantization* (PTQ) accuracy of the same configuration at no extra cost: that un-fine-tuned model is saved as it represents the first candidate checkpoint, and it is replaced only by an epoch that strictly improves the validation accuracy. The selected checkpoint is the one on which every subsequent measurement — accuracy, explanations, firing statistics, energy — is taken. The 30 runs took 13 hours on the same 6 GB consumer GPU used for every training run in this thesis.

Gates before the grid. Two checks were run before any fine-tuning to validate the quantizer and then to validate the PTQ gate. The first verifies the quantizer itself: attaching and removing it restores the original weights exactly, and folding the quantized values into a plain checkpoint and reloading it leaves no drift (0 at 4 bits).

The second targets the PTQ gate: the joint 8-bit model, with no fine-tuning, must indeed match its parent to within the spread of the full-precision reference, four samples out of 1 435. At 8 bits the weight step is of the order of the perturbation that left the spatial correlation at 0.975 in the calibration, so an appreciable loss of accuracy there would effectively indicate a defect of the implementation. Both gates passed (Section 5.9.1).

All 8-bit configurations were run first, then 4-bit, then 2-bit.

Algorithm 3 Quantization-aware fine-tuning and checkpoint selection

Require: parent checkpoint θ_s of seed s, bit-widths b_w and b_u (either may be absent; both absent for the control), learning rate $\eta = 0.1\,\eta_{\text{ref}}$, at most $E = 20$ epochs, patience $P = 5$

Ensure: selected checkpoint $\theta^\star$, state after the last epoch θ_{fin}

load θ_s and seed every generator with s
if b_w given **then**
  parametrize every convolutional and linear weight with the per-channel quantizer
end if
if b_u given **then**
  hook `lif1–lif4`: replace the returned membrane by its b_u-bit image on $[0, 2]$
end if
$a^\star \leftarrow$ validation accuracy {PTQ accuracy, no training}
$\theta^\star \leftarrow \text{fold}(\theta)$; $L^\star \leftarrow \infty$; $p \leftarrow 0$
for $e = 1$ **to** E **do**
  one epoch of Adam, rounding passed through by the straight-through estimator
  $(L, a) \leftarrow$ validation loss and accuracy
  if $a > a^\star$ **then**
    $a^\star \leftarrow a$; $\theta^\star \leftarrow \text{fold}(\theta)$ {strict improvement only}
  end if
  if $L < L^\star$ **then**
    $L^\star \leftarrow L$; $p \leftarrow 0$
  else
    $p \leftarrow p + 1$
  end if
  if $p \geqslant P$ **then**
    stop
  end if
end for
$\theta_{\text{fin}} \leftarrow \text{fold}(\theta)$
return $\theta^\star$, θ_{fin}

4.3 Weight quantizer

Weights are quantized on a *symmetric signed* grid whose outermost level is $q_{max} = 2^{b-1} - 1$ on each side of zero, for a total of $2q_{max} + 1$: 255 levels at 8 bits, 15 at 4 bits and 3 at 2 bits (the values $-\Delta$, 0, $+\Delta$). Zero is exactly representable at every bit-width. The quantizer is applied to every convolutional and linear weight tensor, including the readout.

The scale is computed per output channel. A single scale for the whole tensor is set by its largest weight, so a channel whose own weights are small reaches only the innermost levels of the grid and leaves the external ones unused. The measured distributions of the reference checkpoint make this effect concrete (Table 4.2): indeed in the deeper convolutions the largest weight of the weakest channel falls below half the largest weight of the tensor. This halves the levels that channel can reach and costs it one bit, which at 2 bits it is critical: the signed grid is $\{-\Delta, 0, +\Delta\}$ with Δ the tensor maximum, so every weight of a channel whose maximum lies below $\Delta/2$ rounds to zero and the channel vanishes. A per-channel scale takes Δ from each channel's own maximum, so every channel reaches the outermost level whatever its magnitude. Figure 4.2 draws the two cases for the weakest channel of conv4 at 4 bits: seven of the fifteen levels are reachable under a scale set by the tensor, fifteen of fifteen under the channel's own.

Layer	max $\|w\|$, tensor	max $\|w\|$, weakest channel	Ratio	Bits lost per-tensor
conv1	0.144	0.115	1.25	0.32
conv2	0.126	0.060	2.10	1.07
conv3	0.113	0.054	2.09	1.07
conv4	0.099	0.044	2.25	1.17
linear	0.214	0.146	1.47	0.55

Table 4.2: Weight ranges of the reference checkpoint (seed 1234). "Bits lost" is $\log_2$ of the tensor maximum divided by the weakest channel's maximum: the resolution that channel loses under a single per-tensor scale.

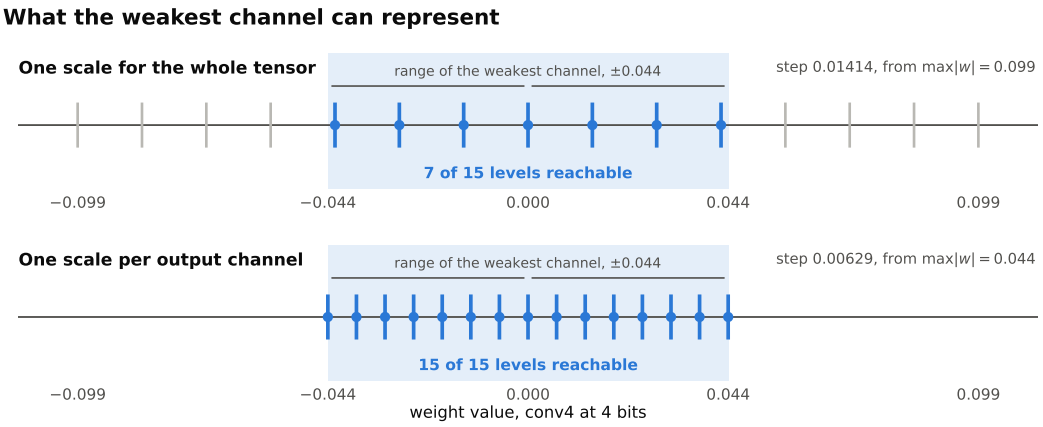


Figure 4.2: Grid levels reachable by the weakest channel of conv4 at 4 bits, under a scale derived from the tensor maximum and from the channel's own.

The global 8-bit step used in the calibration (0.00168, from the largest absolute weight of the network, 0.214) is therefore an upper bound on the per-channel steps actually applied, whose mean is between 0.0005 (conv4) and 0.0013 (readout) at 8 bits, and the floor it defines is conservative. The calibration computes it as $2w_{max}/(2^8 - 1)$, which differs from Equation 2.7 by one unit in the denominator, 0.4 % at 8 bits; at lower widths the two diverge (at 2 bits w_{max} against $2w_{max}/3$), so every multiple of the unit quoted in this thesis

takes the coarser step from Equation 2.7. On the same global grid the 4-bit and 2-bit steps are 18 and 128 times that unit. The per-channel steps average 0.009 to 0.024 at 4 bits and 0.061 to 0.171 at 2 bits, between 5 and 15 and between 37 and 102 times the unit, and the weakest channel of conv4 still has a 2-bit step 26 times larger.

4.4 Membrane quantizer

The membrane potential of the four hidden LIF layers is quantized at every time step. The quantizer acts on the potential returned by each layer after the spike decision, so the quantized value is the *state* carried into the next step: what is stored, and therefore what costs memory, is what is quantized. The membrane of the output layer (four neurons) is left in full precision because its state is negligible in memory and quantizing it would act on the spike decisions of those four neurons, whose summed counts are the logits; its *weights* instead are quantized with every other linear tensor (Section 4.3), so the readout is split between the two arms.

As has been established, a grid for the membrane can not be designed without knowing the range of the quantity, and unlike a weight, the potential has no range until the network is run. Its distribution was therefore measured over the whole validation partition (Section 5.7), and the clipping range was chosen by testing candidate ranges on the full-precision network, so before any quantization.

The weight scheme is the symmetric per-channel formulation of Jacob et al. [20]. For the membrane no comparable convention exists: MINT [43] quantizes the potential of a spiking network, but ties its scale to the weights, which is the coupling Section 4.5 avoids.

The membrane grid is unsigned over $[0, 2]$, twice the firing threshold $U_{\text{thr}} = 1$, with 2^b levels starting exactly at zero. The step is $2/(2^b - 1)$:

Bits	Levels	Step	Step / threshold
8	256	0.0078	0.8 %
4	16	0.1333	13.3 %
2	4	0.6667	66.7 %

Table 4.3: The membrane grid at each bit-width. At 2 bits the representable potentials are 0, 2/3, 4/3 and 2: a single level lies strictly between rest and threshold.

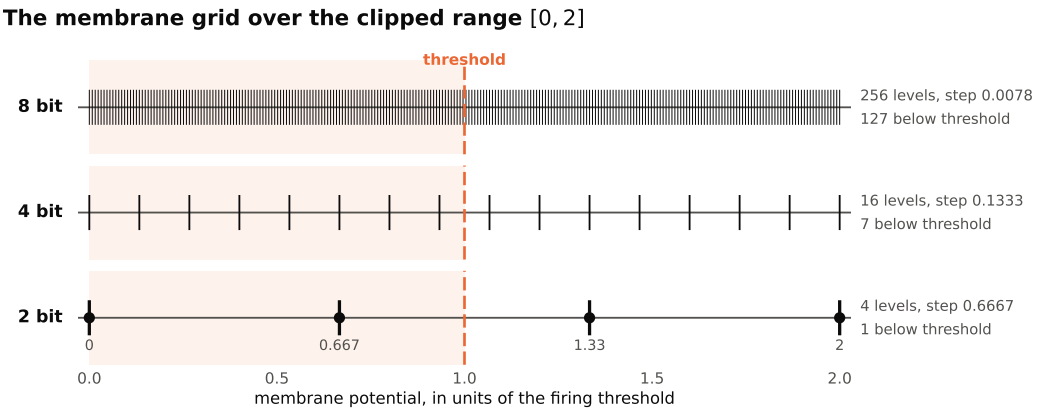


Figure 4.3: The membrane grid at each bit-width over the clipped range, with the firing threshold marked.

Zero is the value to which every neuron is initialized and around which it rests, so it must be exactly representable, which the unsigned grid guarantees by construction, its first level being zero.

Table 4.3 contains a quantitative prediction. The step is 0.8 % of the threshold at 8 bits, 13.3 % at 4 bits and 66.7 % at 2 bits, where it is of the same order as the threshold itself and a neuron's state can encode only whether it is at rest, part-way to threshold, or above it. Figure 4.3 draws the three grids on one axis: the 8-bit row reads as a ruler, the 2-bit row as four states. Smith et al. [38] observe that membrane quantization needs at least four bits and attribute it, without testing it, to the propagation of a membrane error through the recurrent update at every step. The step size is a second account, and the grid tests it directly.

The unsigned grid follows from the clipping study. Potentials do go negative — the median of `lif1` sits at -0.08 and its first percentile at -6.6 — yet clipping them away changes nothing the classifier uses. The symmetric range $[-2, +2]$ and the unsigned range $[0, +2]$ return the same accuracy to the last digit, 0.9826, one validation sample below the unclipped network, so a signed grid would spend half of its levels below zero, where no decision is made. At 8 and 4 bits that costs a factor of two in resolution, one bit. At 2 bits it costs a factor of three: the step grows from $2/3$ to the full 2, and no level at all remains between rest and threshold.

4.5 Joint arm

The joint arm applies both quantizers at once, at the same bit-width, with the per-channel weight scales of Section 4.3 and the fixed membrane grid of Section 4.4. **The two scales are independent.** MINT [43] shares a single scale between weights and membrane so that the inference loop needs no rescaling multiplier, but that choice is an optimization of a specific integer datapath, and imposing it here would couple the two targets that the grid exists to separate. With independent scales the joint arm is mechanically the composition of the two single-target arms: the two quantisers never see each other's grids or state, and that any outcome of the joint arm that the two single-target arms do not account for is therefore an interaction within the network, not a coupling introduced by the instrument.

4.6 Reading the results

For every quantized model, SAM maps are recomputed at the reference layer and γ on twenty validation images per class, and compared with those of **the full-precision parent of the same seed**, using the metrics of Section 3.7.3: spatial PCC, top-20 % IoU, the temporal centre of mass (absolute and signed shift), and the per-timestep correlation curve.

Further measurements characterize the behaviour:

- **cross-faithfulness**: the deletion metric applied to the quantized model with its *own* map and with the map of its *parent*, on fifteen validation images per class and twenty-one deletion fractions. The deletion baseline is recomputed for the parent of each seed, and a class is interpretable for a seed only if its baseline gap between SAM and random deletion is at least 0.05;
- the **diagnostics** of Section 3.5, collected for every model in a single pass over the validation partition: per-layer firing rate, dead and saturated fractions, CV-ISI, and per-sample predictions;
- the **Earth Mover's Distance** between the per-neuron firing-rate distributions of quantized model and parent, for a direct comparison with Smith et al. [38].

The CV-ISI follows the protocol of Section 3.5, applied to the reference in Section 5.3 — 200 positions per layer, seed 42, a coefficient computed only where at least five intervals are available.

Uncertainty is attached to the divergence and faithfulness measurements by a percentile bootstrap over images: the twenty (divergence) or fifteen (deletion) images of a class-and-seed cell are resampled with replacement 2 000 times, and the 95 % interval is read from the resampled means. An interval on the mean of a configuration over its twelve cells resamples each cell independently and treats classes and seeds as fixed. The calibration levels themselves are point estimates and carry no interval, so a bootstrap interval that reaches a level is read as no difference from it.

Five principles, established in Chapter 5 before the grid was run, aid in the reading of those numbers.

1. **Below the numerical floor is noise.** A divergence smaller than the one produced by weight noise of one 8-bit step cannot be deemed a result. The floor is measured per class, and every grid result is read against the floor of its own class; the worst-class values (PCC 0.975, IoU 0.704, CoM difference 0.029 steps) are a summary of the calibration, not a threshold for a mean over classes. A result belongs to the removed bits only where it also exceeds the fine-tuning level of its class (Section 5.6.4).

2. **IoU is an early indicator, and PCC a graded measure.** IoU has a dynamic range of 2.6× its floor and saturates early, while PCC has 34×.

3. **Everything is reported per class and per seed.** The sensitivity ordering Violin_Mode > Scattered_Light > Blip > Extremely_Loud recurs under both interventions that act on a fixed network, changing the kernel and perturbing the weights, so a pooled figure would average a class that responds strongly with one that does not.

4. **A flat result on Extremely_Loud is uninformative.** The calibration moved that class by nothing: PCC 0.9997 under the largest perturbation applied, worst per-timestep correlation is 0.996, a sensitivity in $1 - \text{PCC}$ 94 times below Violin_Mode's. On the other three classes the calibration showed the metrics responding to a change known to have been made,but Extremely_Loud never produced a response to calibrate against. This means that a flat reading is consistent both with an explanation that did not move and with an instrument that could not have seen it move.

5. **Cross-faithfulness is interpretable only where the parent has a baseline signal.** On the reference seed that means Blip and Extremely_Loud; since the baseline is recomputed per seed, the set of interpretable classes is itself reported.

Every quantized model is compared with *its own* parent, and never with the parent of another seed, which is the principle that keeps together the other ones and the whole analysis. Because the three parents differ from each other far more than any quantized model differs from its own (Section 5.6), a comparison against the wrong seed would produce a large spurious divergence. The guard against that error is that in every comparison the 8-bit weight arm, the smallest intervention in the grid, must keep a spatial correlation above 0.98 with its parent in every seed.

The test of the hypothesis. The hypothesis predicts an *ordering*: the membrane arm should displace the temporal centre of mass at a higher bit-width than the one at which the weight arm lowers the spatial correlation. A displacement or a decrease is relevant only where it goes beyond both the numerical floor and the fine-tuning level of its class (principle 1 above), read on the mean over seeds and, as a check, cell by cell against the fine-tuning level of the same seed. Either a reverse order or a simultaneous change falsifies the hypothesis.

4.7 Stratified analysis

Joining the per-sample outputs to the physical descriptors allows the result to be reported as a relation, which an aggregate divergence does not permit.

The hypothesis is that divergence increases as event strength decreases, with a mechanism specific to spiking networks: quantizing the membrane potential flips a neuron's spike decision only where the potential lies within the quantization error of the threshold, and weak events are exactly what produce potentials that sit close to threshold. Because a flipped spike is a discrete event that propagates both to the next layer and forward in time through the decay and reset, and since SAM is constructed from spike trains, the flip changes the object the explanation is made of.

The prediction discriminates between arms: membrane quantization applies a fixed absolute error and should show a steep dependence on event strength, whereas weight quantization perturbs the input current approximately in proportion to the signal and should be flatter. It is tested on the membrane arm at 4 and 2 bits, the configurations in which the arm has an effect and accuracy is intact.

Two preconditions were verified. SNR does translate into image intensity, the Gravity Spy colour scale being fixed, not auto-scaled per image. SNR is also very nearly a proxy for class: on the validation partition the top SNR quartile is 89.7 % Extremely_Loud. Stratification uses **terciles computed within each class**, fifteen images per class and tercile. The summary statistic is the IoU gradient, high-SNR tercile minus low-SNR tercile; a positive gradient means that weak events lose their explanation more.

4.8 Energy and memory

Synaptic operations are counted per layer as

$$\mathrm{SOP}_\ell = \mathrm{f}_{\ell-1} \cdot \mathrm{T} \cdot \mathrm{MAC}_\ell, \tag{4.1}$$

where $\mathrm{f}_{\ell-1}$ is the mean firing rate of the preceding hidden layer and MAC_ℓ the multiply-accumulate count of the equivalent dense layer, and multiplied by the per-operation costs of [17]. The rates are the per-hidden-layer figures measured on the *selected checkpoint* of each configuration, in the same validation pass that produces its diagnostics, so that energy, accuracy and explanations all refer to the same model.

One layer is asymmetric. Under direct encoding the first convolution receives analog pixel values rather than spikes, therefore its operations are multiply-accumulates. Because the input is identical at every time step, the current it produces is constant in time and is computed once rather than T times: it is a standard implementation of constant encoding that avoids recomputing that layer at every step, which would make the spiking network more expensive than its convolutional equivalent (Section 5.8).

Per-operation costs. A multiply-accumulate is costed as one multiplication plus one addition at every precision, the convention of the 4.6 pJ floating-point figure ($3.7 + 0.9$). Full-precision configurations, the membrane arm included, use 4.6 pJ per MAC and 0.9 pJ per accumulate; configurations with quantized weights use the 8-bit integer figures, 0.23 pJ per MAC ($0.2 + 0.03$) and 0.03 pJ per accumulate. Narrower integer operations are not tabulated, so at 4 and 2 bits the costs are extrapolated, the multiplication scaling quadratically with the operand width and the addition linearly: a MAC costs 0.065 pJ and an accumulate 0.015 pJ at 4 bits, 0.020 and 0.0075 pJ at 2 bits.

Two further assumptions are built into these figures. The first convolution is costed at the weight bit-width, which assumes that the input image is quantized to the same width. Every accumulate is costed at the weight bit-width as well, which assumes an adder as narrow as the weights. The sum it builds is wider: a neuron of conv4 adds up to 1 600 weighted inputs per step ($64 \times 5 \times 5$), about eleven bits beyond those of a single weight, and in the weight-only arm the membrane that receives the sum stays in floating point.

Section 5.9.8 shows how much the conclusions depend on each assumption, using the 16- and 32-bit integer additions of the same table (0.05 and 0.1 pJ).

Memory is the number of bits needed to store the 295 088 weights plus those needed to store the membrane state of the 94 080 hidden neurons at the working resolution.

Chapter 5
Results

5.1 Input resolution

5.1.1 Accuracy does not discriminate

Input size	Val. accuracy	Std. dev.	Observed range	Epochs
224 px	0.9854	0.0007	0.9847–0.9861	22.7
112 px	0.9800	0.0035	0.9777–0.9840	17.3
64 px	0.9814	0.0011	0.9805–0.9826	24.7

Table 5.1: Validation accuracy by input resolution, mean and standard deviation over three seeds.

The ordering in Table 5.1 is not monotonic, as 64 px outperforms 112 px, and a genuine resolution effect would produce monotonic degradation. 0.9854 corresponds to 21.0 errors out of 1 435 validation samples and 0.9814 to 26.7, a gap of 5.7 samples, against a 95 % Wilson interval on a single measurement of about 0.7 percentage points. The four classes survive a twelvefold reduction in pixel count, which further confirm their being distinguished by coarse morphology.

Per-class recall is directional but small. Violin_Mode degrades most from 224 px to 64 px, by 1.00 percentage points, which is the direction predicted from morphology since narrow spectral lines are the structure most vulnerable to downsampling — but the magnitude is 3.7 samples out of 365, inside the confidence interval of a single measurement. Extremely_Loud degrades more at 112 px than at 64 px, a second non-monotonicity. The single most relevant observation is that **no class collapses at reduced resolution.**

5.1.2 Network health does discriminate

Since accuracy is uninformative, the resolution was decided on the internal state of the network, with the dead and saturated fractions of Section 3.5: per-neuron rates below 0.01 and above 0.80 on the validation partition. Neither cutoff is derived, and sweeping both on the adopted configuration shows no plateau (Figure 5.1). A factor of two either side of 0.01 moves the dead fraction by 11.2, 9.0, 6.7 and 11.2 percentage points in the four layers, so the distribution of rates holds no gap separating dead units from live ones. The dead fraction is the value of a monotone function read at a conventional point, and it carries meaning only as a comparison at fixed cutoff, which is how Table 5.2 uses it.

Layer 4 carries the resolution effect. The dead fraction falls monotonically from 36.3 % to 19.5 %, while the median per-neuron firing rate rises from 0.0399 to 0.1499. These were the two competing explanations for the higher mean activity observed at lower resolution

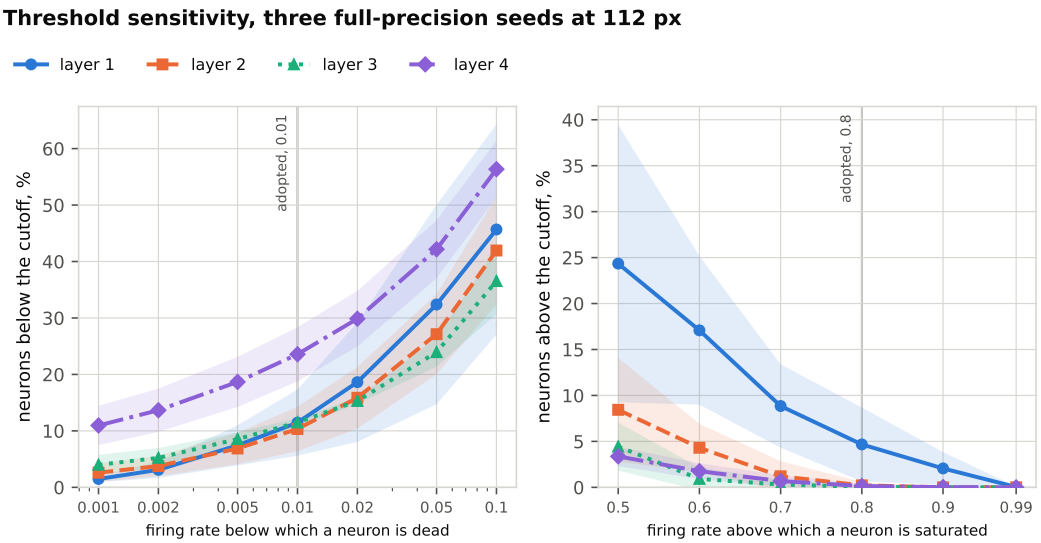


Figure 5.1: Dead and saturated fractions against the cutoff that defines them, mean and standard deviation over the three full-precision seeds at 112 px.

Input size	Layer 1	Layer 2	Layer 3	Layer 4
224 px	12.1 ± 3.3	16.6 ± 4.9	12.6 ± 2.3	$\mathbf{36.3 \pm 4.0}$
112 px	17.8 ± 3.4	11.8 ± 0.8	13.0 ± 1.0	$\mathbf{23.6 \pm 2.9}$
64 px	16.4 ± 4.8	11.2 ± 2.3	11.7 ± 2.3	$\mathbf{19.5 \pm 1.5}$

Table 5.2: Percentage of dead neurons by layer and resolution, mean $\pm$ standard deviation over three seeds, on validation, at the hyperparameters of the unconstrained search ($\beta = 0.4148$, $T = 7$).

— more neurons participating, or the same neurons firing harder — and both occur. Layer 2 also falls monotonically, by 5.4 percentage points against layer 4's 16.8; layers 1 and 3 are not monotonic at all. Saturation instead does not increase to compensate: neurons above 0.80 remain below 2.5 % in layer 4 at every resolution.

The interpretation of this effect is attributed to spatial redundancy. Layer 4 contains 25 088 neurons at 224 px and 6 272 at 112 px, so in absolute terms the number of *live* neurons falls from roughly 15 980 to 4 790. Reducing the resolution does not increase usable capacity: it reduces capacity while preferentially discarding units that were contributing nothing. At 224 px the deepest layer views a 14×14 map derived from a largely empty spectrogram, and a third of its units never encounter signal.

5.1.3 Decision: 112 pixels

Most of the benefit comes from the first halving: 224 to 112 px recovers 12.7 percentage points of dead fraction, the further move to 64 px only recovers 4.1 more. Membrane state at 8 bits falls from 368 to 92 KiB — the state of all four hidden layers held at once, which is the part of the footprint that must sit in fast memory and be rewritten at every one of the T steps, the weights being read-only.

The decisive argument against 64 px is the spatial resolution of the explanations themselves. SAM maps are computed on layer activations, whose side lengths are 56, 28, 14 and 7 at 112 px but 32, 16, 8 and 4 at 64 px. A top-20 % overlap needs enough positions for its selected set to describe a region: at 112 px the second layer offers 784 positions, a selected set of about 157, and the third 196; at 64 px the second layer offers 256, a set of about 51, and the third 64, a set of about thirteen. The temporal centre of mass sums over positions and is unaffected; the spatial metrics are not. Since the contribution of this thesis is the fidelity of explanations, the resolution that keeps a usable spatial map deeper in the network is preferred, whatever the lower one saves in compute.

5.2 Hyperparameters and the full-precision reference

The constrained search selected $\beta = 0.75$, $T = 8$, **learning rate** 1.047×10^{-4} at a best validation accuracy of 0.9847 within the 15-epoch budget.

The constraint on β is not equivalent to rounding. The unconstrained search, run at 224 px, selects $\beta = 0.4148$, whose admissible neighbours are 0.25 and 0.5. The constrained search at the working resolution selects 0.75, a value that rounding would never have reached. Rounding the unconstrained optimum after the fact would have produced 0.5, a value the constrained search had available and did not select.

$T = 8$ **is an optimum at the imposed ceiling**: the search was forbidden from exploring $T > 8$. The mitigating evidence is that an unconstrained search with range up to 15 selected $T = 7$, so the preference for long integration windows is mild.

5.2.1 The operating point improves the measurement instrument

Holding resolution and partition fixed and moving from the unconstrained configuration ($\beta = 0.4148$, $T = 7$) to the adopted one ($\beta = 0.75$, $T = 8$), the layer-4 dead fraction is *unchanged* at 23.6 %, while layers 1 to 3 lose between 1.5 and 6.3 percentage points (17.8/11.8/13.0 to 11.5/10.3/11.5 %, three seeds each). The change of configuration, β and T together, leaves the deep-layer deficit where it was, and resolution moves it. This supports the reading of Section 5.1: the deep-layer problem is spatial redundancy, not a dynamical pathology curable by a better decay constant.

What the change affects is saturation, by an order of magnitude and most where it matters. Neurons above 0.80 fall from 9.3/3.5/2.7/1.8 % across the four layers to 4.7/0.22/0.06/0.14 %. The layer-2 figure (where the SAM map is computed) is the consequential one. A saturated neuron fires at nearly every step. It is then almost always

firing at t, and always arrives there with a long spike history, so it accumulates the maximum possible contribution score on every image and therefore contributes a constant, non-discriminative floor to every SAM map. **Selecting on validation accuracy alone happened to reduce that contamination by a factor of sixteen at the reference layer.**

5.2.2 Three-seed reference

Quantity	Mean	Std
Validation accuracy	0.9833	0.0014
Epochs to early stop	21.0	1.7
Recall, Blip	0.9915	0.0028
Recall, Extremely_Loud	0.9855	0.0000
Recall, Scattered_Light	0.9839	0.0027
Recall, Violin_Mode	0.9726	0.0099

Table 5.3: Full-precision reference at 112 px, $\beta = 0.75$, $T = 8$, over three seeds on validation.

The reference is statistically indistinguishable from the 224 px configuration (0.9854): three samples out of 1 435 separate them, against a four-sample spread across seeds. This result means that **reducing the pixel count by a factor of four, and constraining β and T to hardware-realizable values costs no measurable accuracy.**

A tabular baseline measures how much of the task the released descriptors solve on their own. Trained on the four retained descriptors of the training partition, logistic regression reaches 0.900 on validation and a random forest or gradient boosting 0.948 (Table 5.4). The spiking network, which sees only the spectrogram, makes 24 errors on average against their 75, and the gap is widest on Violin_Mode, 0.973 against 0.917 in recall.

Model	Accuracy	Blip	Extremely_Loud	Scattered_Light	Violin_Mode	Errors
Majority class	0.2411	0.000	1.000	0.000	0.000	1089
Logistic regression	0.9003	0.946	0.919	0.957	0.781	143
Random forest	0.9477	0.967	0.953	0.955	0.917	75
Gradient boosting	0.9477	0.960	0.954	0.965	0.912	75
`GWGlitchSNN`, FP32	0.9833	0.991	0.986	0.984	0.973	24

Table 5.4: Tabular baseline on the four retained descriptors (duration, peak frequency, SNR and bandwidth, with $\log(1 + x)$ of duration and SNR), trained on the training partition and evaluated on validation over three seeds: accuracy, per-class recall and errors out of 1 435. The random-forest accuracy ranges over 0.9470–0.9484 across seeds; the other models do not vary.

Accuracy is reproducible, while the internal dynamics are not. At fixed accuracy the mean firing rate of layer 1 varies by a factor of 2.6 across seeds, and the coefficient of variation falls monotonically with depth: 46.9 % at layer 1, 28.6 % at layer 2, 15.3 % at layer 3, 9.5 % at layer 4. Furthermore, this regime is established within the first epoch and persists, which indicates that the initialization selects the firing regime and training does not subsequently equalize it. In the units used for the grid comparison of Section 5.9.7, the Wasserstein-1 distance between the per-neuron firing-rate distributions of two full-precision models differing only in seed falls with depth: 0.160, 0.066, 0.036 and 0.022 from layer 1 to layer 4, averaged over the three pairs, with the pairs spanning 0.088–0.240 in layer 1 and 0.018–0.024 in layer 4. Section 5.9.7 reads every quantized model against those figures, layer by layer. This is the observation that motivates Section 5.6.

5.3 Validation of the dynamics

At the adopted configuration the confusion matrix diagonal reads 0.994/0.986/0.984/0.970, consistent with Table 5.3; the only non-negligible errors are on Violin_Mode, the class that in the tabular domain partially overlaps both Blip and Scattered_Light (Figure 5.2).

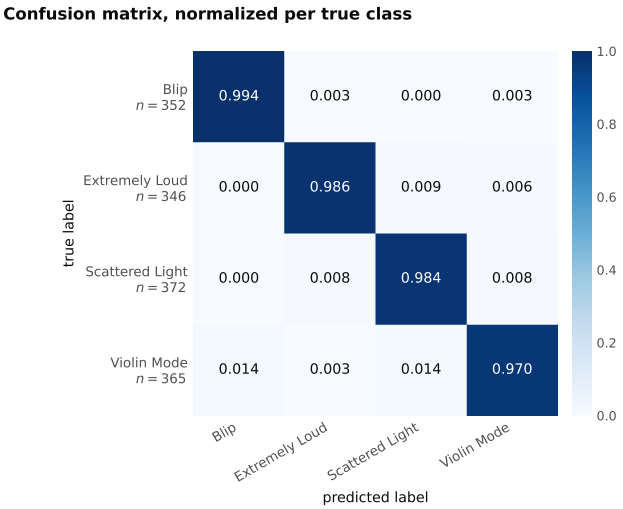


Figure 5.2: Confusion matrix of the reference checkpoint (seed 1234) on validation, normalized per true class.

The dead fraction is 16.1/12.6/11.3/22.8 % across the four layers (Figure 5.3). Saturation is confined to the first layer and to 24 of its 50 176 neurons, 0.05 %, with none above the threshold in the other three. The reference checkpoint is the quiet one. Across the three seeds the layer-1 saturated fraction is 4.67 ± 4.02 %, and in two of them neurons fire on more than 99 % of steps: 13 at seed 2345, 40 at seed 3456. Layer 1 moves in both directions at once, holding the fewest dead units in the seed where it holds the most saturated ones. These figures reproduce to every digit the unclipped row of Table 5.13, which is produced by a different script from a separate forward pass, so both are verified at no cost.

A note worth mentioning is that the layer-1 distribution is **bimodal**, with a secondary population near a firing rate of 0.65–0.75 that has no counterpart in the deeper layers. A high-activity subpopulation therefore exists in the first layer only, which may be related to its being the layer with the largest coefficient of variation across seeds.

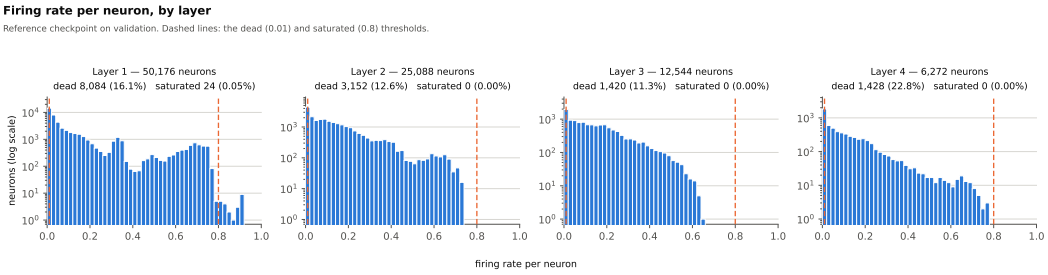


Figure 5.3: Distribution of per-neuron firing rates in the four hidden layers, on validation, with the dead (0.01) and saturation (0.80) thresholds.

The coefficient of variation of inter-spike intervals, pooled over the partition, concentrates between 0.35 and 0.43 in all four layers. Firing is regular rather than Poisson-like, which is the expected outcome under constant direct encoding. The input indeed does

not vary over time, and any irregularity observed is generated entirely by the membrane dynamics and by interaction between layers. This is an evidence about internal dynamics.

With $T = 8$ a neuron can produce at most seven intervals, so the five-interval threshold used for the estimate is close to the ceiling, which is why the accumulation runs over the whole partition and not on a single image. Between 184 and 196 of the 200 sampled neurons per layer met the threshold (Figure 5.4).

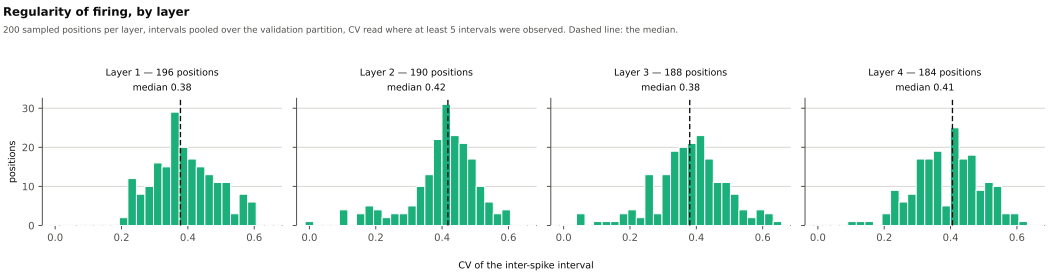


Figure 5.4: CV of inter-spike intervals for 200 sampled neurons per layer, accumulated over the validation partition.

5.4 What the explanations show

Figure 5.5 shows the SAM sequence of a single Blip at the reference layer. The map is zero at $t = 0$ by definition and builds up over the first steps as spike histories accumulate: it can be observed that from the third step onward the attended region is the vertical stroke of the glitch.

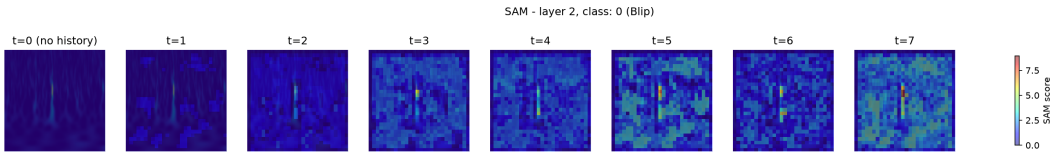


Figure 5.5: SAM at layer 2 for one Blip, one panel per time step, on a shared colour scale.

The class statistics (Figure 5.6) show four recognizably different attention patterns, each matching the morphology of its class. For Blip the mean map concentrates on the narrow vertical stroke and so does the variability, which is the stroke shifting slightly between instances. For Extremely_Loud instead the attended region is large and saturated, and it covers the lower and central part of the spectrogram. For Scattered_Light the attention lies on a horizontal band at low frequency. Violin_Mode produces the weakest and most diffuse mean map, with variability concentrated in a few localized spots along the line structure. It is also the class with the lowest recall, and the one whose explanations respond most when the kernel or the weights of a fixed network are perturbed (Sections 5.5.1 and 5.6.3).

Across depth (Figure 5.7) the progression is the expected one: from fine localization in the early layers to coarse localization in the deep ones, with maps of side 56, 28, 14 and 7. At layer 4 a 7×7 map leaves only 49 positions, which are too few for a top-20 % overlap or a structural comparison to be meaningful. Layer 2 instead resolves the morphology of every class, which is why **Layer 2 is adopted as the reference layer**. It is the deepest layer at which the metrics of Section 3.7.3 remain well-defined and the class morphologies remain legible.

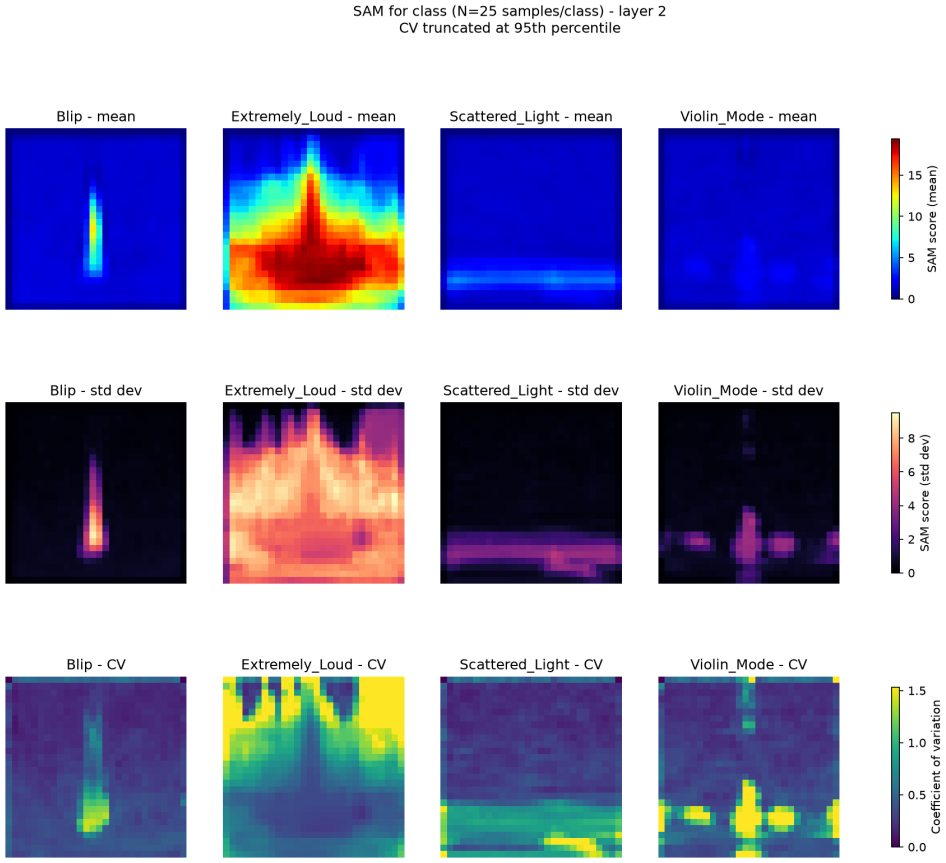
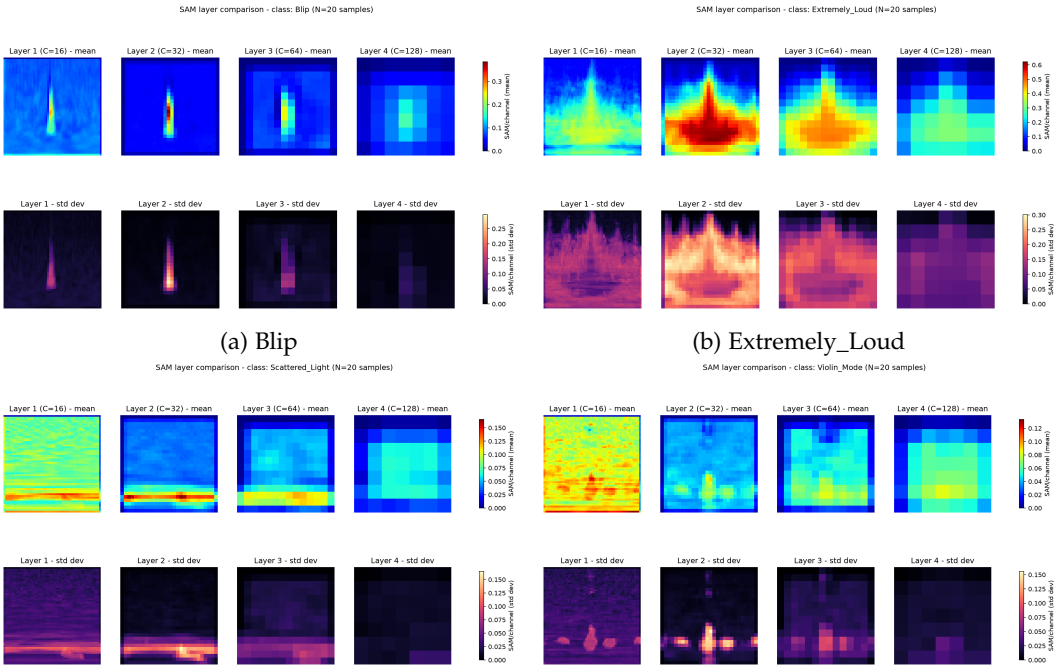


Figure 5.6: SAM statistics per class at layer 2, 25 validation images per class: mean (top), standard deviation (middle) and coefficient of variation (bottom, truncated at the 95th percentile). Mean and standard deviation share a colour scale across classes.



(a) Blip

(b) Extremely_Loud

(c) Scattered_Light

(d) Violin_Mode

Figure 5.7: SAM mean and standard deviation across the four hidden layers, 20 images per class, normalized by the channel count of each layer.

5.5 Fixing the explainer

5.5.1 Choice of the decay rate

SAM has one free parameter, γ, which represents **the decay of a temporal kernel, so it must be evaluated on the temporal axis.** Comparing maps averaged over time removes precisely the axis on which γ acts, and a high spatial correlation is then close to guaranteed regardless of γ. Three quantities are therefore measured: the spatial Pearson correlation between time-aggregated maps, retained for continuity with the literature; the *temporal centre of mass* of the SAM energy; and the *per-timestep correlation* against a reference γ.

These metrics were checked analytically: on a fully firing spike train with $T = 8$ the centre of mass moves from 4.86 at $\gamma = 0.1$ to 4.02 at $\gamma = 3.0$, so the metric does respond. A second check comes free from the definition. At $t = 1$ the score is $e^{-\gamma} S_0$, so changing γ multiplies the map by a constant, and Pearson correlation is affine-invariant. The measured per-timestep correlation at $t = 1$ is exactly 1.000 for every γ.

Class	Spatial PCC, γ 0.1 vs 3.0	CoM at 0.1	at 3.0	Shift
Extremely_Loud	0.9993	4.973	4.144	0.829
Blip	0.9890	5.192	4.339	0.852
Scattered_Light	0.9758	5.271	4.352	0.919
Violin_Mode	0.9598	5.260	4.327	0.933

Table 5.5: γ sensitivity at layer 2, mean over fifteen validation images per class.

Two main conclusions follow from Table 5.5. **The temporal shift is kernel arithmetic, and not explanation structure.** The measured shift of 0.83–0.93 steps is what a *fully firing* spike train produces, 0.84. It is also what a random train produces at any density: 0.84 at a firing rate of 0.05, 0.83 at 0.19, the rate measured in layer 2, and 0.84 at 0.70. The

independence is exact because with Bernoulli trains at rate p the expected SAM energy at step t is p^2 times a function of γ alone, and the p^2 cancels in the ratio that defines the centre of mass. Changing γ therefore moves the centre of mass by an amount that carries no information about which neurons fired or when. For this reason the centre of mass against γ is *not* a criterion for selecting γ — though it remains valid as a divergence metric, where models are compared at fixed γ and the trivial component cancels.

Sensitivity is strongly class-dependent. In terms of $1 - \text{PCC}$, Violin_Mode is 57 times more sensitive than Extremely_Loud. The ordering Violin_Mode > Scattered_Light > Blip > Extremely_Loud recurs under the weight perturbation of Section 5.6.3.

Within $\gamma \in [0.3, 0.9]$ all per-timestep correlations remain above 0.98 even in the worst class, and only $\gamma = 3.0$ diverges appreciably. $\gamma = 0.5$ **is adopted**, the value of the original formulation [22], which is also supported by evidence on the axis γ governs (Figure 5.8).

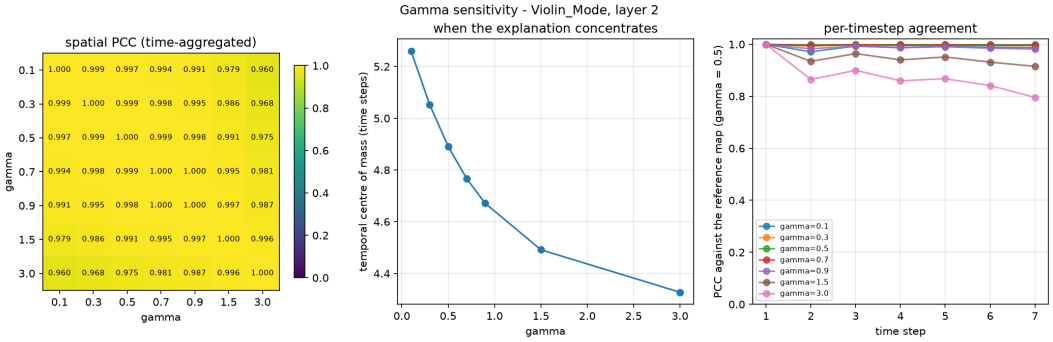


Figure 5.8: γ sensitivity for Violin_Mode, the most sensitive class, at layer 2. The other three classes show the same behaviour with smaller amplitude, in the proportions of Table 5.5.

5.5.2 Faithfulness

Robustness to γ is necessary but not sufficient, since a stable explanation could still be stably wrong. Faithfulness is measured with the deletion metric of Section 3.7.2: pixels are removed in decreasing order of SAM score, or in random order, and the confidence in the class predicted on the unmodified image is recorded at each fraction.

Class	AUC-SAM	AUC-random	Δ	95 % interval	Relative	Reading
Blip	0.3119	0.4826	−0.1706	[−0.257, −0.081]	−35.4 %	faithful
Extremely_Loud	0.6121	0.8936	−0.2815	[−0.345, −0.215]	−31.5 %	faithful
Scattered_Light	0.3835	0.3484	+0.0351	[−0.036, +0.087]	+10.1 %	not faithful
Violin_Mode	0.4528	0.4297	+0.0231	[−0.030, +0.066]	+5.4 %	not faithful

Table 5.6: Deletion metric by class on the reference checkpoint, 15 validation images each. A faithful explanation gives AUC-SAM below AUC-random, so a negative Δ is expected. The interval is the 95 % bootstrap interval of Δ over the fifteen images (Section 4.6).

The metric separates SAM from the baseline on two classes and not on the other two, and the split is systematic (Table 5.6, Figure 5.9). Blip and Extremely_Loud show reductions of 35 and 32 % in area. On Scattered_Light and Violin_Mode deleting the pixels SAM ranks highest destroys confidence *more slowly* than deleting pixels at random, by 10.1 and 5.4 % of area, which is roughly a third and a sixth of the 35 and 32 % by which it runs faster on the other two classes. The bootstrap intervals of Δ over the fifteen images exclude zero for Blip and Extremely_Loud and include it for Scattered_Light and Violin_Mode: on these two classes SAM order is not distinguishable from random order in either direction, which is not the same as pointing away from the evidence.

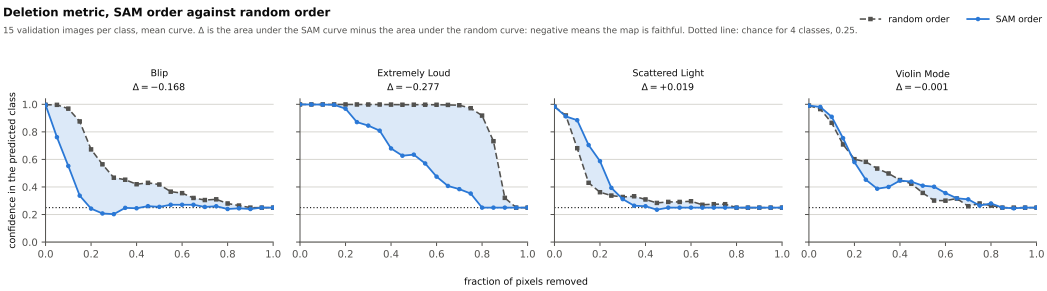


Figure 5.9: Deletion curves, SAM order (blue) against random order (grey), 15 validation images per class. The curves are drawn by `xai/deletion_metric.py`, whose random order is drawn with a different generator from the one behind Table 5.6; the Δ printed on the panels (-0.168, -0.277, $+0.019$, -0.001) differ from the table by at most 0.024, the sampling noise of the random baseline, and lie inside its intervals.

The pattern follows again the spatial geometry of the classes. Blip is a compact vertical line and Extremely_Loud a dominant saturated structure, so removing the high-scoring region removes the defining evidence. On the other hand Scattered_Light is an extended horizontal band, and Violin_Mode a set of thin distributed lines: the interpretation is that random deletion breaks the pattern across the whole image, whereas SAM-guided deletion concentrates removal in one region and leaves the rest — which still carries sufficient evidence — intact.

The random baseline of Extremely_Loud, 0.8936, is the highest of the four: its confidence survives the random removal of most of the image, which is consistent with evidence distributed across the whole spectrogram.

5.6 Calibrating the divergence metrics

The metrics used to compare two explanations return numbers between zero and one whatever is fed to them, so a few questions must be answered before any quantization result can be interpreted: what they report when nothing has changed, when something negligible has changed, and when everything has changed.

All measurements use fifteen validation images per class at layer 2 with $\gamma = 0.5$, on the three checkpoints of Table 5.3.

5.6.1 Verification of the instrument

The verification comes from the control arm, whose selected checkpoint is bit-identical to its parent (Section 5.9.1): compared against that parent the metrics return PCC $= 1$, IoU $= 1$ and a centre-of-mass difference of 0, in every class and every seed. This means that two models that are the same model therefore produce the same explanation under the full comparison pipeline, image sampling included, which is what the measurement assumes. The network is evaluated with gradients disabled, the images are the first of each class in the fixed order of the validation loader, identical for every model, and under direct encoding the forward pass involves no sampling, so the cross-model differences below cannot be run-to-run noise.

5.6.2 Two independent solutions

The spatial metrics collapse. Two models agreeing to within one validation sample in accuracy produce maps correlating at 0.153 on Scattered_Light, with a top-20 % overlap

Class	PCC mean	worst	IoU mean	worst	CoM diff, max
Extremely_Loud	0.917	0.869	0.520	0.460	0.076
Violin_Mode	0.844	0.816	0.597	0.543	0.115
Blip	0.612	0.336	0.593	0.503	0.083
Scattered_Light	0.477	**0.153**	0.376	0.234	0.104

Table 5.7: Divergence between full-precision models differing only in seed, over the three seed pairs. Chance-level IoU at $k = 0.20$ is 0.111.

of 0.234 against a chance level of 0.111. For comparison, a thirtyfold change in γ reduces the correlation only to 0.960: so initialization variance is 21 times larger than that effect.

The collapse is also not uniform across pairs. On Blip the three seed pairs correlate at 0.34, 0.72 and 0.78, so which two networks are compared matters more than which class is compared.

The collapse is not a relabelling of channels. SAM sums the contribution scores over the channel axis (Equation 2.11), so two networks whose layer-2 channels were the same features in a different order would produce identical maps. Positions, unlike channels, cannot be permuted: they are tied to the input image. A low correlation between seeds therefore means that the two networks **concentrate their layer-2 activity on different regions of the same spectrogram**. Three models of equal accuracy are not three copies of one solution: the firing rate of their first layer differs by a factor of 2.6 (Section 5.2), and their spatial explanations differ as well. The class ordering of Section 5.5.1 does not survive the change of seed either: across seeds the most disturbed class is Scattered_Light, ahead of Blip and then Violin_Mode, whereas within a fixed network Violin_Mode is always the most sensitive. A cross-seed comparison measures the distance between two solutions, which is not the effect of an intervention on one of them.

The temporal centre of mass survives. The maximum difference across seed pairs is 0.115 time steps, against the 0.93 produced by a thirtyfold change in γ in the worst class, a factor of eight. The metric sums the map over every position before reading its time axis, so it is blind to where the activity sits, which is precisely the property on which the two solutions differ. Independently trained networks place their activity in different regions of the input but accumulate it over the eight steps on nearly the same schedule.

5.6.3 A negligible perturbation

The grid compares a quantized model with its own parent, so the operative threshold is what a perturbation far smaller than quantization already produces. For this scope, gaussian noise was added to the weight tensors of one checkpoint — biases and LIF buffers excluded, since perturbing β would confound membrane dynamics with weight quantization — at magnitudes expressed as fractions of the measured 8-bit step of those weights (0.00168, from a maximum absolute weight of 0.214).

This is a graded form of the model randomization test of Adebayo et al. [4]: where they randomize weights to verify that a metric responds at all, this instead perturbs towards zero to find where it stops responding.

The spatial metrics are usable for same-parent comparisons. PCC remains above 0.975 under a perturbation as large as the entire 8-bit step (Table 5.8). This further supports that the collapse of Table 5.7 is a consequence of comparing different initializations, and not a fragility of the metric.

The metrics differ sharply in dynamic range. Measuring the distance from perfect agreement, PCC starts 0.025 away from 1 and can reach 0.847: 34 times its own floor. IoU starts 0.296 away and reaches 0.766, a factor of only 2.6, and is already at 0.704 under a perturbation 128 times smaller than the 2-bit step of the same global grid, and at least 26 times smaller than the 2-bit step of any channel. IoU is therefore read as an early indicator that a change has occurred, and PCC as the graded measure of how large it is.

Noise / 8-bit step	Extremely_Loud	Blip	Scattered_Light	Violin_Mode
0.01	1.0000	0.9997	0.9996	0.9991
0.1	0.9999	0.9981	0.9977	0.9930
0.5	0.9998	0.9948	0.9938	0.9824
1.0	0.9997	0.9921	0.9901	0.9749

Table 5.8: Spatial PCC against the unperturbed model, seed 1234, fifteen validation images per class. On the same global grid the 4-bit and 2-bit steps are 18 and 128 times the 8-bit step; the per-channel steps the quantizer applies are, layer by layer, 5 to 15 and 37 to 102 times it on average (Section 4.3).

The temporal centre of mass sits between them, with a factor of 3.9 between its numerical floor (0.029 steps) and the seed level (0.115).

Extremely_Loud does not respond. Its PCC is 0.9997 under the largest perturbation and its worst per-timestep correlation is 0.996. Combined with its behaviour under γ, this class registers no explanation change under any calibration measure tested.

5.6.4 Fine-tuning alone

Every quantized model in the grid is its parent with bits removed *and* then fine-tuned, which means that part of any divergence measured against that parent belongs to the fine-tuning. The levels above cannot separate the two contributions: the numerical floor is a perturbed copy that was never trained, and the independent-solutions level is two networks grown from different initializations. To answer the question a model that went through the whole procedure with nothing removed is needed.

The control arm is that model, **taken after its last fine-tuning epoch**, not at the selected checkpoint, which for the control coincides with the parent (Section 5.9.1). Its parameter vector moves by an L_2 distance of 0.395, 0.736 and 0.399 over the three seeds, nine tenths of it in the last convolution and the linear readout.

Seed	Epochs	$\|\Delta w\|_2$	Accuracy	IoU	PCC	CoM shift	Worst class IoU
1234	6	0.395	0.9819	0.711	0.985	−0.032	0.671
2345	14	0.736	0.9791	0.792	0.989	−0.035	0.779
3456	6	0.399	0.9805	0.748	0.987	−0.031	0.705
Mean			0.9805	**0.750**	0.987	−0.033	0.718

Table 5.9: The control arm after its last fine-tuning epoch, against its own full-precision parent, twenty validation images per class at layer 2, $\gamma = 0.5$. CoM shift is the signed difference of the temporal centre of mass in time steps, negative when the SAM energy moves earlier. No bits are removed in any of these runs.

Six to fourteen epochs with the learning rate scaled by 0.1, and no bits removed, move the top-20 % overlap to 0.750, with a seed range of 0.711 to 0.792 and a worst class of 0.671 (Table 5.9). Validation accuracy falls by 0.3 points over the same runs, so the movement is not a model that degraded, but rather a model that kept training.

A relevant consideration is that the seed whose weights moved furthest, 2345 at 0.736, is the seed whose explanation moved *least*, at 0.792: over three runs this is an observation and not a result itself, but distance in parameter space does not predict distance in explanation space, which is the premise the grid rests on. The Pearson correlation of this anchor is 0.987 against an IoU of 0.750: the metric disagreement that recurs throughout the grid appears here too, where there is no quantization to attribute it to.

The centre of mass moves as well, and always the same way. In eleven of the twelve class-and-seed cells continued training shifts the SAM energy earlier, by 0.033 steps on

average and at most 0.048, beyond the floor of every class; the twelfth, Extremely_Loud on seed 1234, moves by 0.001. A shift toward earlier steps in a quantized model is therefore expected from the training alone, and only its excess over this level, or a shift in the opposite direction, can be charged to the removed bits.

5.6.5 What the calibration establishes

Level	What it measures	PCC	IoU	CoM
Numerical floor	Weight noise at 1× the 8-bit step	0.975	0.704	0.029
Fine-tuning alone	The procedure with no bits removed	0.987	0.750	0.033
γ reference	A thirtyfold change in the kernel	0.960	—	0.93
Independent solutions	Two FP32 models, different seeds	0.153	0.234	0.115

Table 5.10: The four calibration levels at 112 px, layer 2, $\gamma = 0.5$, on validation. The floor, the γ reference and the independent-solutions level are the worst class in each case; the fine-tuning level is the mean over classes and seeds, its worst class being 0.671 in IoU and 0.048 in centre-of-mass shift. The first, third and fourth use fifteen images per class, the fine-tuning level twenty, matching the grid. Per-class values are in Table 5.11.

	PCC		IoU		CoM $\lvert\Delta\rvert$	
Class	floor	fine-tuning	floor	fine-tuning	floor	fine-tuning
Blip	0.9921	0.9864	0.704	0.745	0.023	0.041
Extremely_Loud	0.9997	0.9976	0.841	0.777	0.003	0.009
Scattered_Light	0.9901	0.9852	0.760	0.730	0.017	0.038
Violin_Mode	0.9749	0.9793	0.724	0.749	0.029	0.045
Mean	0.9892	0.9871	0.757	0.750	0.018	0.033

Table 5.11: Numerical floor and fine-tuning level class by class, the form in which the grid is read. Floor: weight noise of one 8-bit step on the seed-1234 checkpoint, fifteen validation images per class. Fine-tuning: the control arm after its last epoch against its own parent, three seeds, twenty images per class. A grid result is attributed to the removed bits only where it exceeds both levels of its class: below both in PCC and IoU, above both in CoM.

Table 5.10 collects the four calibrations and Table 5.11 gives the floor and the fine-tuning level class by class. The *numerical floor* marks measurement noise: a divergence smaller than the one a negligible weight perturbation already produces cannot be read. *Fine-tuning alone* is what the procedure costs when nothing is quantized. Neither level contains the other. Fine-tuning moves the centre of mass more than the floor in every class, the correlation more in three, and the overlap less on Blip and Violin_Mode but more on Extremely_Loud and Scattered_Light. A grid result is therefore attributed to the removed bits only where it exceeds both levels of its own class, and the worst-class values of Table 5.10 serve as a summary, never as the threshold for a mean over classes. The γ *reference* gives a scale for the quantization effect: a PCC of 0.90 would mean that quantization does more than a thirtyfold change of the explainer itself. The *independent-solutions* level is a pessimistic upper bound and a result in its own right.

The class sensitivity ordering — Violin_Mode > Scattered_Light > Blip > Extremely_Loud — recurs identically under the two interventions that act on a fixed network: changing the kernel and perturbing the weights. Nonetheless, it does not survive a change of initialization, where Scattered_Light is the most disturbed class and Violin_Mode only the third. Within a given network, explanation sensitivity is a property of the class morphology, so every grid result should be analysed and reported per class.

5.7 Preparing the membrane arm

5.7.1 The range of the membrane potential

The membrane potential of every LIF layer was recorded at every time step over the full validation partition (1 435 images, eight steps) on the reference checkpoint. Table 5.12 summarizes the distributions.

Layer	min	$p_{0.1}$	median	$p_{99.9}$	max	std	$> U_{thr}$
1	−12.75	−9.43	−0.08	6.40	8.76	1.98	14.9 %
2	−15.73	−11.10	0.25	5.63	11.64	1.42	15.1 %
3	−47.03	−36.82	0.00	11.27	20.23	4.22	15.3 %
4	−26.04	−17.01	−0.62	6.34	14.67	2.59	11.5 %

Table 5.12: Membrane potential of the four hidden layers over the validation partition, reference checkpoint (seed 1234). The last column is the fraction of values above the firing threshold $U_{thr} = 1$, which equals the mean firing rate of the layer.

The analysis shows that the distributions are wide and strongly skewed towards negative values: in layer 3 the potential reaches −47, almost fifty thresholds below rest. If the grid would cover the full observed range it would be dominated by values far from the threshold, which is the only point at which a potential affects a spike decision. At 4 bits, a symmetric grid over $[-47, +47]$ would have a step of more than six thresholds, and the network could not be represented at all.

5.7.2 Clipping the potential costs nothing

Clipping is an operation that sets a maximum and minimum threshold in order to truncate extreme values or outliers. Before any quantization, the potential of the four hidden layers was *clipped* at every time step to a candidate range, on the full-precision network, and the effect on accuracy and dynamics was then measured. Clipping isolates the choice of range from the choice of bit-width: if a range already changes the network in full precision, no grid built on it could be faithful.

Clipping range	Val. accuracy	Δ samples	Layer-4 CoM
none	0.98328	—	3.9469
per layer, $[p_{0.1}, p_{99.9}]$	0.98328	0	3.9469
per layer, $[p_1, p_{99}]$	0.98328	0	3.9469
fixed $[-8, +4]$	0.98328	0	3.9469
fixed $[-4, +4]$	0.98328	0	3.9469
fixed $[-2, +2]$	0.98258	−1	3.9415
unsigned $[0, +4]$	0.98258	−1	3.9483
unsigned $[0, +2]$	**0.98258**	**−1**	**3.9431**
sanity $[-0.5, +0.5]$	0.84808	−194	4.0684

Table 5.13: Effect of clipping the hidden membrane potentials on the full-precision reference (seed 1234), on validation. The tolerance is the spread of the three-seed reference, four samples out of 1 435. The last column is the temporal centre of mass of the layer-4 spike counts.

Three observations follow from Table 5.13.

Negative potentials carry no information the network uses. Clipping to $[0, 2]$ removes every negative value — specifically in layer 3 a range of 47 thresholds — and costs one validation sample out of 1 435, so a quarter of the spread between seeds. The per-layer

dead fractions move by at most 0.3 percentage points (16.1 / 12.5 / 11.3 / 22.5 % against 16.1 / 12.6 / 11.3 / 22.8 %), and the largest shift in the temporal centre of mass of any layer's spike counts is 0.004 steps, meaning that is well below the 0.029-step floor measured for the SAM centre of mass in Section 5.6 (the two quantities are related but not identical). Below rest, how far below no longer matters: a neuron at -20 and a neuron at 0 both need more than one threshold of input to fire.

The upper bound can be tight. This comes from the observation that potentials above twice the threshold are rare and can be clipped at no measurable cost. Since the clipping acts after the spike decision, it only alters the carried state of neurons that have just fired well above threshold; the spike they emitted at that step is unaffected.

The clipping is not inert. The sanity range $[-0.5, +0.5]$, which keeps the potential permanently below threshold after each step, drops accuracy to 0.848 and raises the layer-3 firing rate from 0.153 to 0.227. This guarantees that the hook acts on the state the network actually carries, and the null results above are genuine.

The unsigned range $[0, 2]$ is adopted: it is the tightest range tested that stays within tolerance and it centres the threshold in the grid. What it saves against the symmetric range $[-2, 2]$, which performs identically, is given in Section 4.4.

5.8 Energy and memory of the reference

Table 5.14 applies the accounting of Section 4.8 to the full-precision reference, using the per-hidden-layer firing rates averaged over the three seeds (0.258/0.186/0.185/0.126).

Layer	MAC (dense)	Operations	Energy (µJ)
conv1 (analog input, once)	3.76×10^6	3.76×10^6 MAC	17.31
conv2	10.04×10^6	20.72×10^6 SOP	18.65
conv3	10.04×10^6	14.93×10^6 SOP	13.44
conv4	10.04×10^6	14.83×10^6 SOP	13.34
linear	0.03×10^6	0.03×10^6 SOP	0.02
SNN, conv1 once			**62.8**
SNN, conv1 at every step			183.9
Equivalent CNN (all MAC)	33.89×10^6		155.9

Table 5.14: Estimated energy per inference of the FP32 reference at 112 px, $T = 8$, with 45 nm costs of 4.6 pJ per MAC and 0.9 pJ per accumulate.

The spiking network costs 62.8 µJ per inference: 17.31 µJ for the 3.76×10^6 multiply-accumulates of the first convolution, computed once, and 45.45 µJ for the 50.5×10^6 accumulates that the spikes of the four hidden layers trigger downstream, at 0.9 pJ each.

A convolutional network of identical shape executes all 33.89×10^6 operations as multiply-accumulates in a single pass, at 4.6 pJ each, for 155.9 µJ. This means that the spiking counterpart costs 2.5 times less. The gap comes from the price of each operation, not from their number.

The saving depends primarily on the treatment of the first layer, as it is the only layer that multiplies. Recomputing its constant current at every step would cost 183.9 µJ, more than the convolutional equivalent, and the two readings differ by a factor of 2.9. Even computed once, the first convolution is 28 % of the budget.

The dependence of the firing regime on the seed, established in Section 5.2, reaches the energy as well. Computed seed by seed, the estimate is 50.0, 78.1 and 60.1 µJ for seeds 1234, 2345 and 3456, where the difference is coming almost entirely from the first two layers, where the firing regime varies the most. Three models indistinguishable in accuracy therefore differ by more than half in estimated energy, so every energy comparison in the grid is made at matched seed.

In memory, the 295 088 weights occupy 1 152.7 KiB in 32-bit float (the 244 biases, kept in full precision, are negligible), and the membrane state of the 94 080 hidden neurons another 367.5 KiB. At 8, 4 and 2 bits these values drop to 288, 144 and 72 KiB for the weights and 92, 46 and 23 KiB for the state.

5.9 The quantization grid

All thirty fine-tuning runs were completed without failure. Unless stated otherwise, every figure that will be shown below is a mean over the three seeds, each quantized model being compared with the full-precision parent of its own seed, on validation. Figure 5.10 summarizes the grid.

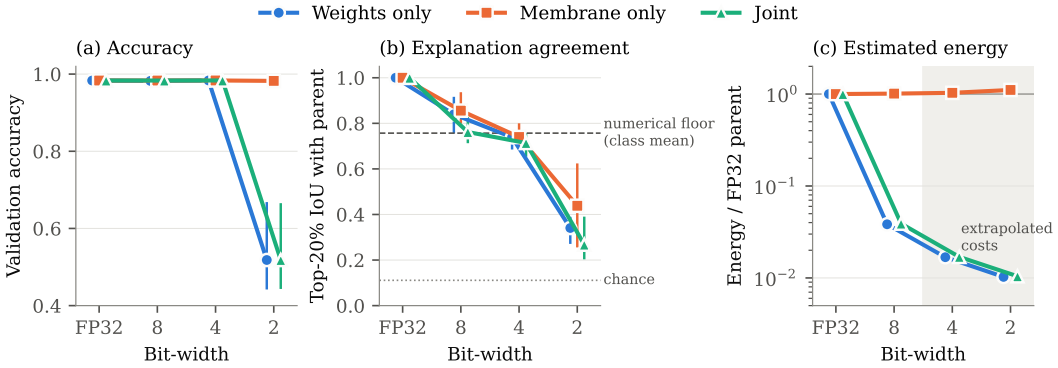


Figure 5.10: The grid on a common bit-width axis. (a) Validation accuracy after fine-tuning. (b) Top-20 % IoU between the SAM maps of each quantized model and those of its parent, mean over the four classes; the dashed line is the class-mean numerical floor (0.757), the dotted line chance. (c) Estimated energy per inference relative to the parent, logarithmic scale; the shaded region uses extrapolated per-operation costs. Points are means over three seeds, bars span the seeds. Markers are offset horizontally for legibility.

5.9.1 Gates and control

Attaching and removing the quantizer restores the original weights exactly, and a quantized checkpoint reloads with zero drift. With no fine-tuning, the joint 8-bit model of seed 1234 reaches 0.98328 validation accuracy, which is identical to its parent; indeed, the 8-bit weight arm loses three samples out of 1 435, so inside the four-sample tolerance.

On none of the three seeds did fine-tuning with no quantizer improve on the parent's validation accuracy, so the selected control checkpoint is the parent itself (Section 4.1) and returns PCC and IoU of exactly 1 and a centre-of-mass shift of exactly 0 in every class and seed. The model at the end of fine-tuning does move, by the amount measured in Section 5.6.4.

The guard of Section 4.6 holds as well: the 8-bit weight arm keeps a spatial correlation of at least 0.982 with its parent in every class at every seed.

5.9.2 Accuracy

At 8 and 4 bits accuracy is indistinguishable from full precision in every arm (Table 5.15); on average no arm differs from the reference by more than 1.3 validation samples, inside the four-sample spread of the reference itself. Fine-tuning, however, matters at 4 bits: post-training quantization of the weights costs about 40 samples, and fine-tuning recovers all of them.

Arm	Bits	PTQ	After fine-tuning	Δ samples	Epochs
Weights	8	0.9819	0.9823 ± 0.0008	-1.3	9.7
Membrane	8	0.9814	0.9833 ± 0.0000	0.0	8.3
Joint	8	0.9819	0.9833 ± 0.0007	0.0	8.0
Weights	4	0.9547	0.9835 ± 0.0016	$+0.3$	11.7
Membrane	4	0.9807	0.9835 ± 0.0011	$+0.3$	10.3
Joint	4	0.9508	0.9835 ± 0.0015	$+0.3$	14.0
Weights	2	0.3591	0.5182 ± 0.1300	-667	18.3
Membrane	2	0.9517	$\mathbf{0.9823 \pm 0.0004}$	-1.3	13.0
Joint	2	0.3652	0.5182 ± 0.1275	-667	19.0

Table 5.15: Validation accuracy across the grid: post-training quantization (PTQ, no fine-tuning) and after fine-tuning, mean $\pm$ standard deviation over three seeds. Δ is the difference from the FP32 reference (0.9833 ± 0.0014) in validation samples out of 1 435.

At 2 bits the arms separate completely. The membrane arm keeps the parity, 0.9823, within three samples of its parent in every seed. The weight and joint arms collapse to 0.518, and the collapse is a silence of the readout. On 800 to 1 145 of the 1 435 validation images the 2-bit weight models emit no output spike at all; an argmax over an all-zero count returns the first class, Blip, and every Blip prediction of these six models comes from such an image. The reported accuracy therefore counts as correct the 345 to 351 Blip images on which the network says nothing: without them it is 0.20 to 0.43. On the images where the readout fires without a tie, 96.2 to 98.3 % are classified correctly. Fine-tuning recovers 16 points over post-training quantization but cannot recover a ternary network in weights: a 2-bit membrane, with a single level between rest and threshold, is enough to classify, while ternary weights silence the output layer.

For the membrane, the result inverts the finding of Smith et al. [38] recalled in Section 4.4, that membrane quantization needs at least four bits: here a 2-bit membrane keeps accuracy at parity. For the weights the two studies do not conflict. The weights that Smith et al. find viable at two bits are trained with a learned quantizer; the grid used here is uniform and scaled to the largest weight of each channel, the kind of quantizer that collapses in their CIFAR-100 experiments as well. Whether the dynamics survive is a separate question, assessed below.

5.9.3 Explanation divergence

At 8 bits the explanation is preserved; at 4 bits it moves slightly beyond fine-tuning alone. Every reading below is made class by class against the levels of Table 5.11; the per-class values are in Tables 5.16 and 5.18.

At 8 bits the weight-only and membrane-only arms stay within the floor of every class on all three metrics, with one exception: the PCC of Extremely_Loud, 0.9993 and 0.9991 against a floor of 0.9997 that leaves almost no room below it. The joint arm passes the floor in most classes, on PCC in three, on IoU in two and on the centre of mass in all four, yet stays within the fine-tuning level of every class on every metric. Nothing has moved at 8 bits by an amount the experiment can attribute to the removed bits.

At 4 bits the overlap falls below the fine-tuning level in three classes for the weight arm and in all four for the membrane and joint arms, and below the floor in two, two and four classes respectively. The margins are small. No 4-bit arm falls more than 0.05 below the fine-tuning level of a class, less than the 0.08 that separates the fine-tuning controls of different seeds (0.711 to 0.792), and all of it remains far above the independent-solutions level of 0.234. The 95 % bootstrap intervals of the mean overlap over the twelve cells, $[0.726, 0.753]$ for the weight arm and $[0.725, 0.755]$ for the membrane arm, reach the mean fine-tuning level of 0.750; only the joint arm, $[0.698, 0.729]$, stays below it.

Arm	Bits	Blip	Extremely_Loud	Scattered_Light	Violin_Mode	Mean	Worst cell
Weights	8	0.821	0.854	0.841	0.841	0.839	0.716
Membrane	8	0.845	0.876	0.843	0.858	0.856	0.766
Joint	8	0.753	0.779	0.749	0.766	0.762	0.674
Weights	4	0.719	0.787	0.717	0.733	0.739	0.649
Membrane	4	0.726	0.773	0.719	0.743	0.740	0.628
Joint	4	0.696	0.752	0.691	0.715	0.714	0.595
Weights	2	0.269	0.335	0.415	0.344	0.341	0.127
Membrane	2	**0.300**	**0.559**	**0.490**	**0.404**	**0.438**	**0.102**
Joint	2	0.217	0.339	0.344	0.171	0.268	0.057
Numerical floor		0.704	0.841	0.760	0.724	0.757	—
Fine-tuning alone		0.745	0.777	0.730	0.749	0.750	0.671

Table 5.16: Top-20 % IoU between the SAM maps of each quantized model and those of its parent, layer 2, $\gamma = 0.5$, twenty validation images per class, mean over three seeds. "Worst cell" is the lowest class-and-seed value. The last two rows are the levels of Table 5.11; chance is 0.111.

Arm	Bits	PCC mean	PCC worst	CoM $\lvert\Delta\rvert$ mean	CoM $\lvert\Delta\rvert$ worst	CoM signed
Weights	8	0.995	0.982	0.007	0.015	−0.005
Membrane	8	0.995	0.989	0.010	0.025	−0.009
Joint	8	0.989	0.973	0.022	0.049	−0.022
Weights	4	0.982	0.960	0.049	0.094	−0.048
Membrane	4	0.985	0.963	0.044	0.064	+0.043
Joint	4	0.977	0.949	0.021	0.043	+0.018
Weights	2	0.827	0.748	0.211	0.369	−0.211
Membrane	2	0.923	0.799	0.068	0.137	−0.053
Joint	2	0.780	0.580	0.235	0.367	−0.235

Table 5.17: Spatial Pearson correlation and temporal centre-of-mass shift (in time steps) against the parent, same protocol as Table 5.16. A negative signed shift means that the spike energy of the explanation moves earlier. Worst-class numerical floors: PCC 0.975, CoM 0.029; independent-solutions level: PCC 0.153, CoM 0.115. Per-class values at 8 and 4 bits are in Table 5.18.

		PCC				CoM $\lvert\Delta\rvert$			
Arm	Bits	Blip	E. Loud	S. Light	V. Mode	Blip	E. Loud	S. Light	V. Mode
Weights	8	0.9950	0.9993	0.9946	0.9907	0.007	0.003	0.007	0.009
Membrane	8	0.9938	0.9991	0.9941	0.9924	0.012	0.003	0.011	0.015
Joint	8	0.9879	0.9989	0.9870	0.9821	0.025	0.008	0.024	0.032
Weights	4	**0.9808**	**0.9959**	**0.9796**	**0.9736**	0.059	0.018	0.056	0.060
Membrane	4	0.9843	0.9983	0.9827	0.9766	**0.058**	**0.014**	**0.052**	**0.055**
Joint	4	0.9746	0.9940	0.9721	0.9675	0.027	0.009	0.023	0.026
Numerical floor		0.9921	0.9997	0.9901	0.9749	0.023	0.003	0.017	0.029
Fine-tuning alone		0.9864	0.9976	0.9852	0.9793	0.041	0.009	0.038	0.045

Table 5.18: Spatial PCC and absolute centre-of-mass shift per class at 8 and 4 bits, mean over three seeds, with the levels of Table 5.11. In bold, the two quantities the hypothesis orders: at 4 bits the weight-arm PCC lies below both levels of every class, and the membrane-arm shift above both levels of every class.

The temporal centre of mass moves in both single-target arms at 4 bits, by 0.014 to 0.060 steps across classes, above the floor and the fine-tuning level of every class and below the 0.115 that separates independent solutions. The joint arm does not: its shifts, 0.009 to 0.027, stay within the fine-tuning level of every class. The direction separates the arms. Fine-tuning alone moves the SAM energy earlier (-0.033 on average); weight quantization moves it further the same way (-0.048), membrane quantization reverses it ($+0.043$), and in the joint arm the two partly cancel ($+0.018$). Part of the weight-arm shift therefore belongs to the training, while the membrane arm moves the explanation by about 0.08 steps relative to the control.

At 2 bits the weight and joint arms explain a different classifier. Their IoU falls to 0.34 and 0.27 and their centre of mass shifts by more than the independent-solutions level. Since those models no longer classify, the divergence of their explanations is a consequence of the collapse, and not a measurement of explanation fidelity.

The membrane arm at 2 bits is exactly the result the grid was built to find. While accuracy is at parity, the IoU with the parent's explanation is 0.438, between 34 and 57 % below the floor of each class and far below anything fine-tuning produces; its 95 % bootstrap interval over the twelve cells is $[0.408, 0.467]$. In all twelve class-and-seed cells the upper end of the interval lies below the floor of the class, the highest mean being 0.642. The two lowest, Blip and Violin_Mode on seed 1234 (0.102, interval $[0.080, 0.124]$, and 0.110, interval $[0.064, 0.170]$), are indistinguishable from the chance level of 0.111: for those, the 20 % of the image that SAM highlights in the quantized model overlaps the parent's no more than two random selections would do.

Figure 5.11 shows, for one seed, the image of median overlap in each class.

The comparison that carries the thesis is between two models with the same accuracy. The 4-bit weight model and the 2-bit membrane model both reach 0.983 (0.9835 and 0.9823). The first highlights nearly the same region as its parent, with an IoU of 0.739 (interval $[0.726, 0.753]$); the second reaches only 0.438 ($[0.408, 0.467]$). Selected on accuracy, the two are interchangeable but their explanations are not.

The spatial correlation alone would have given a different answer. The 2-bit membrane model scores a PCC of 0.923, inside the range of 0.86 to 0.99 that Rabbi et al. [33] report for convolutional models performing well at INT8, and would be read as a surviving explanation; it is below the floor of every class, by an amount that understates what changed. Pearson correlation weighs every position by its value and is dominated by the broad structure that every map of a class shares; the top-k overlap reads only the ranking, and only above the $(1 - k)$ quantile. This is the disagreement between metrics that Bylinskii et al. [8] warn of, and it is why both are reported.

The effect depends on the seed. The 2-bit membrane IoU is 0.255, 0.624 and 0.435 for the three seeds, a spread of 0.37, of the size of the largest effect measured within a seed (the 0.443 SNR gradient of Scattered_Light on seed 1234, Table 5.21). The class ordering is also not the one of the calibration: Blip, and not Violin_Mode, loses the most (0.300), and Extremely_Loud, which registered no perturbation in the calibration, now moves (0.559).

The effect also depends on k. On seed 3456, the one model for which the overlap was computed at several mask sizes (Figure 5.12), the 2-bit membrane keeps 0.54 of the parent's top 2 % on Blip and 0.47 on Scattered_Light, 53 and 47 times chance, against 0.17 and 0.50 at the reported $k = 0.20$. Extremely_Loud and Violin_Mode go the other way, from 0.16 and 0.10 at $k = 0.02$ to 0.61 and 0.46. For the compact Blip stroke the positions SAM ranks highest survive quantization, and what changes is the ranking of the background that completes the 20 % set (Figures 5.11 and A.3). The saturated Extremely_Loud maps peak on a broad region, where a small k chooses among near-equal values. How much the explanation changes, and which class it hits hardest, depend on k as well as on the model.

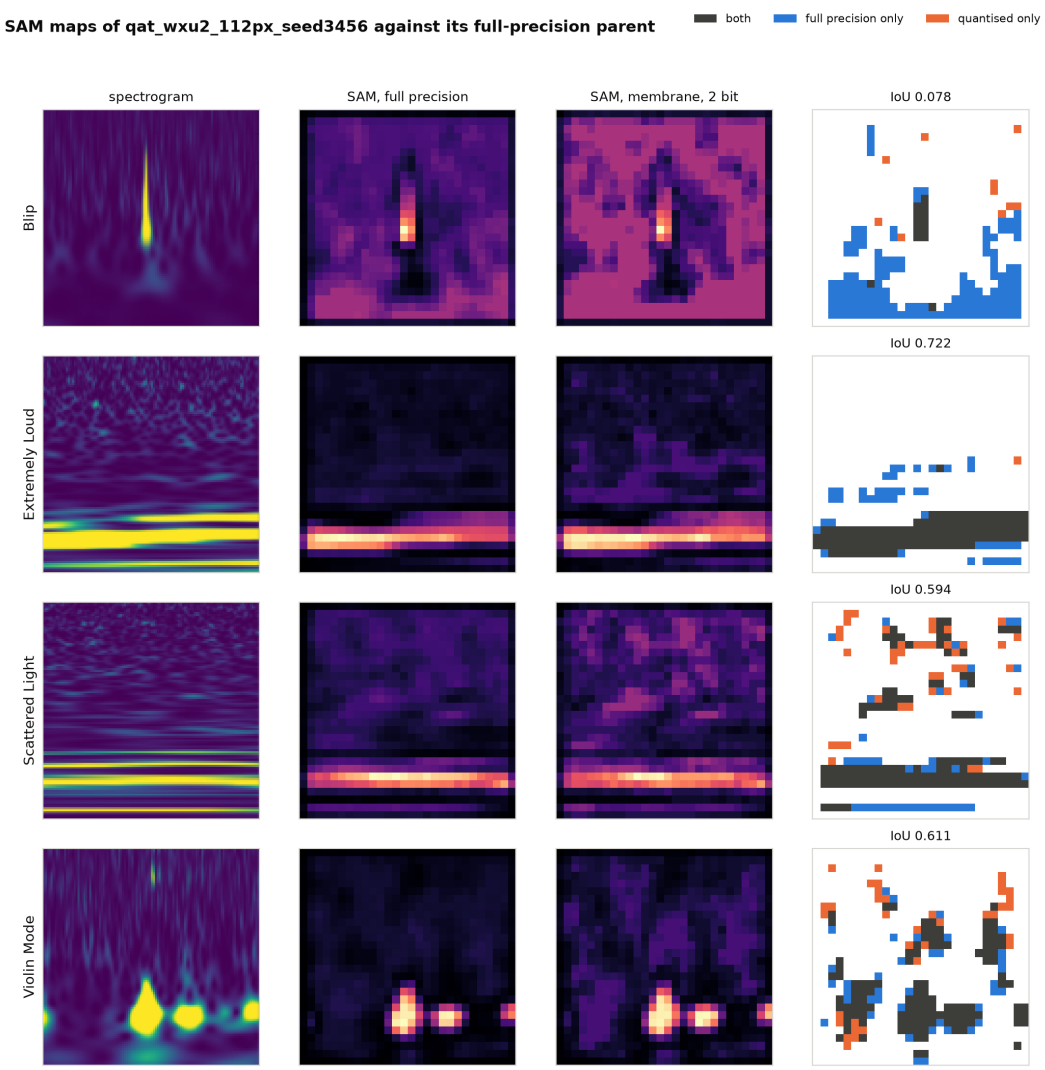


Figure 5.11: SAM at layer 2 ($\gamma = 0.5$, summed over time) of the 2-bit membrane model of seed 3456, the seed closest to the arm mean, against its parent. Each row is the image of median IoU among the twenty of its class. Right column: top-20 % sets, shared (dark), parent only (blue), quantized model only (orange), with the IoU of the image. The Blip image (IoU 0.078) keeps the stroke in both maps, and the two top-20 % sets diverge on the low-intensity background. The Extremely_Loud image is labelled so by the Gravity Spy classifier with a score of 0.93, but its SNR of 49 and peak frequency of 25 Hz are atypical of the class, and in the 0.5 s window it shows the horizontal low-frequency bands of Scattered_Light, an instance of the label noise discussed in Section 2.4.

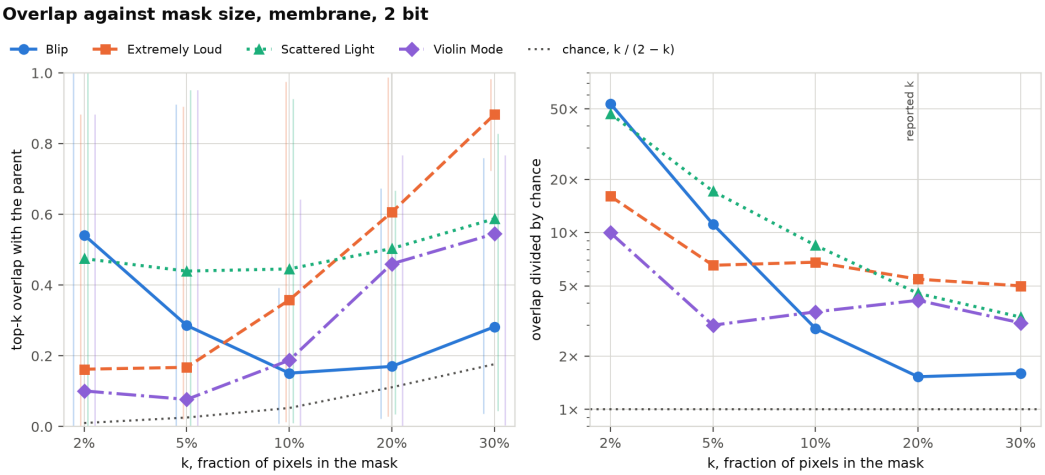


Figure 5.12: Top-k IoU between the 2-bit membrane model of seed 3456 and its parent as a function of k, twenty validation images per class. Left: mean, bars from minimum to maximum, dotted line the chance level $k/(2-k)$. Right: the same values divided by chance. The reported $k = 0.20$ is marked.

5.9.4 The hypothesis

The hypothesis predicted an ordering: the membrane arm displaces the temporal centre of mass at a higher bit-width than the one at which the weight arm lowers the spatial correlation. The grid does not show that ordering. Both effects first appear at the same bit-width, 4 bits, and in every class on the mean over seeds (Table 5.18): the weight arm lowers the PCC below both the floor and the fine-tuning level, and the membrane arm moves the centre of mass beyond both. At 8 bits neither arm crosses the fine-tuning level of any class on either metric.

Read cell by cell, against the floor of the class and the fine-tuning level of the same seed, the reading does not change. At 8 bits the two quantities the hypothesis orders cross both levels in none of the twelve class-and-seed cells. At 4 bits the membrane arm moves the centre of mass beyond both in nine of twelve, the exceptions being Scattered_Light and Violin_Mode on seed 1234 and Extremely_Loud on seed 3456; the weight arm lowers the PCC below both in ten of twelve, the exceptions being Violin_Mode on seeds 2345 and 3456, where it stays above the floor of that class. The two effects remain simultaneous.

At 2 bits, where the arms do separate, the visible effect of membrane quantization on the explanation is *spatial* — which pixels are highlighted — while its centre-of-mass shift (0.068) stays below the independent-solutions level. **As formulated in terms of SAM metrics, the hypothesis is not supported.**

The per-model validation pass measures a second temporal property, the regularity of firing, and there the 2-bit membrane differs from every other configuration.

A 2-bit membrane makes the firing of layer 1 regular. The first evidence is the coefficient of variation of the inter-spike intervals, pooled over the partition (Table 5.19), which in the three parents lies between 0.35 and 0.39. With a 2-bit membrane it falls by about a third, to 0.24–0.25 in the membrane-only arm and 0.21–0.26 in the joint arm. A 4-bit membrane leaves it within 0.01 of the parent, and ternary (2-bit) weights alone lower it as well, but by less than half as much.

The pooled value mixes intervals from different neurons and different images, so the coefficient is also computed inside each image, over the neurons that emit at least three spikes in the eight steps. In all six models with a 2-bit membrane it is *exactly zero* for every such neuron in every image: each one fires at perfectly regular intervals. Exact regularity is not rare at $T = 8$ even in full precision: in the parents 42 to 53 % of the same neuron-image pairs are exactly regular, and the mean coefficient is 0.153 to 0.188. The 2-bit membrane

Configuration	Seed 1234	Seed 2345	Seed 3456
FP32 parent	0.391	0.347	0.359
Membrane, 4 bit	0.385	0.340	0.353
Weights, 2 bit	0.362	0.297	0.327
Membrane, 2 bit	**0.242**	**0.252**	**0.246**
Joint, 2 bit	**0.235**	**0.213**	**0.256**

Table 5.19: Coefficient of variation of the inter-spike intervals in layer 1, pooled over the validation partition as in Section 5.3. Between 98 and 100 % of the 200 sampled positions reach the five-interval threshold in every configuration, so the comparison is between the same population of neurons.

takes that fraction to 100 %. The two sets of neurons are comparable in size: 20 to 53 % of neuron-image pairs reach three spikes in the quantized models, 18 to 51 % in the parents. The two controls locate the effect in the 2-bit membrane grid. A 4-bit membrane raises the fraction of exactly regular pairs by only 3 to 6 points over the parent of the same seed, to 45 to 59 %, with a mean coefficient of 0.133 to 0.179, while ternary weights alone raise it to 67 to 77 %, with a mean coefficient of 0.070 to 0.096, without making every neuron periodic.

The cause is in the grid of Table 4.3. At 2 bits the stored state takes one of four values, and under direct encoding layer 1 receives the same input current at every step, so the next stored state is a function of the current one alone. Every trajectory therefore falls into a cycle within a few steps, and the spike train repeats with it. Deeper layers receive spike trains that vary from step to step and are not constrained in the same way.

Membrane quantization does act on the temporal dynamics, as the hypothesis argued, but not before the spatial explanation changes: the regularity appears at 2 bits, where the overlap of the explanation also collapses, and not at 4 bits, where the pooled coefficient stays within 0.01 of the parent and the fraction of exactly regular neurons rises by a few points. What a 2-bit membrane changes is the *regularity* of the spike trains. The temporal centre of mass measures where in time the SAM energy falls, not how regularly the spikes arrive, and cannot detect the change.

The firing statistics separate the two arms (Figure 5.13). In the membrane-only arm the 2-bit membrane raises the firing rate of every layer, from 0.258/0.186/0.185/0.126 to 0.273/0.218/0.229/0.151, and moves the dead fraction by at most four points. Its regularizing effect is confined to layer 1: in layers 2 to 4 the pooled coefficient of variation rises by 3, 9 and 6 %. The 2-bit weights double the dead fraction of layer 4, from 23.6 to 43.1 %, and raise the saturated fraction of layer 1 from 4.7 % in the parents to 13 % on average, higher in every seed and 22 % in the worst. They lower the coefficient of variation in every layer, by 21 % in layer 4, but there only 60 % of the sampled positions keep five intervals, against 88 % in the parents: activity disappearing rather than becoming regular.

5.9.5 Faithfulness

The deletion metric was repeated on the 2-bit membrane models, which was the only configuration with intact accuracy and a changed explanation, with the baseline recomputed on the parent of each seed (Table 5.20).

The explanation is changed, and not invalidated. In five of the seven interpretable cells the quantized model's own map separates from random deletion with an interval that excludes zero, and the median gap is 0.94 of the parent's. In the other two, Blip and Scattered_Light on seed 3456, the gap falls to 0.043 and 0.055, from 0.113 and 0.075 in the parent, and its interval reaches zero. Explaining the quantized model with its parent's maps works about as well as with its own: the paired difference between the two gaps averages +0.012 over the seven cells, and its interval includes zero in five of them; in the remaining two, Blip on seed 1234 and Scattered_Light on seed 3456, the model's own map

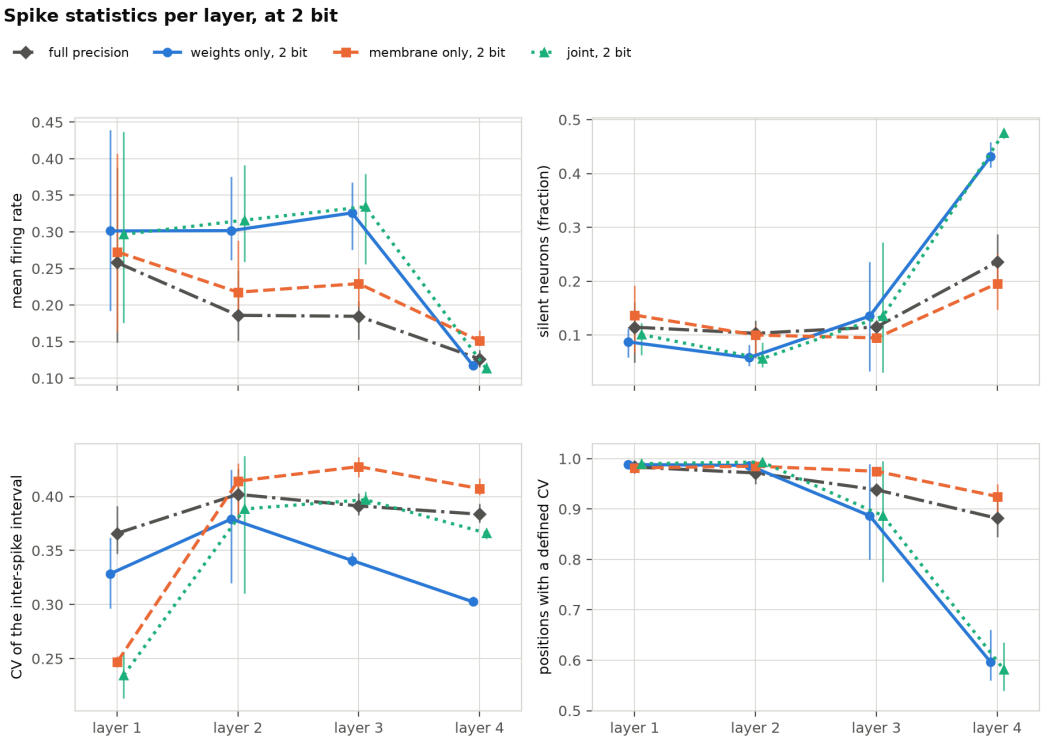


Figure 5.13: Spike statistics of the 2-bit arms against the full-precision parents, per layer: mean firing rate, dead fraction, pooled CV-ISI and fraction of the 200 sampled positions with a defined CV, mean over seeds with bars spanning the seeds. In layer 1 every configuration keeps a defined CV on 98 to 100 % of the positions.

Seed	Class	Parent baseline	Own map	Parent's map	Own − parent's
1234	Blip	0.171 [0.081, 0.257]	0.189 [0.116, 0.261]	0.140 [0.059, 0.229]	+0.049 [0.013, 0.085]
1234	Extremely_Loud	0.282 [0.215, 0.345]	0.272 [0.204, 0.344]	0.266 [0.195, 0.340]	+0.007 [−0.009, 0.028]
2345	Extremely_Loud	0.198 [0.139, 0.251]	0.134 [0.097, 0.172]	0.143 [0.107, 0.183]	−0.009 [−0.019, 0.001]
2345	Violin_Mode	0.084 [0.038, 0.137]	0.163 [0.115, 0.222]	0.168 [0.126, 0.217]	−0.005 [−0.035, 0.024]
3456	Blip	0.113 [0.046, 0.190]	**0.043** [−0.007, 0.096]	0.041 [−0.017, 0.103]	+0.002 [−0.025, 0.027]
3456	Extremely_Loud	0.222 [0.164, 0.285]	0.209 [0.155, 0.264]	0.199 [0.146, 0.254]	+0.010 [−0.002, 0.028]
3456	Scattered_Light	0.075 [0.010, 0.141]	**0.055** [−0.001, 0.112]	0.025 [−0.029, 0.077]	+0.030 [0.005, 0.053]

Table 5.20: Deletion gap (AUC of random deletion minus AUC of SAM-guided deletion; larger means more faithful) for the 2-bit membrane models, fifteen validation images per class, with 95 % bootstrap intervals over the images. The last column is the paired difference between the quantized model's own map and its parent's map. Only the seven class-and-seed cells whose parent baseline reaches 0.05 are shown; the other five have no signal to degrade. In bold, the two cells where the own map no longer separates from random deletion.

is the more faithful, by 0.049 and 0.030. This holds although the top-20 % regions of the two maps overlap with an IoU of only 0.438.

The two measurements are therefore not in contradiction, but different, and they answer different questions. The deletion metric rewards any map that points at evidence the model uses, and when the evidence is redundant — like in a Blip where stroke covers many positions— several disjoint sets of pixels satisfy it. The IoU instead says which of those sets a reader is shown.

Which classes SAM can explain at all depends on the seed. Only seven of the twelve class-and-seed cells have a baseline signal: Extremely_Loud in all three parents, Blip in two, Scattered_Light and Violin_Mode in one each. Even Blip's baseline gap ranges from 0.029 to 0.171 across networks of equal accuracy. Therefore the set of classes on which faithfulness can be assessed is a property of the individual network, not of the class.

5.9.6 Stratification by event strength

	Membrane, 2 bit			Membrane, 4 bit		
Class	1234	2345	3456	1234	2345	3456
Blip	+0.121	+0.038	+0.164	+0.060	+0.012	+0.039
Scattered_Light	+0.443	−0.021	+0.138	+0.242	+0.017	+0.109
Violin_Mode	+0.134	−0.038	−0.124	+0.093	+0.022	+0.028
Mean of the three	+0.233	−0.007	+0.059	+0.132	+0.017	+0.059
Extremely_Loud	−0.469	−0.202	−0.354	−0.234	−0.101	−0.142

Table 5.21: IoU gradient with event strength: IoU in the high-SNR tercile minus IoU in the low-SNR tercile, terciles computed within each class, fifteen images per cell. A positive value means that weak events lose their explanation more.

The data obtained do not support the hypothesis that degradation concentrates on weak glitches. The mean gradient over Blip, Scattered_Light and Violin_Mode is +0.233, −0.007 and +0.059 in the three seeds at 2 bits (Table 5.21): positive on average, but the spread between seeds is as large as the effect. The between-seed range of the 2-bit IoU itself, 0.369, is of the same size as the largest within-seed gradient, 0.443 on Scattered_Light, so which network is measured matters as much as how strong the event is.

It can be seen that in Extremely_Loud the gradient is negative in all six measurements: the *strongest* events lose their explanation most. These are the events that saturate the spectrogram, so that most of the image carries evidence and the top-20 % region is least constrained, so a small change of the network is enough to move it.

5.9.7 Firing-rate distance and the disagreement subset

The two measurements read the grid on the network's own terms (against the seed-to-seed reference of Section 5.2) rather than on the explanation's.

The Wasserstein-1 distance between the per-neuron firing-rate distributions of a quantized model and its parent is read layer by layer against the distance between full-precision seeds. At 8 bits every layer stays below 0.015 on average. At 4 bits the values reach 0.030 in layer 3 and 0.023 in layer 4, never above the largest seed-pair distance, below the whole seed range in layer 1 and, in layer 4, at the seed mean. The 2-bit membrane stays well below the seed level in layers 1 and 2 (0.019 and 0.032) and exceeds its mean in layers 3 and 4 (0.045 and 0.025), remaining inside the range of the pairs in layer 3 and above it in layer 4. The 2-bit weight arms reach 0.148 and 0.161 in layer 3, about three times the largest distance between seeds. Down to 4 bits, then, quantization moves the firing regime

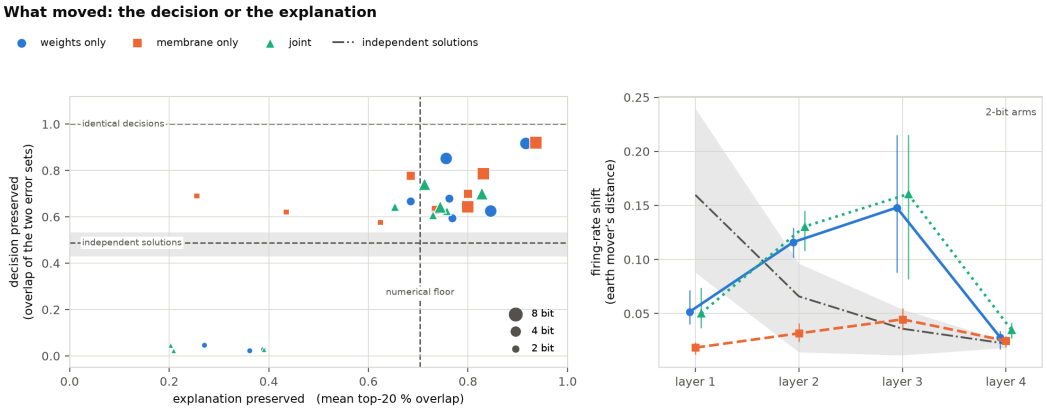


Figure 5.14: Left: each point is one quantized model; abscissa, mean top-20 % IoU with its parent; ordinate, Jaccard overlap between the error sets of model and parent; marker size, bit-width. Dashed horizontal line and band: the same overlap between full-precision models of different seed (mean 0.49, range 0.43–0.53); dashed vertical line: the worst-class numerical floor. Right: Wasserstein-1 distance between the per-neuron firing-rate distributions of the 2-bit models and their parents, per layer; dash-dot line and band: the same distance between full-precision seeds, mean and range over the three pairs.

no more than a change of initialization does, the comparison Smith et al. [38] draw; at 2 bits the membrane does so only in the deep layers.

The disagreement subset says the same of the decisions. The Jaccard overlap between the error sets of a quantized model and its parent is 0.78–0.80 at 8 bits in the single-target arms and 0.62–0.70 at 4 bits and at the 2-bit membrane, against 0.49 between full-precision models of different seed and 0.03 in the collapsed arms. Of the 1 435 validation images, three to six on average are correct in the parent and wrong after quantization at 8 and 4 bits, six at the 2-bit membrane, against 671 in the 2-bit weight arms. **The 2-bit membrane model keeps most of its parent's decisions and explains them with different pixels**: about six images go from correct to wrong and five the other way, and its error set overlaps the parent's with a Jaccard index of 0.58 to 0.69 across seeds, so roughly a third of the errors of the two models are not shared. Figure 5.14 places every model on both axes: the 2-bit membrane models keep the decision overlap of the 4-bit models and lose most of the explanation overlap.

5.9.8 Energy and memory

The membrane arm does not save energy; at 2 bits it costs more. Its weights stay in full precision, so the cost of each operation never falls, and the operation count rises: +1.5 % at 8 bits, +4.4 % at 4 and +15.3 % at 2 against the parent, and the energy with it, at 1.011, 1.031 and 1.108 times the parent (Table 5.22). What it does save is the state memory, 344 KiB at 2 bits, almost a quarter of the total footprint.

Part of that rise belongs to the training rather than to the bits. Fine-tuning alone, measured on the control after its last epoch, raises the operation count by 5.0 to 6.2 %, from 1–3 % more spikes in layer 1 to 10–22 % more in layer 4. Measured against that state, with every run also taken at its last epoch (`results/energy_final.json`), the 8-bit configurations change the count by −0.3 to +0.5 %, the 4-bit ones by +2.5 to +6.6 %, the 2-bit membrane by +11 % and the 2-bit weights by +48 and +51 %, the collapsed networks firing more. At 8 bits the removed bits change the operation count by nothing. The 2-bit membrane costs 1.08 times the control in energy, a rise that belongs to the quantized state. In the weight and joint arms the saving comes entirely from the cheaper operations.

Arm	Bits	SOP (10^6)	Energy (µJ)	× FP32	Memory (KiB)
FP32	32	50.50	62.76	1.000	1520.2
Weights	8	51.02	2.40	0.038	655.7
Membrane	8	51.34	63.52	1.011	1244.6
Joint	8	52.29	2.43	0.039	380.0
Weights	4	54.21	1.06*	0.017	511.6
Membrane	4	52.62	64.67	1.031	1198.6
Joint	4	54.84	1.07*	**0.017**	**190.0**
Weights	2	74.60	0.63*	0.010	439.5
Membrane	2	57.81	69.34	1.108	1175.7
Joint	2	76.06	0.65*	0.010	95.0

Table 5.22: Synaptic operations, estimated energy and memory per inference, mean over three seeds, from the firing rates of each selected checkpoint. "× FP32" is the mean of the per-seed ratios to the parent. The first convolution and every accumulate are costed at the weight bit-width. *Extrapolated per-operation costs. Memory is weights plus membrane state.

This grid did not find a trade-off between efficiency and explainability. The configuration that changes the explanation at constant accuracy, the 2-bit membrane, is dominated on every axis by the 4-bit weight model: same accuracy, IoU 0.739 against 0.438, energy 0.017 against 1.108 times the parent, memory 512 against 1 176 KiB.

The configuration to deploy is the **joint 4-bit model**: accuracy at parity (0.9835), an explanation that moves slightly beyond what continued training alone produces (IoU 0.714 against 0.750, below the fine-tuning level of every class by at most 0.05 and of every seed by 0.02 to 0.06, against 0.438 for the 2-bit membrane), estimated energy about 1/60 of the parent under the costing of Table 5.22, and one eighth of its memory (190 KiB).

A few assumptions of Section 4.8 determine the size of the saving. The first convolution receives the analog image, and Table 5.22 costs it at the weight bit-width, which assumes the input is quantized to the same width. At 4 bits that layer is 24 % of the joint model's energy. Instead, costed in floating point, it lowers the joint 4-bit saving factor from 59 to 2.8, 4.2 and 3.3 on the three seeds, whose firing regimes differ. The accumulates are costed at the weight width too. With 16-bit integer additions the joint 4-bit factor becomes 20 to 23 across seeds, with 32-bit additions 10 to 12, and with the floating-point first layer and 32-bit additions together 2.4 to 3.2. Under 16-bit additions the 2-bit models, which fire more, are no longer the cheapest: the joint 2-bit factor falls to 16, below the 21 of the joint 4-bit model.

The operation counts and the memory column depend on neither assumption, and neither does the conclusion that the membrane arm is dominated, since its weights and accumulates stay in floating point under every costing. The factor by which the deployable model is cheaper does depend on them, by more than an order of magnitude.

5.10 Test-set evaluation

The test partition was opened once, after every decision in this thesis had been taken on validation, and all thirty-three models of the grid were evaluated on it in a single pass. The test accuracy was computed twice by independent routes: from the aggregated evaluation loop and from the per-sample predictions, and the two agree to within 10^{-6} for every model.

Neither quantized configuration costs measurable accuracy on held-out data (Table 5.23), as both sit marginally above the full-precision reference, by 1.6 and 3.0 samples

Model	Test accuracy	95 % Wilson interval
FP32 reference	0.9745	[0.9650, 0.9815]
Joint, 4 bit	0.9756	[0.9663, 0.9824]
Membrane, 2 bit	0.9766	[0.9674, 0.9832]

Table 5.23: Accuracy on the test partition (1 436 images), opened once, mean over three seeds. The interval is computed on the mean.

out of 1 436, inside a between-seed spread of 0.9714 to 0.9770 in the reference itself. The drop from validation to test is of 0.9 points for the reference, 0.8 for the joint 4-bit model and 0.6 for the 2-bit membrane. It is of the same size in all three, so it is not a cost of quantization. Part of it is expected from the protocol: the hyperparameters, the stopping epoch and the selected checkpoint of every model were all chosen on the validation partition, whose accuracy is therefore an optimistic estimate, and the test partition is the only unbiased one.

Configuration	Blip (359)	Extremely_Loud (362)	Scattered_Light (343)	Violin_Mode (372)
FP32 reference	0.9824	0.9724	0.9874	0.9570
Joint, 4 bit	0.9842	0.9724	0.9854	0.9615
Membrane, 2 bit	0.9842	0.9761	0.9874	0.9597

Table 5.24: Per-class recall on the test partition, mean over three seeds, with the number of test images of each class.

Per-class recall keeps Violin_Mode as the weakest class in all three configurations (Table 5.24), and no class moves by more than half a percentage point between the reference and either quantized model. The order of the other three differs from validation: on test Scattered_Light leads and Blip follows, the same in all three configurations. What the network decides survives the quantization that changes its explanations; what it looks at does not.

Chapter 6
Discussion and Conclusions

6.1 The instrument

This thesis set out to measure whether a quantized spiking neural network reasons like its full-precision parent, and first found that the measurement could not be interpreted until the instrument was calibrated.

A four-block convolutional spiking network classifies four families of transient noise (*glitches*) in LIGO data at 0.9833 ± 0.0014 validation accuracy over three seeds, under constraints that keep the configuration deployable by construction: a shift-implementable membrane decay, at most eight time steps, and a 112-pixel input. None of the constraints costs measurable accuracy, and the resolution choice lowered the dead fraction of the deepest layer by 12.7 percentage points.

The Spike Activation Map was implemented, its summation range corrected, and its free parameter fixed on the temporal axis it governs instead of on time-aggregated maps that cannot see it. Its faithfulness was tested quantitatively: on the reference checkpoint it holds on two classes of four, with a geometric interpretation for the other two, and across the three parents on seven of the twelve class-and-seed cells.

The calibration established four levels of divergence: weight noise as large as an entire 8-bit quantization step (PCC 0.975 in the worst class), continued training with no bits removed (PCC 0.987 and IoU 0.750 on average), a thirtyfold change in the explainer (PCC 0.960), and initialization alone (PCC 0.153). The last is far larger than expected, and it shaped the rest of the study.

Spatial metrics cannot compare independently trained models. Two networks that agree to within one validation sample in accuracy produce explanations correlating at 0.153. The metric is not at fault: SAM sums over channels, so a mere reordering of features would leave the map unchanged, and the low correlation means that the two networks concentrate their activity on different regions of the same input.

They can compare a model with its own parent. Weight noise as large as an entire 8-bit step, at least 26 times smaller than the 2-bit step of any channel, leaves the PCC above 0.975 in every class. A divergence beyond the floor of its class is therefore a change in the explanation, not noise of the instrument. The two results answer different questions, which is why every quantized model is fine-tuned from, and compared with, its own parent.

Resulting figures are not interpretable without these levels. Rabbi et al. [33] describe models whose explanations correlate with the full-precision ones at 0.86 to 0.99 (Grad-CAM) and 0.81 to 0.97 (LIME) as performing well at INT8. For this architecture a correlation of 0.85 falls between the numerical floor and the independent-solutions level,

so it cannot be read without knowing which comparison produced it. The floor and the independent-solutions level need no training beyond a multi-seed baseline, which is good practice regardless; the fine-tuning level needs one run per seed with the quantizer left out.

The class sensitivity ordering is a property of the class within a network. Changing the kernel and perturbing the weights, the two interventions that act on a fixed network, rank the classes identically: Violin_Mode, then Scattered_Light, Blip and Extremely_Loud. A change of initialization does not preserve the ordering, and neither does 2-bit quantization, where Blip loses most. Extremely_Loud, saturated and spatially extensive, registers no change under any calibration, yet moves once the membrane is quantized to 2 bits.

The preparation of the membrane arm added a result of its own: the network makes no use of negative membrane potentials. Clipping the hidden state to $[0, 2]$, which removes up to 47 thresholds of negative range, costs one validation sample and leaves the dynamics unchanged, so the membrane can be stored on an unsigned grid. At 8 and 4 bits that saves one bit against a symmetric grid; at 2 bits it keeps the one level between rest and threshold that a symmetric grid would lose.

6.2 Quantization and explanation fidelity

Accuracy is not a sufficient criterion for spiking networks. The finding of Rabbi et al. [33] on convolutional networks carries over, and in a sharper form. Two configurations of this network classify equally well, a 4-bit weight model and a 2-bit membrane model, and their explanations differ by a factor of almost two in agreement with the original. Selected on accuracy alone, the second could be deployed: a model that reaches the right labels while its highlighted region overlaps the original's with an IoU of 0.438. In the detector-characterization setting of Chapter 1 that is the scenario that matters. How deep the change goes depends on the class: on the one model measured at several mask sizes, the most salient positions of a Blip survive and the change lies in the region around them, while for Violin_Mode it reaches the most salient positions as well.

The divergence is present but bounded. The changed explanations are still faithful in five of the seven cells where the parent's maps are, with bootstrap intervals that exclude zero, and no longer separate from random deletion in the other two. Deleting the pixels they highlight degrades the model's confidence almost as much as deleting the parent's highlights does, a median of 94 % of the parent's gap, and in five of the seven cells the quantized model can be explained with its parent's maps as well as with its own. The quantized network has not stopped using spectrogram evidence; among several redundant sets of evidence it now highlights a partly different one. A faithfulness test alone would have declared the explanation intact and an overlap test alone would have suggested it was lost: two valid metrics measuring different properties of the same map.

The hypothesis is not supported; its mechanism shows only in the dynamics. The prediction that the membrane would move the temporal axis of the explanation before the weights moved its spatial axis is not supported: both arms first cross the floor and the fine-tuning level of every class at 4 bits, the membrane arm in nine of the twelve class-and-seed cells and the weight arm in ten, and at 2 bits the membrane changes where the explanation points more than when. The membrane does reach the timing of the network itself, at 2 bits. With a single representable level between rest and threshold, every first-layer neuron that fires at least three times within an image does so at exactly regular intervals, in all six models with a 2-bit membrane, against about half of such neurons in the full-precision parents. At 4 bits the pooled regularity of layer 1 stays within 0.01 of the parent, so the dynamics do not change before the spatial explanation does. The temporal centre of mass,

built to capture the time axis of the explanation, is blind to the change, because it measures where the spike mass falls and not how it is spaced. A temporal explanation metric for spiking networks should see regularity as well as timing, and the present instrument does not.

The comparison with Smith et al. inverts for the membrane. Smith et al. [38] report that the membrane needs at least four bits, losing 4–8 % of accuracy at two, while learned weight quantization remains viable at two bits. The weight result agrees with theirs: a uniform 2-bit grid scaled to the largest weight of each channel collapses the classifier here, silencing its readout on most images, as their uniform quantizer with a wide clipping range collapses on CIFAR-100. The membrane result does not: with a grid fitted to the measured range, a 2-bit membrane keeps accuracy at parity.

Weak events are not where the explanation fails. Quantization was expected to disturb spike decisions mostly in weak events, whose potentials sit near threshold. The data give no consistent support: at 2 bits the IoU gradient with event strength is $+0.233$, -0.007 and $+0.059$ on the three seeds, a spread as large as the effect. The one consistent effect has the opposite sign: the strongest Extremely_Loud events lose their explanation most, in all six measurements. A plausible reading is that their evidence covers most of the image, so the highlighted region is the least constrained.

Efficiency and explainability do not come to a trade off. The configuration that changes the explanation at constant accuracy is the 2-bit membrane, and it is the one that saves no energy: with weights in full precision the per-operation cost stays the same while firing rates rise, at 2 bits by 11 % beyond what fine-tuning alone adds. The deployable configuration, joint 4-bit, keeps accuracy at parity and an explanation that moves only slightly beyond what continued training alone produces, with one eighth of the memory and an estimated energy of about one sixtieth of the parent's when the input and the accumulators are quantized with the weights, about one third when they are not.

Seed variance is the dominant source of variation. Three full-precision networks of equal accuracy differ by a factor of 2.6 in the firing rate of their first layer, by more than half in estimated energy (50.0 to 78.1 µJ), and in which classes SAM can explain faithfully at all. The 2-bit membrane IoU ranges from 0.26 to 0.62 across seeds, a spread as large as any effect measured within a seed.

6.3 Limitations

Generalization of the levels. The four calibration levels are specific to this architecture, this dataset and this explainer. That independently trained networks of equal accuracy can attend to different regions of the same input is unsurprising; the size of the gap measured here is not general, and a different architecture could place the levels elsewhere. What generalizes is the claim that the levels must be measured, not the values measured here.

Labels are model output. Accuracy is agreement with the Gravity Spy convolutional classifier, whose output the labels are, and not with a physical ground truth. A model that reproduced that classifier's systematic errors would score well here. The per-class confidences are released with the labels and would allow accuracy to be stratified by the reference classifier's own certainty, which was not done.

T **sits at the imposed ceiling.** The search was forbidden from exploring more than eight time steps, so it is not known whether it would have preferred a longer integration window. An unconstrained search, at 224 px, selected seven, which bounds the likely cost without removing the question.

The deletion metric does not separate SAM from random everywhere. On the reference checkpoint it fails on Scattered_Light and Violin_Mode, and across the three parents on five of the twelve class-and-seed cells. The geometric explanation offered in Section 5.5.2, that with extended, redundant evidence the region SAM ranks highest is not necessary because the rest of the pattern still suffices, is an interpretation and not a measurement. Deletion tests whether the highlighted pixels are necessary; the insertion curve of the same work [32], which starts from the mean image and restores pixels in order of attributed importance, tests whether they are sufficient. If the interpretation holds, insertion should separate SAM from random order on the classes where deletion does not.

The explanation is class-agnostic. SAM, as formulated and as implemented here, uses no labels, so the deletion metric measures faithfulness to the model's *predicted* class. This is the correct reading of the metric, but it makes misclassified examples informative, and they should be inspected separately rather than pooled.

The temporal metric is partly kernel arithmetic. A thirtyfold change of γ moves the centre of mass by 0.83 to 0.93 steps whatever the spike trains, the amount a fully firing train produces (Section 5.5.1). Between models compared at the same γ this component cancels, so the centre of mass carries information only as a difference between models, which is the only form in which it is reported.

Quantization is simulated. Fake quantization reproduces the values an integer implementation would hold, but not its arithmetic: the membrane is quantized as a stored state after each spike decision, while on integer hardware the comparison with threshold would itself be made on integers. The energy figures are analytical, at a 45 nm reference node, and below eight bits they rest on an extrapolation of the per-operation costs.

The overlap is read at a single k**.** Every IoU of the grid is computed at $k = 0.20$, the value of Rabbi et al. [33]. The sweep over k exists for one model only, the 2-bit membrane of seed 3456, and there the class ranking reverses between $k = 0.02$ and $k = 0.20$: whether the headline overlap measures the most salient evidence or the ranking of the region around it is settled for that model alone.

Sample sizes of the explanation analyses. Divergence and the fine-tuning level are measured on twenty validation images per class; the numerical floor, the independent-solutions level, the γ analysis and faithfulness on fifteen; stratification on fifteen per class and tercile. The numerical floor and the membrane range come from a single checkpoint. The divergence and faithfulness figures carry bootstrap intervals over images, while the calibration levels they are read against do not. These sizes suffice for the effects reported here, which are large, but limit the resolution with which small differences between configurations can be ranked: at 4 bits the intervals of the single-target arms reach the fine-tuning level.

The fine-tuning level is not matched in epochs. The control arm is measured after its last epoch, six to fourteen epochs from the parent, while each quantized model is the checkpoint selected at the epoch of its best validation accuracy, which can come earlier. The level therefore bounds what continued training does to an explanation, but it is not the effect of exactly the epochs each quantized model received.

Checkpoints are selected on the partition on which they are analyzed. Selection uses accuracy and the analysis uses explanations, so the dependence is weak, but it exists; the test-set evaluation remains the only fully independent estimate.

The saving factor rests on two assumptions. The analog first convolution and every accumulate are costed at the weight bit-width, which gives a joint 4-bit saving of 59. Costing the first convolution in floating point gives 2.8 to 4.2 across seeds; costing the accumulates with 16- or 32-bit integer additions gives 20 to 23 or 10 to 12; both changes together give 2.4 to 3.2. Under wider accumulators the 2-bit models, which fire more, cost more than the 4-bit ones. The operation counts, the memory and the dominance of the membrane arm do not depend on either assumption.

6.4 Future work

Deployment. Hardware deployment was left out of scope. Exporting the selected integer configuration to a neuromorphic platform or a microcontroller would measure the latency, energy and memory that are estimated analytically here.

Normalization. The baseline contains no batch normalization, because quantizing its parameters is a problem distinct from quantizing weights and potentials. Deeper spiking networks need it, and it can be approximated with shifts or folded into the preceding convolution before deployment; repeating the study on a normalized architecture would test whether the conclusions survive.

The temporal axis of the input. Under direct encoding every temporal structure is generated by the network. Feeding the spectrogram column by column, so that time in the input corresponds to time in the signal, would give the temporal explanation a physical meaning.

Wider benchmark. Extending the four classes to the full Gravity Spy taxonomy, with a split by observing run, would make the classification results comparable with the published benchmark.

6.5 Closing remark

Compressed models are increasingly deployed where their explanations are acted upon. This thesis asked whether a quantized spiking network preserves the internal dynamics and reasoning of the network it came from, and found that the question has two prerequisites: calibrating the comparison metrics against the values they report in the absence of any real change, and comparing every model with its own parent. With both in place, the answer for this network is that it largely does down to four bits in every arm, where explanations move about as far as continued training moves them. The one configuration that does not, the 2-bit membrane, shows no sign of it in accuracy.

Correlation and the deletion test register the change only weakly: a PCC of 0.923 would still read as a preserved explanation, and the quantized model's own maps keep a median 94 % of the parent's deletion gap. The change is plain only in the overlap of the highlighted regions, an IoU of 0.438, below the floor of every class (0.70 to 0.84).

A deployment chosen on accuracy alone could have picked this configuration, and that is why the question was worth asking.

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Appendix A
Additional figures

A.1 Data and explainer

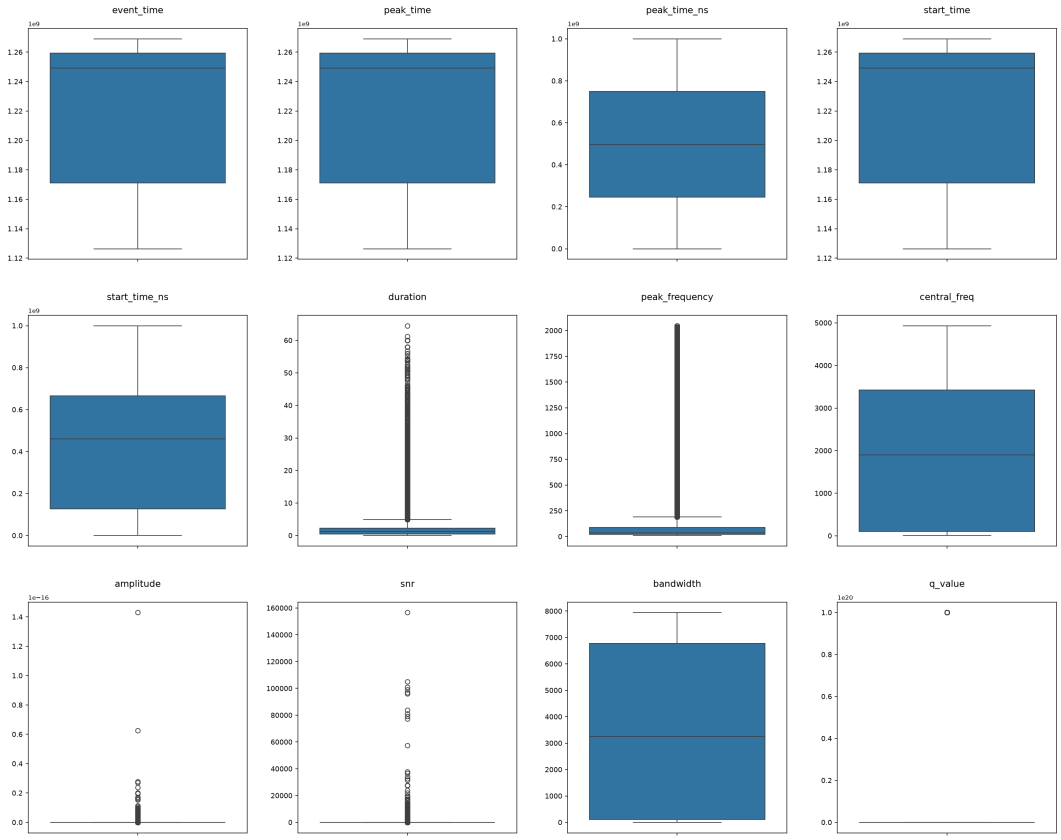


Figure A.1: Distributions of the twelve numerical Omicron descriptors over the full release. The quality factor collapses to a sentinel value of order 10^{20}; amplitude and SNR are dominated by a long tail of outliers.

A.2 The quantization grid

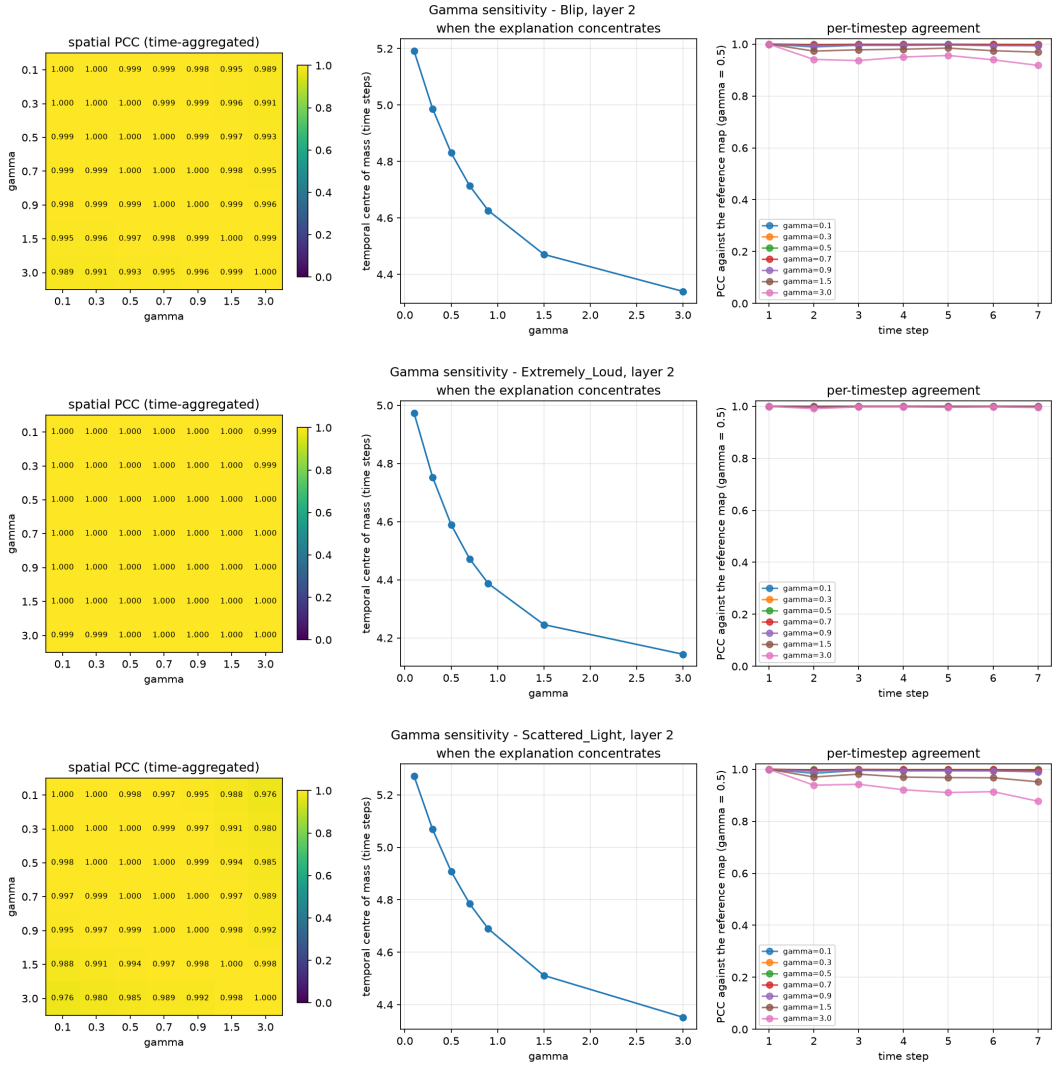


Figure A.2: γ sensitivity at layer 2 for Blip, Extremely_Loud and Scattered_Light, fifteen validation images per class (Violin_Mode is in Figure 5.8).

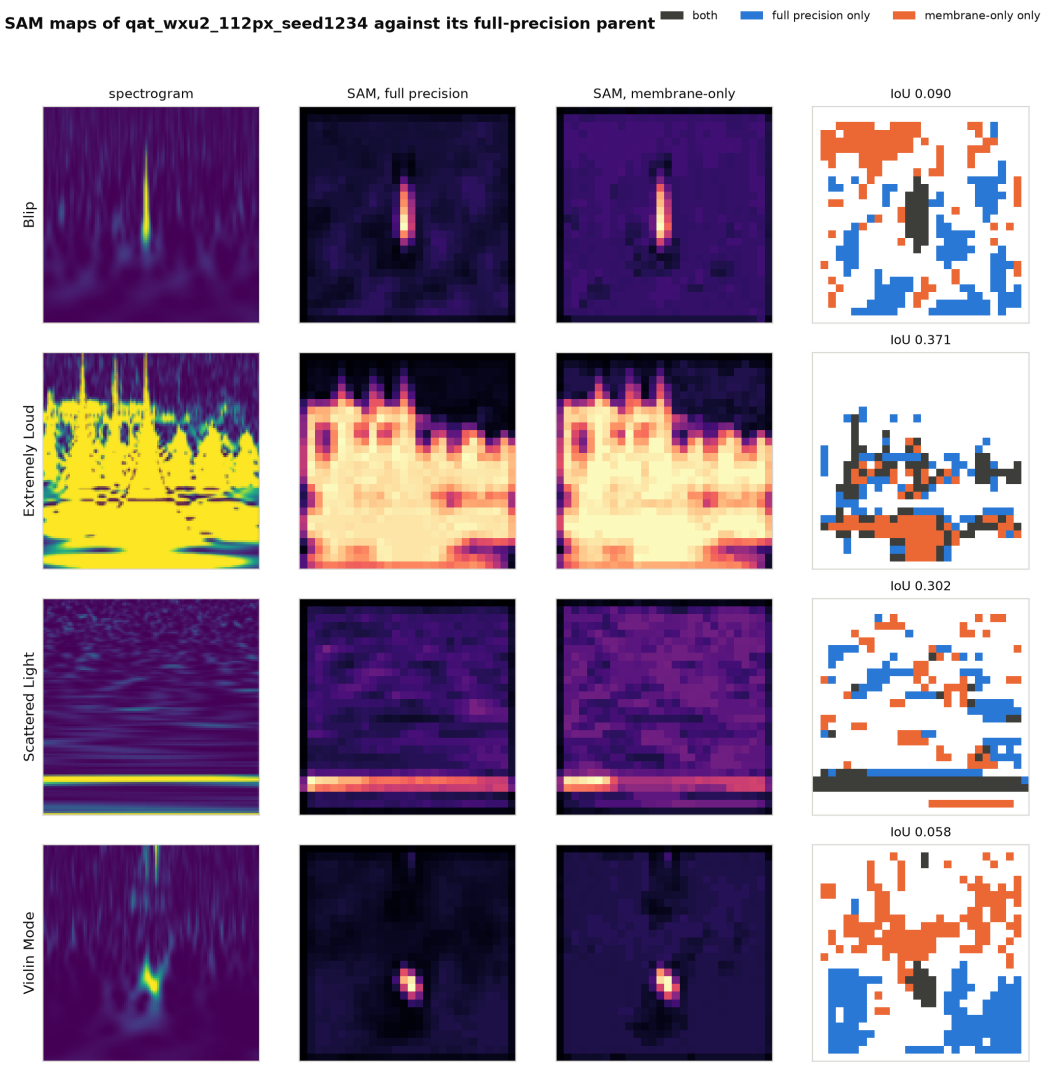


Figure A.3: As Figure 5.11, seed 1234, the seed with the lowest overlap of the arm. In every row the highest-scoring positions coincide; the disagreement lies in the lower-ranked positions that complete the 20 % set.

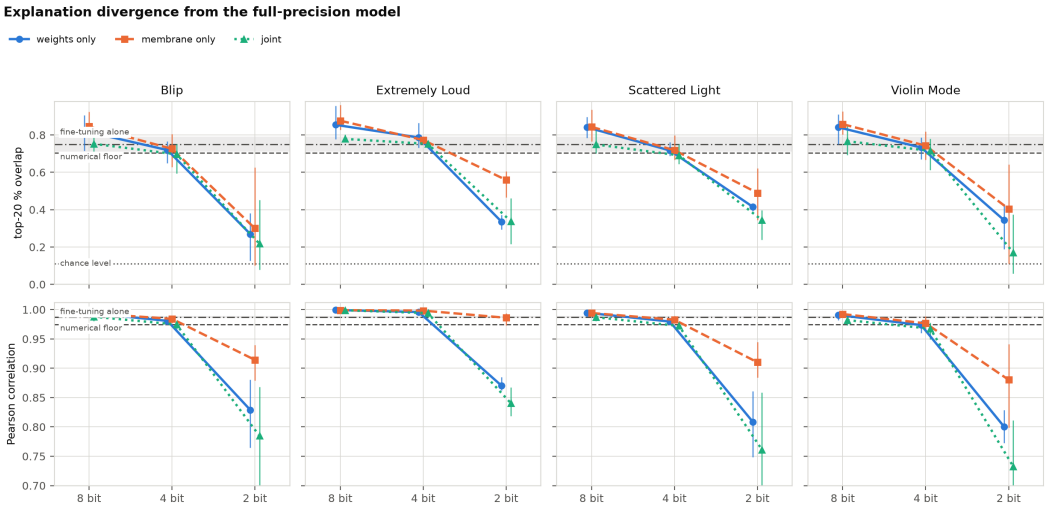


Figure A.4: Top-20 % IoU (top) and spatial PCC (bottom) per class against bit-width, mean over three seeds, bars spanning the seeds. Dashed line: numerical floor of the worst class; dash-dot line and grey band: fine-tuning level, mean over classes and seeds and its seed range. The per-class levels against which the grid is read are in Table 5.11, the values in Tables 5.16 and 5.18.

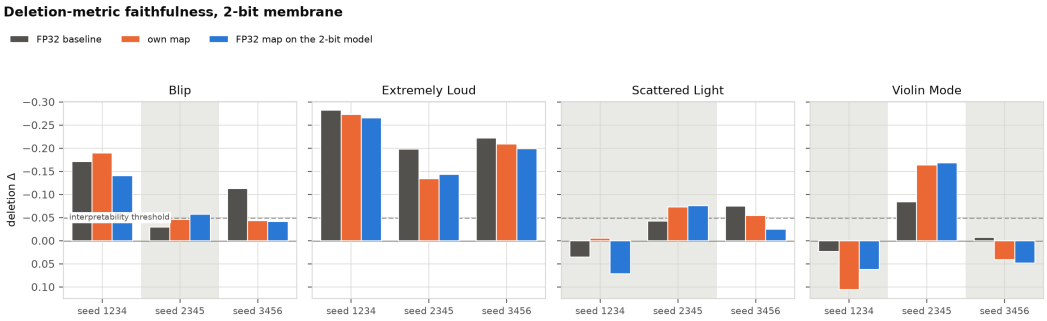


Figure A.5: Deletion gap Δ (inverted axis: upward is more faithful) of the parent with its own map, of the 2-bit membrane model with its own map, and of the 2-bit membrane model with the parent's map, per class and seed, fifteen validation images per cell. Shaded panels: the five cells whose parent baseline does not reach the interpretability threshold of 0.05 (dashed line), omitted from Table 5.20.

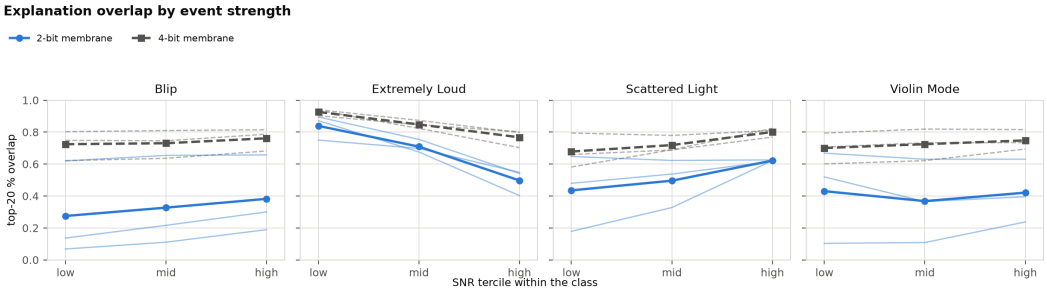


Figure A.6: Top-20 % IoU by SNR tercile computed within each class, membrane arm at 2 and 4 bits; thick lines are the mean of the three seeds, thin lines single seeds, fifteen images per cell. The gradients are those of Table 5.21.

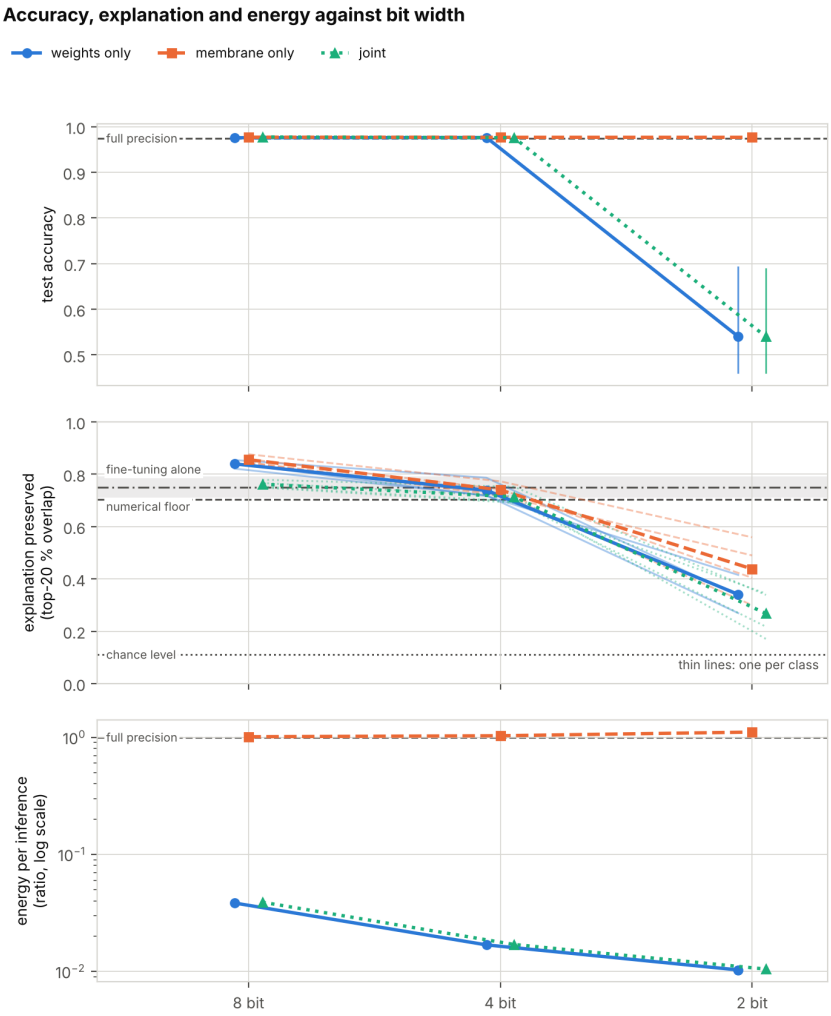


Figure A.7: Test-set counterpart of Figure 5.10: test accuracy (top), top-20 % IoU measured on validation with one thin line per class (middle), estimated energy relative to the parent (bottom), mean over three seeds.

Appendix B
Algorithms

List of Algorithms

1 Spike Activation Map of one image at one layer 28
2 Deletion metric for one image 29
3 Quantization-aware fine-tuning and checkpoint selection 35
4 Explanation divergence between a quantized model and its parent 83
5 Numerical floor by weight perturbation 84
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7 Energy and memory of one operating point 84

The procedures of Algorithms 4–7 are implemented, in order, in `xai/divergence_metrics.py`, `xai/perturbation_floor.py`, `scripts/evaluate_model.py` together with `scripts/per_image_cv.py`, and `xai/energy.py`. SAM denotes Algorithm 1.

Algorithm 4 Explanation divergence between a quantized model and its parent

Require: parent f_A, quantized model f_B of the same seed, images X_c (first 20 of class c in loader order), layer $\ell = 2$, $\gamma = 0.5$, $k = 0.20$
Ensure: per-class PCC, IoU_k, signed and absolute CoM shift, per-step PCC
for $x \in X_c$ **do**
  $M^A \leftarrow \mathrm{SAM}(f_A, x, \ell, \gamma)$, $M^B \leftarrow \mathrm{SAM}(f_B, x, \ell, \gamma)$ {$[T, H, W]$}
  $A \leftarrow \sum_t M_t^A$, $B \leftarrow \sum_t M_t^B$
  $\rho_x \leftarrow \mathrm{PCC}(A, B)$ {undefined if either map is constant}
  $A_k \leftarrow \{i : A_i > Q_{1-k}(A)\}$, $B_k \leftarrow \{i : B_i > Q_{1-k}(B)\}$
  $J_x \leftarrow |A_k \cap B_k| / |A_k \cup B_k|$
  $\delta_x \leftarrow \mathrm{CoM}(M^B) - \mathrm{CoM}(M^A)$ {Eq. 3.5; $\delta_x < 0$: earlier}
  for $t = 1$ **to** $T - 1$ **do**
    $r_{x,t} \leftarrow \mathrm{PCC}(M_t^A, M_t^B)$
  end for
end for
return means over X_c, ignoring undefined values, of ρ_x, J_x, δ_x, $|\delta_x|$ and $r_{x,t}$

Algorithm 5 Numerical floor by weight perturbation

Require: reference checkpoint f (seed 1234), fractions $\Phi = \{0.01, 0.1, 0.5, 1\}$, noise seed 999, images X_c (15 per class)
Ensure: per-class PCC, IoU_k, $|\Delta\mathrm{CoM}|$, per-step PCC at every fraction
$w_{\max} \leftarrow \max |w|$ over all convolutional and linear weights
$s_8 \leftarrow 2w_{\max}/(2^8 - 1)$ {global 8-bit step, 0.00168}
for $\phi \in \Phi$ **do**
  $\tilde{f} \leftarrow$ copy of f; add $\mathcal{N}(0, (\phi s_8)^2)$ to every weight tensor {biases and LIF buffers untouched}
  compare $\tilde{f}$ with f on X_c as in Algorithm 4
end for

Algorithm 6 Coefficient of variation of the inter-spike intervals

Require: model f, validation partition $\mathcal{V}$, $T = 8$, 200 positions per layer, seed 42, minimum of 5 intervals
Ensure: pooled CV per layer and its defined fraction; per-image CV statistics
draw 200 neuron positions P_ℓ per hidden layer, once, at the first batch
$\mathcal{I}_p \leftarrow \emptyset$ for every $p \in P_\ell$
for $x \in \mathcal{V}$ **do**
  $S \leftarrow$ hidden spike trains of $f(x)$
  for every hidden neuron n **do**
    $\tau \leftarrow \{t : S_{n,t} = 1\}$, $d \leftarrow$ successive differences of τ
    if $n \in P_\ell$ **then**
      $\mathcal{I}_n \leftarrow \mathcal{I}_n \cup d$ {pooled over images}
    end if
    if $|\tau| \geqslant 3$ **then**
      $\mathrm{CV}_{n,x} \leftarrow \sigma(d)/\mu(d)$ {per-image; two intervals at least}
    end if
  end for
end for
$\mathrm{CV}_p \leftarrow \sigma(\mathcal{I}_p)/\mu(\mathcal{I}_p)$ for every p with $|\mathcal{I}_p| \geqslant 5$
return per layer: mean of CV_p, fraction of P_ℓ where it is defined; mean of $\mathrm{CV}_{n,x}$, fraction of neuron-image pairs with $|\tau| \geqslant 3$, fraction with $\mathrm{CV}_{n,x} = 0$

Algorithm 7 Energy and memory of one operating point

Require: per-hidden-layer firing rates $f_1, \ldots, f_4$ of the selected checkpoint, T, dense MAC counts MAC_ℓ, weight bit-width b_w, membrane bit-width b_u (absent: 32-bit float), $N_w = 295\,088$ weights, $N_u = 94\,080$ hidden neurons
Ensure: synaptic operations, energy, memory
if b_w absent **then**
  $c_{\mathrm{MAC}} \leftarrow 4.6\,\mathrm{pJ}$, $c_{\mathrm{AC}} \leftarrow 0.9\,\mathrm{pJ}$
else
  $c_{\mathrm{AC}} \leftarrow 0.03\,(b_w/8)\,\mathrm{pJ}$, $c_{\mathrm{MAC}} \leftarrow 0.2\,(b_w/8)^2 + c_{\mathrm{AC}}\,\mathrm{pJ}$ {exact at 8 bits, extrapolated below}
end if
$E \leftarrow \mathrm{MAC}_{\mathrm{conv1}} \cdot c_{\mathrm{MAC}}$ {analog input, computed once}
for $(\ell, g) \in \{(\mathrm{conv2}, 1), (\mathrm{conv3}, 2), (\mathrm{conv4}, 3), (\mathrm{linear}, 4)\}$ **do**
  $\mathrm{SOP}_\ell \leftarrow f_g \cdot T \cdot \mathrm{MAC}_\ell$, $E \leftarrow E + \mathrm{SOP}_\ell \cdot c_{\mathrm{AC}}$
end for
$\mathrm{Mem} \leftarrow N_w \cdot (b_w \text{ or } 32)/8 + N_u \cdot (b_u \text{ or } 32)/8$ bytes
return $\sum_\ell \mathrm{SOP}_\ell$, E, Mem

Appendix C
Reproducibility

All code, configuration and result files are in the public repository

`https://github.com/devs-glitch/Tesi`

Every analysis reads the model, the hyperparameters, the SAM layer and γ, and the partition from `config/baseline_manifest.json`.

C.1 Environment and data

The code requires Python $\geqslant$ 3.12 and is installed from the repository root with `pip install -e .` followed by `pip install -r requirements.txt` (PyTorch $\geqslant$ 2.5, torchvision $\geqslant$ 0.20, snnTorch $\geqslant$ 0.9, Optuna $\geqslant$ 4.0, Weights & Biases $\geqslant$ 0.18, NumPy $\geqslant$ 2.0, pandas, scikit-learn, umap-learn, matplotlib, seaborn). Every script is run from the repository root as a module, `python -m <package>.<script>`. All training and evaluation ran on a single 6 GB consumer GPU; the thirty fine-tuning runs of the grid took 46 924 s (13.0 h).

The exact versions of every package used for the reported runs are recorded in `requirements-lock.txt`; the runs used PyTorch 2.6, whose `torch.clamp` passes a gradient of 1 at the ends of its range. More recent versions (2.14 among them) pass 0 there, so the quantizer sets the gradient on its grid explicitly, and `python -m model.quantization` checks it.

The metadata are the Gravity Spy release on Zenodo, record 5649212 [15]. The selected events are written to `data/processed/selected_dataset.csv` and the spectrograms to `data/processed/<class>/`. The partition file `data/split_assignment.csv` lists 9 568 files; the two that are unavailable at source are listed in `data/corrupted_files.txt` and dropped from the training partition at load time. The first sixteen hexadecimal digits of its SHA-256 digest, `754b2515b6509075`, are recorded in the manifest and checked by `scripts/verify_manifest.py`.

C.2 Seeds

Partition: 42. Class balancing and UMAP subsample: 42. Training, reference and fine-tuning: 1234, 2345, 3456. CV-ISI positions: 42. Weight perturbation of the numerical floor: 999. Random deletion order: a generator seeded with 0 for Table 5.6 and cross-faithfulness, the global seed 42 for the curves of Figure 5.9. Bootstrap resampling: seed 0, 2 000 resamples. Image samples for every SAM analysis are the first n validation images of each class in loader order, so they are identical across models.

C.3 From result to script

Tables C.1 and C.2 list the scripts in the order in which they must be run. Trained checkpoints are written to `runs/<tag>/` (`best_model.pt`, `final_model.pt`, `run.json`) and are not versioned; every step from the per-model validation pass onwards needs them.

Result	Module	Output
Metadata download	`scripts.dataset_downloader`	raw CSV files
Class and feature selection, balancing	`scripts.data_prep`	`data/processed/selected_dataset.csv`, Figs. 3.1–3.4
Spectrogram download	`scripts.image_downloader`	`data/processed/<class>/`
Frozen partition	`scripts.freeze_split`	`data/split_assignment.csv`
Tabular baseline	`scripts.tabular_baseline`	`results/tabular_baseline.json`
Resolution study	`scripts.resolution_study`	`runs/resolution_study.csv`, copied to `results/`
Dead neurons by resolution	`scripts.dead_neurons_by_resolution -out results/dead_neurons.json`	`results/dead_neurons.json`
Hyperparameter search (v4, v5)	`scripts.tune`	`config/best_hyperparams*.json`
FP32 reference, manifest	`scripts.fp32_reference`	`config/baseline_manifest.json`
Cutoff sweep	`scripts.dead_neuron_cutoff_sweep`	`results/tables/dead_neuron_sweep.csv`, Fig. 5.1
Confusion matrix, firing rates, CV-ISI	`scripts.diagnostics_figures`	Figs. 5.2, 5.3, 5.4
SAM per time step	`scripts.sam_visualization`	Fig. 5.5
SAM class and layer statistics	`xai.sam_class_comparison`, `xai.sam_layer_comparison`	Figs. 5.6, 5.7
γ sensitivity	`xai.gamma_sensitivity_analysis`	`results/gamma_sensitivity.json`, Figs. 5.8, A.2
Deletion metric	`xai.cross_faithfulness` (table), `xai.deletion_metric` (curves)	Table 5.6, Fig. 5.9
Independent solutions	`xai.noise_floor`	`results/noise_floor.json`
Numerical floor	`xai.perturbation_floor`	`results/perturbation_floor.json`

Table C.1: Reproducibility, full-precision stages. Steps up to the resolution study read the input size from the environment variable `SNN_INPUT_SIZE`; from the reference onwards it is read from the manifest.

Result	Module	Output
Membrane range	`scripts.membrane_range`	`config/membrane.json`
Clipping study	`scripts.test_clipping`	`results/clipping_test.json`
Quantizers	`model.quantization`	—
PTQ gate	`scripts.ptq_check`	`results/ptq_check.json`
Fine-tuning grid	`scripts.grid_runner` (calls `scripts.train_quantised`)	`runs/qat_*`, `results/grid_summary.json`
Per-model validation pass	`scripts.evaluate_model`	`results/grid/<tag>.json`, `_per_sample.csv`, `_rates.npz`
Fine-tuning level	`scripts.finetuning_anchor`	`results/finetuning_anchor.json`
SAM divergence	`xai.divergence_metrics`	`results/divergence.json`
Cross-faithfulness	`xai.cross_faithfulness`	`results/cross_faithfulness.json`
SNR stratification	`xai.stratify`	`results/stratified.json`
EMD, disagreement subset	`scripts.grid_analysis`	`results/grid_analysis.json`
Per-image CV-ISI	`scripts.per_image_cv`	`results/per_image_cv.json`
Energy and memory	`xai.energy`	`results/energy.json`
Energy against the control	`xai.energy -rates final -baseline qat_wxux_{input_size}px_seed{seed} -out results/energy_final.json`	`results/energy_final.json`
SAM maps after quantization	`xai.sam_qat_comparison -run <tag>`	Figs. 5.11, 5.12, A.3
Grid figures and tables	`scripts.figures.result_figures`	`results/tables/*.csv`, Figs. 5.13, 5.14, A.4–A.7
Methods figures	`scripts.architecture_figure`, `scripts.figures.*`	Figs. 3.6, 4.1, 4.2, 4.3
Grid summary figure	`scripts.figures.grid_headline_figure`	Fig. 5.10
Readout ties, CV controls, bootstrap intervals	`scripts.robustness_checks all`	`results/robustness/`
Test set (opened once)	`scripts.evaluate_test`	`results/test/`, `results/test_opened.json`

Table C.2: Reproducibility, quantization study. The test partition was opened once, on 7 October 2026, for all thirty-three models; the opening is logged in `results/test_opened.json` and the script refuses a second one unless forced.